EXAM PACK

EEE459

Compound Semiconductor and Hetero-Junction Devices



Exam Pack

Session 2020–21 *Solutions and formula sheet of the 4th Year 1st Semester Final Examination, 2024*

IN THIS PACK

1 PAST PAPER SOLUTIONS
Question Paper Solutions 2020–21 — pages 3--98 — 96 pages

2 FORMULA SHEET
Formula Sheet — pages 99--108 — 10 pages

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Prepared by Nur Mohammad

November 2026

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1 PAST PAPER SOLUTIONS
Question Paper Solutions 2020–21

pages 3--98

Question Paper 5
1(a) Electron Mobility in Si and GaAs 11
1(b) Velocity Saturation and the Gunn Effect 15
1(c) Light and Heavy Hole Bands of Silicon 19
1(d) The GaAs and AlGaAs Heterojunction 23
2(a) Semiconductors for the Near-Infrared Region . 29
2(b) Lattice Matching, Strain and Critical Thickness 32
2(c) Isoelectronic Traps in GaAsP 36
2(d) Composition of a Quaternary Alloy from Fig. 1 38
3(a) Band Diagrams of a Biased Schottky Junction 43
3(b) Density of States in 3D, 2D and 1D 46
3(c) Tungsten on n-Type Silicon Schottky Diode . . 50
3(d) Image-Force Barrier Lowering in GaAs 54
4(a) Fields and Potentials in a Straddling Heterojunction 58
4(b) Non-Ideal Effects on the Schottky Barrier Height 64
4(c) Formation of a Linearly Graded Junction 68
5(a) JFET Channel Profiles in the Three Regions . 72
5(b) Internal Pinch-Off Voltage of an n-Channel JFET 75
5(c) Gate Voltage to Open a GaAs JFET Channel . 79
5(d) Ohmic Metal-Semiconductor Contact 81
6(a) Ebers-Moll Model and the Saturation Voltage . 85
6(b) High-Level Injection and Current Crowding . . 90
6(c) Maximum Current and Transconductance of a Silicon JFET 94

2 FORMULA SHEET
Formula Sheet

pages 99--108

1 Carrier transport 102
2 Bands and density of states 102
3 Compound alloys 103
4 Heterojunctions 103
5 Metal–semiconductor contacts 104
6 Graded junction 104
7 JFET 105
8 Bipolar transistor 105
9 Key formulas 106
10 Symbol index 107

PAST PAPER SOLUTIONS

EEE459

Compound Semiconductor and Hetero-Junction Devices

Question Paper Solutions

Session 2020–21

4th Year 1st Semester Final Examination, 2024

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

EACH ANSWER

Theory → Answer → Keyword Explanation → Notes and Remarks → Final Concise Answer

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November 2026

BACK OF THE COVER

The paper at a glance

3 h	100	4 of 6	45 min
Total time	Total marks	Questions to answer	Per question

Time = marks × 1.8 min (180 min for 100 marks). Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
Part-A · Answer any two of the following questions				
Q1	(a) Electron Mobility in Si and GaAs	6	11 min	☐
	(b) Velocity Saturation and the Gunn Effect	5	9 min	
	(c) Light and Heavy Hole Bands of Silicon	6	11 min	
	(d) The GaAs and AlGaAs Heterojunction	8	14 min	
Q2	(a) Semiconductors for the Near-Infrared Region	5	9 min	☐
	(b) Lattice Matching, Strain and Critical Thickness	9	16 min	
	(c) Isoelectronic Traps in GaAsP	4	7 min	
	(d) Composition of a Quaternary Alloy from Fig. 1	7	13 min	
Q3	(a) Band Diagrams of a Biased Schottky Junction	5	9 min	☐
	(b) Density of States in 3D, 2D and 1D	6	11 min	
	(c) Tungsten on n-Type Silicon Schottky Diode	8	14 min	
	(d) Image-Force Barrier Lowering in GaAs	6	11 min	
Part-B · Answer any two of the following questions				
Q4	(a) Fields and Potentials in a Straddling Heterojunction	12	22 min	☐
	(b) Non-Ideal Effects on the Schottky Barrier Height	8	14 min	
	(c) Formation of a Linearly Graded Junction	5	9 min	
Q5	(a) JFET Channel Profiles in the Three Regions	6	11 min	☐
	(b) Internal Pinch-Off Voltage of an n-Channel JFET	8	14 min	
	(c) Gate Voltage to Open a GaAs JFET Channel	7	13 min	
	(d) Ohmic Metal-Semiconductor Contact	4	7 min	
Q6	(a) Ebers-Moll Model and the Saturation Voltage	9	16 min	☐
	(b) High-Level Injection and Current Crowding	8	14 min	
	(c) Maximum Current and Transconductance of a Silicon JFET	8	14 min	

Formulas and constants follow Neamen, Semiconductor Physics and Devices (4th ed.); compound-semiconductor data follow Sze and Ng, Physics of Semiconductor Devices.

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
4th Year 1st Semester Final Examination, 2024
Course Code: EEE 459, Session: 2020-2021
Course Title: Compound Semiconductor and Hetero-Junction Devices

Full Marks: 100 **Credit: 3.0** **Total Time: 3 Hours**

Part-A

*[Answer any **two** of the following questions]*

1. (a) Between Si and GaAs, which has higher electron mobility and why? 6
   (b) Discuss Velocity saturation and Gunn effect . 5
   (c) Prove that along the [100] and [111] direction in k-space, for Si, the variation of electron energy with respect to wave number is higher for light holes than for heavy holes. 6
   (d) For a $GaAs - Al_xGa_{1-x}As$ heterojunction, determine the following:
      (i) Energy-Band diagram: Case-1: Before contact, Case-2: After contact
      (ii) Show that the built-in potential (using hole concentration) $eV_{bi} = -\Delta E_v + kT \ln\left(\frac{P_P\, N_{VN}}{P_N\, N_{VP}}\right)$. Given, GaAs is doped P type, $Al_xGa_{1-x}As$ is doped N^+ type. Assume: $x = 0.25$ & $\Delta E_c = \frac{1}{2}\Delta E_g$ P_p = hole concentration in P type region, P_N = hole concentration in N type region, N_{VN} and N_{VP} are density of states of electron and hole respectively. 8

2. (a) Which semiconductors are most extensively used in near infra-red region? Explain the reasons behind it. 5
   (b) Why is lattice matching of semiconductor important for hetero-structured devices? Discuss the effect of strain and critical thickness in epitaxial growth. 9
   (c) Discuss the effect of isoelectric trap in the ternary compound semiconductor $GaAs_{1-x}P_x$. 4
   (d) Fig. 1 shows the Bandgap Vs. lattice constant for different binary compounds. Assume that a quaternary compound $In_xGa_{1-x}As_ySb_{1-y}$ is grown on InP substrate. Find the composition if the bandgap is 1 eV and lattice parameter is 5.93 Å. 7

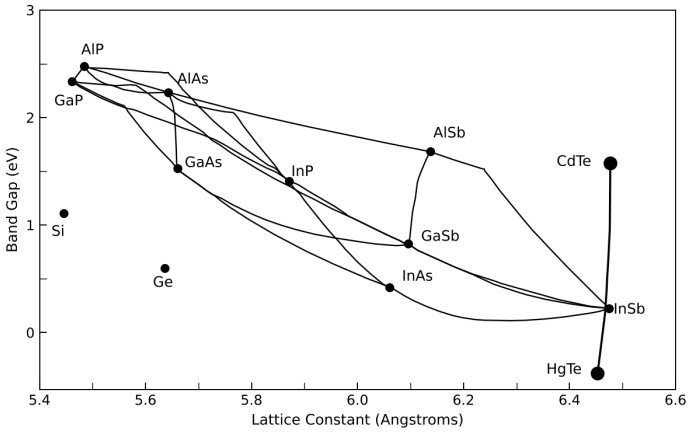


Fig. 1: Band gap vs. lattice constant of binary compounds (redrawn from the paper)

3. (a) Draw the ideal energy band diagram of a metal semiconductor junction under:

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(i) Reverse bias condition
(ii) Forward bias condition 5

(b) Explain how the density of states changes qualitatively as the dimensionality reduces from $3D \rightarrow 2D \rightarrow 1D$. Sketch $D(E)$ vs E for each case. 6

(c) Consider a contact between tungsten and n-type silicon doped to $N_d = 1 \times 10^{16}$ cm^{-3} at $T = 300$ K. Determine the

(i) theoretical barrier height
(ii) built-in potential barrier
(iii) maximum electric field in the metal-semiconductor diode for zero applied bias, and
(iv) the junction capacitance. Given that, the metal work function for tungsten (W) is 4.55 V and the electron affinity for silicon is 4.01 V. Assume $N_c = 2.8 \times 10^{19}$ cm^{-3}. 8

(d) Consider a gallium arsenide metal-semiconductor contact in which the electric field in the semiconductor is assumed to be $E = 6.8 \times 10^4$ V cm^{-1}. Calculate the Schottky barrier lowering (due to image force induced lowering) and the position of the maximum barrier height. 6

Part-B

*[Answer any **two** of the following questions]*

4. (a) A narrow bandgap N-type and a wide bandgap P-type material are used to form a heterojunction. If they are straddling type, then draw the energy band diagram of the heterojunction before contact and after contact at thermal equilibrium. With necessary assumption, derive expression for electric field and electrostatic potential in both regions. 12

(b) Discuss the following non-ideal effects on barrier height of a metal-semiconductor heterojunction:

(i) Image-force-induced lowering
(ii) Interface states 8

(c) Describe how a linearly graded junction is formed. 5

5. (a) Draw the channel space charge region profiles of a single gate n-channel JFET for the following operating regions: (where V_p, V_g, and V_d are pinch-off, gate, and drain voltage respectively)

(i) $V_g > V_p$ and $V_d < V_{d,sat}$
(ii) $V_g > V_p$ and $V_d = V_{d,sat}$
(iii) $V_g > V_p$ and $V_d > V_{d,sat}$ 6

(b) Derive the formula of internal pinch-off voltage in an n-channel JFET. 8

(c) Consider a GaAs n-channel pn JFET at $T = 300$ K with $N_a = 1 \times 10^{18}$ cm^{-3}, $N_d = 3 \times 10^{15}$ cm^{-3}, $\epsilon_s = 13.1$, and $a = 0.70$ μm. Determine the forward-bias gate voltage required to open a channel region that is 0.10 μm thick with zero drain voltage. 7

(d) Sketch the ideal energy band diagram of a metal semiconductor junction in which $\Phi_m < \Phi_s$. Explain why this is an ohmic contact. 4

6. (a) Draw the equivalent circuit for basic Ebers-Moll model as used in BJT. Also derive the expression of $V_{CE}(sat)$ from this model. 9

(b) Explain high level injection and current crowding effects in a BJT with necessary figures. How can the problem of current crowding be solved? 8

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(c) Consider a silicon n-channel JFET at $T = 300$ K with the following parameters: $N_a = 1 \times 10^{18}$ cm^{-3}, $N_d = 1 \times 10^{16}$ cm^{-3}, $a = 0.75$ μm, $L = 10$ μm, $W = 30$ μm, and $\mu_n = 1000$ cm^2/(V s)

(i) Calculate the maximum current in an n-channel JFET.

(ii) Determine the maximum transconductance of an n-channel depletion mode JFET biased in the saturation region. $I_{P1} = 0.522$ mA, $V_{bi} = 0.814$ V, and $V_{po} = 4.35$ V. 8

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Contents

1(a) Electron Mobility in Si and GaAs **9**
1 Theory . . . 9
2 Answer . . . 12
3 Keyword Explanation . . . 13
4 Notes and Remarks . . . 13

1(b) Velocity Saturation and the Gunn Effect **13**
1 Theory . . . 13
2 Answer . . . 16
3 Keyword Explanation . . . 16
4 Notes and Remarks . . . 17

1(c) Light and Heavy Hole Bands of Silicon **17**
1 Theory . . . 17
2 Answer . . . 18
3 Keyword Explanation . . . 20
4 Notes and Remarks . . . 21

1(d) The GaAs and AlGaAs Heterojunction **21**
1 Theory . . . 21
2 Answer . . . 23
3 Keyword Explanation . . . 25
4 Notes and Remarks . . . 26
5 Final Concise Answer . . . 26

2(a) Semiconductors for the Near-Infrared Region **27**
1 Theory . . . 27
2 Answer . . . 29
3 Keyword Explanation . . . 29
4 Notes and Remarks . . . 30

2(b) Lattice Matching, Strain and Critical Thickness **30**
1 Theory . . . 30
2 Answer . . . 32
3 Keyword Explanation . . . 33
4 Notes and Remarks . . . 33

2(c) Isoelectronic Traps in GaAsP **34**
1 Theory . . . 34
2 Answer . . . 35
3 Keyword Explanation . . . 35
4 Notes and Remarks . . . 35

2(d) Composition of a Quaternary Alloy from Fig. 1 **36**
1 Theory . . . 36
2 Answer . . . 38
3 Keyword Explanation . . . 39
4 Notes and Remarks . . . 39
5 Final Concise Answer . . . 40

3(a) Band Diagrams of a Biased Schottky Junction **41**

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1 Theory . . . 41
2 Answer . . . 43
3 Keyword Explanation . . . 44
4 Notes and Remarks . . . 44

3(b) Density of States in 3D, 2D and 1D **44**
1 Theory . . . 44
2 Answer . . . 46
3 Keyword Explanation . . . 47
4 Notes and Remarks . . . 48

3(c) Tungsten on n-Type Silicon Schottky Diode **48**
1 Theory . . . 48
2 Answer . . . 50
3 Keyword Explanation . . . 51
4 Notes and Remarks . . . 52

3(d) Image-Force Barrier Lowering in GaAs **52**
1 Theory . . . 52
2 Answer . . . 53
3 Keyword Explanation . . . 54
4 Notes and Remarks . . . 55
5 Final Concise Answer . . . 55

4(a) Fields and Potentials in a Straddling Heterojunction **56**
1 Theory . . . 56
2 Answer . . . 59
3 Keyword Explanation . . . 61
4 Notes and Remarks . . . 62

4(b) Non-Ideal Effects on the Schottky Barrier Height **62**
1 Theory . . . 62
2 Answer . . . 65
3 Keyword Explanation . . . 66
4 Notes and Remarks . . . 66

4(c) Formation of a Linearly Graded Junction **66**
1 Theory . . . 66
2 Answer . . . 68
3 Keyword Explanation . . . 69
4 Notes and Remarks . . . 69
5 Final Concise Answer . . . 69

5(a) JFET Channel Profiles in the Three Regions **70**
1 Theory . . . 70
2 Answer . . . 72
3 Keyword Explanation . . . 72
4 Notes and Remarks . . . 73

5(b) Internal Pinch-Off Voltage of an n-Channel JFET **73**
1 Theory . . . 73
2 Derivation / Explanation . . . 74

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3 Answer . . . 75
4 Keyword Explanation . . . 76
5 Notes and Remarks . . . 77

5(c) Gate Voltage to Open a GaAs JFET Channel 77
1 Theory . . . 77
2 Answer . . . 78
3 Keyword Explanation . . . 79
4 Notes and Remarks . . . 79

5(d) Ohmic Metal-Semiconductor Contact 79
1 Theory . . . 79
2 Answer . . . 81
3 Keyword Explanation . . . 82
4 Notes and Remarks . . . 82
5 Final Concise Answer . . . 82

6(a) Ebers-Moll Model and the Saturation Voltage 83
1 Theory . . . 83
2 Answer . . . 85
3 Keyword Explanation . . . 87
4 Notes and Remarks . . . 88

6(b) High-Level Injection and Current Crowding 88
1 Theory . . . 88
2 Answer . . . 90
3 Keyword Explanation . . . 91
4 Notes and Remarks . . . 92

6(c) Maximum Current and Transconductance of a Silicon JFET 92
1 Theory . . . 92
2 Answer . . . 94
3 Keyword Explanation . . . 95
4 Notes and Remarks . . . 95
5 Final Concise Answer . . . 95

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1.(a) Between Si and GaAs, which has higher electron mobility and why? Marks: 6

1. Theory

1. What is mobility? (Core Idea)

A semiconductor has free electrons. Even with no voltage applied, they are never still: at room temperature each one moves fast in a random direction, hits something, changes direction, and so on. This is **thermal motion**. Because the directions are random, the electrons go nowhere on average, and no current flows.

When we apply an electric field E, a small push is added to this random motion. On average the electrons now creep in one direction (against the field, because their charge is negative). This slow average motion is called **drift**, and its average speed is the **drift velocity** v_d.

For small fields, the drift velocity is proportional to the field:

$$v_d = \mu_n E$$

where

- v_d is the drift velocity of the electrons (cm/s)
- E is the electric field (V/cm)
- μ_n is the **electron mobility** ($\text{cm}^2/\text{V s}$): how easily an electron moves in the material

So mobility answers a simple question: *for each volt per centimetre of field, how fast do the electrons drift?* A high mobility means a large current for a small field, and also a fast device, because the electrons cross it quickly. The drift current density is $J_n = en\mu_n E$ (Neamen), with e the electron charge and n the electron concentration.

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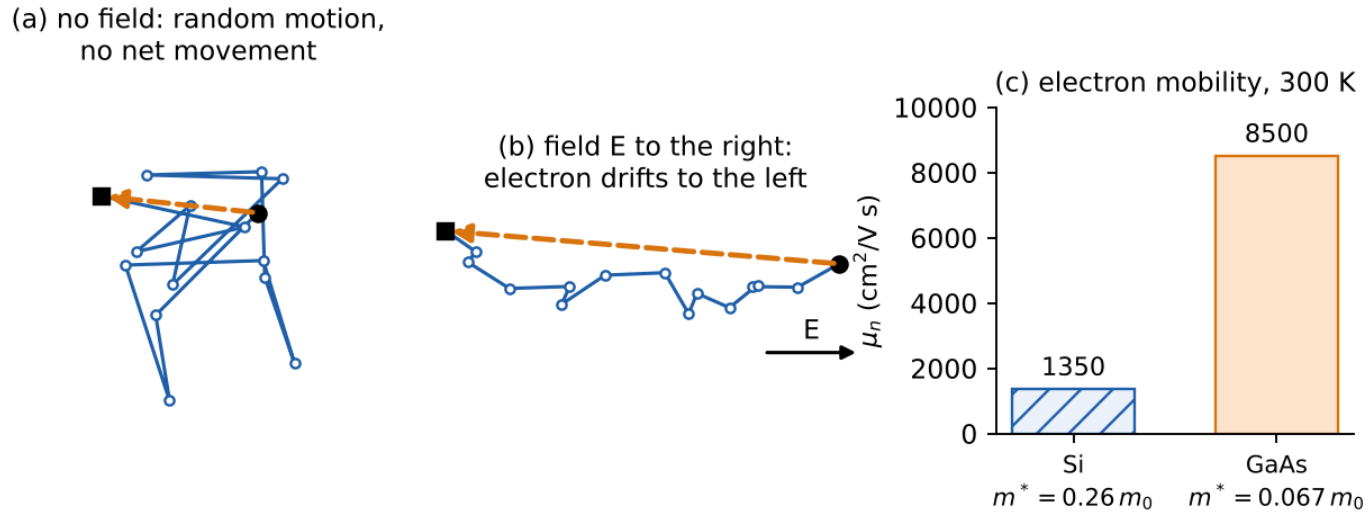


Figure 1: Drift of an electron: random thermal motion with no field, a slow net drift with a field, and the electron mobilities of Si and GaAs at 300 K

Reading the figure. In panel (a) the solid blue zig-zag line is the path of one electron with no field; each small circle is a collision. The dashed orange arrow from the start (filled circle) to the end (filled square) is short and points anywhere: there is no net movement. In panel (b) a field points to the right. The path is still a zig-zag, but every free flight is bent a little to the left, so the dashed orange arrow (the net movement) is long and points to the left. Panel (c) compares the mobilities: the hatched bar for Si (1350 cm^2/V s) and the plain filled bar for GaAs (8500 cm^2/V s).

2. Why do electrons not speed up for ever? (Scattering)

A force $-eE$ acts on the electron, so between collisions it accelerates. But after a short time it hits something and its direction is randomised again. This event is called **scattering**. The two main scatterers are:

- **Lattice (phonon) scattering:** the atoms of the crystal vibrate with heat. These vibrations (their quanta are called **phonons**) disturb the perfect periodic crystal and deflect the electron. It grows with temperature.
- **Ionized impurity scattering:** the dopant atoms are charged ions. Their Coulomb force bends the path of a passing electron. It grows with doping.

The average time between two collisions is the **mean free time** τ. It is very short, a fraction of a picosecond.

3. Where does the mobility formula come from?

Newton's law for the electron, using its effective mass m^* (Section 4), is

$$m^* \frac{dv}{dt} = -eE$$

Starting from a random direction after a collision, the electron gains an extra velocity $-eEt/m^*$ during a free flight of time t. Averaged over all flights, the free time is τ, so the average extra velocity, the drift velocity, is

$$v_d = -\frac{e\tau}{m^*}E \quad \Rightarrow \quad \boxed{\mu_n = \frac{e\,\tau}{m^*}}$$

where

- $e = 1.602 \times 10^{-19}$ C is the electron charge
- τ is the mean free time between collisions (s)
- m^* is the conductivity effective mass of the electron (kg)

This formula tells us exactly what makes a high mobility: **a long time between collisions** (large τ)

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and **a light electron** (small m^*). A light particle gains more speed from the same push before the next collision.

4. What is the effective mass?

An electron inside a crystal is not free: it moves through the electric potential of the atoms. Quantum mechanics shows that we can still use Newton's law if we replace the real mass m_0 with an **effective mass** m^* that already includes the effect of the crystal.

The effective mass is read from the **E–k diagram** of the material: a plot of electron energy E against wave number k (which is the electron's momentum divided by $\hbar$). Near the bottom of the conduction band the plot is a parabola,

$$E = E_c + \frac{\hbar^2 k^2}{2m^*}$$

so a **sharply curved** band means a **small** m^* (a light electron), and a **flat** band means a **large** m^* (a heavy electron). This idea is built in full in 1(c).

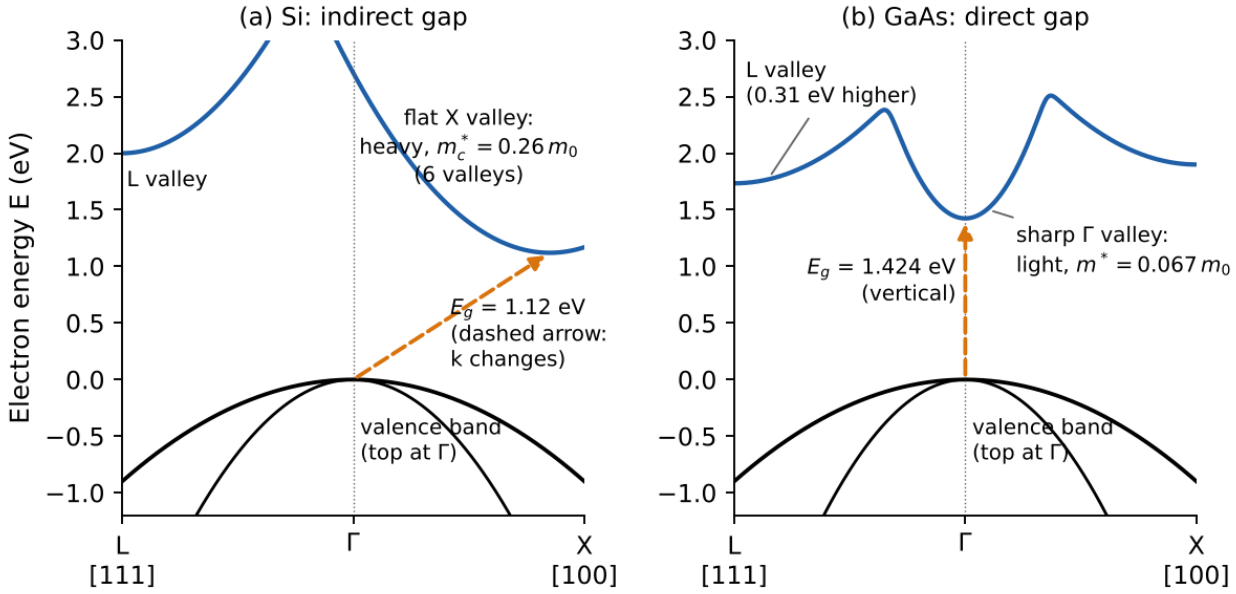


Figure 2: Simplified E-k diagrams of Si (indirect gap, flat X valleys) and GaAs (direct gap, sharp Gamma valley)

Reading the figure. The solid blue curve is the bottom of the conduction band; the solid black curves are the top of the valence band (heavy and light holes). The horizontal axis runs from the zone edge L (along [111]) through the centre Γ ($k = 0$) to the zone edge X (along [100]).

- **Si (panel a):** the lowest conduction-band point is not at Γ but near X, at about 0.85 of the way to the zone edge. So Si is an **indirect-gap** semiconductor (the dashed orange arrow from the valence-band top to the conduction-band bottom is slanted: k changes). There are six such X valleys (one along each of $\pm x, \pm y, \pm z$). Each valley is shaped like a cigar: heavy along its axis ($m_l = 0.98\,m_0$) and light across it ($m_t = 0.19\,m_0$).
- **GaAs (panel b):** the lowest point is at Γ, directly above the valence-band top, so GaAs is a **direct-gap** semiconductor (the dashed orange arrow is vertical). The Γ valley is single, round (the same in every direction) and very sharply curved: $m^* = 0.067\,m_0$. The next valley (L) is 0.31 eV higher.

5. The conductivity effective mass of Si

In Si an electron in one cigar-shaped valley is heavy in one direction and light in two. For current in any direction the six valleys together act like one average mass, the **conductivity effective mass**:

$$m_{cn}^* = \frac{3}{\frac{1}{m_l} + \frac{2}{m_t}} = \frac{3}{\frac{1}{0.98} + \frac{2}{0.19}}\,m_0 = \frac{3}{11.55}\,m_0 = 0.26\,m_0$$

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This is the mass that goes into $\mu_n = e\tau/m^*$. (Neamen also lists $m_n^* = 1.08\,m_0$ for Si; that is the **density-of-states** mass, used for counting states, not for mobility.)

6. How long is the mean free time in each material?

Turning the mobility formula round, $\tau = \mu_n m^*/e$. With the Neamen mobilities (μ_n in m^2/V s):

$$\tau_{Si} = \frac{(0.135)(0.26 \times 9.11 \times 10^{-31})}{1.602 \times 10^{-19}} = 2.0 \times 10^{-13}\ \text{s} = 0.20\ \text{ps}$$

$$\tau_{GaAs} = \frac{(0.85)(0.067 \times 9.11 \times 10^{-31})}{1.602 \times 10^{-19}} = 3.2 \times 10^{-13}\ \text{s} = 0.32\ \text{ps}$$

So the two mean free times differ only by a factor of 1.6, while the masses differ by a factor of $0.26/0.067 = 3.9$. The mobility ratio $8500/1350 = 6.3 = 3.9 \times 1.6$ comes **mostly from the lighter mass** and partly from the longer time between collisions.

2. Answer

GaAs has the higher electron mobility. At 300 K (Neamen, Table 5.1):

$$\mu_n(\text{GaAs}) = 8500\ \text{cm}^2/\text{V s}, \qquad \mu_n(\text{Si}) = 1350\ \text{cm}^2/\text{V s}$$

so electrons in GaAs are about $8500/1350 = 6.3$ times more mobile.

Why. Mobility is set by the mean free time τ between collisions and the effective mass m^*:

$$\mu_n = \frac{e\,\tau}{m^*}$$

1. Much smaller effective mass in GaAs. GaAs is a direct-gap semiconductor: its conduction-band minimum is the single Γ valley at $k = 0$, which is very sharply curved, so $m^* = 0.067\,m_0$. Si is indirect: its electrons sit in six flat, cigar-shaped X valleys ($m_l = 0.98\,m_0$, $m_t = 0.19\,m_0$) whose conductivity mass is $m_{cn}^* = 0.26\,m_0$. The GaAs electron is $0.26/0.067 = 3.9$ times lighter, so the same field gives it 3.9 times more drift velocity in the same free time (E–k diagrams in the Theory).

2. Less intervalley scattering in GaAs. In Si an electron can be scattered by phonons between the six equivalent X valleys, which shortens τ. In GaAs at low field all electrons are in the single Γ valley; the next valley (L) is 0.31 eV higher, about $12\,kT$, so intervalley scattering hardly occurs.

3. Mean free time. From $\tau = \mu_n m^*/e$: $\tau_{Si} = 0.20$ ps and $\tau_{GaAs} = 0.32$ ps. GaAs does have strong **polar optical phonon** scattering (its Ga–As bond is partly ionic), but its τ is still a little longer, and the factor that dominates is the light mass:

$$\frac{\mu_{GaAs}}{\mu_{Si}} = \frac{\tau_{GaAs}}{\tau_{Si}} \cdot \frac{m_{Si}^*}{m_{GaAs}^*} = 1.6 \times 3.9 = 6.3$$

Practical note: this high mobility is why GaAs is used for high-frequency transistors (MESFETs, HEMTs). The advantage holds at low fields; at high fields both materials reach about the same saturation velocity (1(b)).

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3. Keyword Explanation

1. Mobility

The drift velocity per unit electric field, $\mu = v_d/E$, in $\mathrm{cm}^2/\mathrm{V\ s}$. It measures how easily a carrier moves through the crystal.

2. Effective mass

The mass that makes Newton's law work for an electron inside a crystal. It is set by the curvature of the E–k diagram: $1/m^* = (1/\hbar^2)\, d^2E/dk^2$.

3. Direct and indirect band gap

Direct: the conduction-band minimum and the valence-band maximum are at the same k (GaAs). Indirect: they are at different k (Si).

4. Intervalley scattering

Scattering of an electron from one conduction-band valley to another, with a phonon supplying the large change of k.

4. Notes and Remarks

- **Holes are the other way round:** the hole mobility of GaAs (400 $\mathrm{cm}^2/\mathrm{V\ s}$) is a little lower than that of Si (480 $\mathrm{cm}^2/\mathrm{V\ s}$, Neamen), because the GaAs heavy holes are heavy. The advantage of GaAs is for electrons only, which is why GaAs fast devices are n-channel.
- **Doping and temperature** lower both mobilities (more impurity and phonon scattering); the values above are for pure material at 300 K.
- **Use the right mass:** the density-of-states mass of Si ($1.08\, m_0$) is often wrongly used here; the mobility depends on the conductivity mass ($0.26\, m_0$).

1.(b) Discuss Velocity saturation and Gunn effect . Marks: 5

1. Theory

The basics

A semiconductor conducts because electrons move through it. An electric field E pushes them, and they keep colliding with the crystal, so they settle at an average **drift velocity**. The ratio $\mu_n = v_d/E$ is the **mobility**.

1. Drift velocity at low field (Core Idea)

From 1(a): in a small field the electrons drift with $v_d = \mu_n E$, because between collisions they gain a little velocity and the collisions keep the average fixed. Doubling E doubles v_d. This is the **ohmic** region, where Ohm's law holds.

2. What happens at high field? (Hot electrons)

The field also gives the electrons energy. At low fields they hand this energy to the lattice through collisions as fast as they get it, so they stay at the lattice temperature. At high fields they gain energy faster than they lose it. Their average energy rises above $\frac{3}{2}kT$: they become **hot electrons**.

A hot electron has enough energy to make the lattice vibrate strongly, by emitting an **optical phonon** (a high-energy lattice vibration, about 0.06 eV in Si). Each such emission throws away most of the energy and momentum the electron gained from the field. The harder we push, the more often this happens. So beyond a certain field the drift velocity stops growing.

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3. Velocity saturation in Si

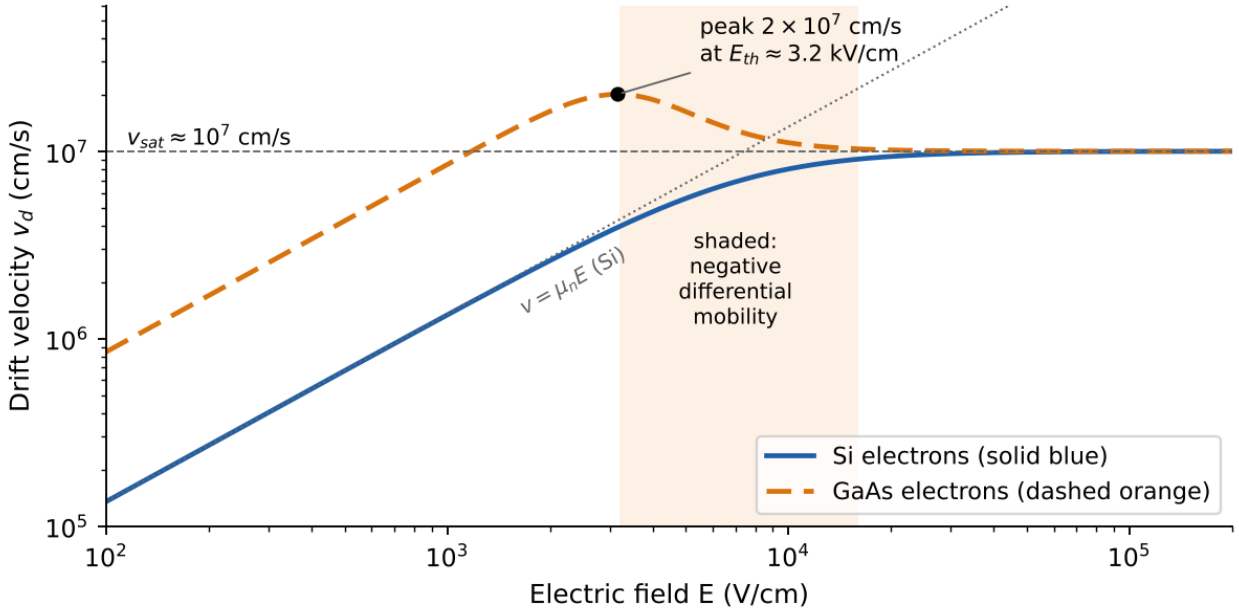


Figure 3: Drift velocity against electric field for electrons in Si and GaAs at 300 K, log scales

Reading the figure. The solid blue curve (Si) follows the dotted grey line $v = \mu_n E$ at low field, bends over above about 10^4 V/cm, and becomes flat at the **saturation velocity** $v_{sat} \approx 10^7$ cm/s (horizontal dashed grey line) for fields above about 3×10^4 V/cm. The dashed orange curve (GaAs) rises much faster at first (higher mobility), reaches a **peak** of about 2×10^7 cm/s at the threshold field $E_{th} \approx 3.2$ kV/cm, then *falls* as the field grows (the shaded band), and finally settles near 10^7 cm/s too. The region where the velocity falls as the field rises is called **negative differential mobility**: $dv_d/dE < 0$. (Curves drawn with standard empirical fits; compare Neamen Fig. 5.7.)

4. Why does GaAs have a peak? (The two valleys)

The conduction band of GaAs has a low, sharp Γ valley and, 0.31 eV higher, a set of broad L valleys (1(a) shows both).

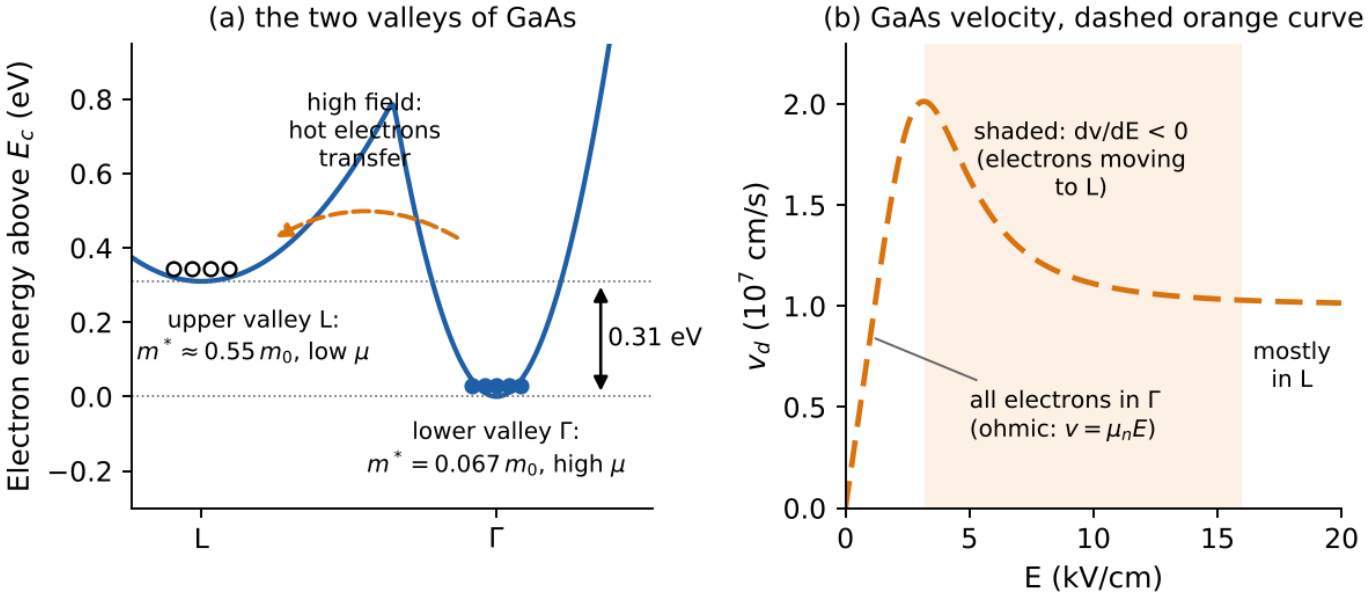


Figure 4: The two conduction-band valleys of GaAs and the velocity-field curve that results

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Reading the figure. Panel (a): at low field all electrons (filled dots) sit in the lower Γ valley, where they are light ($m^* = 0.067\,m_0$) and fast. At high field they become hot; when their energy reaches 0.31 eV they are scattered into the upper L valley (dashed orange arrow, open dots), where they are heavy ($m^* \approx 0.55\,m_0$) and slow. There are also more L valleys and more states in them, so once electrons get there, most of them stay. Panel (b) is the GaAs velocity on linear axes (dashed orange curve): ohmic at first, the drop in the shaded region as more electrons move to the slow valley, and a nearly flat tail when almost all are in L.
This mechanism is called the **transferred-electron effect** (Ridley, Watkins, Hilsum). It needs three conditions, all met by GaAs:

- the upper valley is well above the lower one compared with kT (0.31 eV $\approx 12\,kT$), so at low field it is empty;
- the separation is smaller than the band gap (0.31 < 1.424 eV), so electrons transfer before they can break bonds (impact ionization);
- electrons in the upper valley are much heavier and slower.

5. What is the Gunn effect?

In 1963 J. B. Gunn found that a plain bar of n-type GaAs (no p-n junction) with a d.c. voltage above a threshold gives a current that **oscillates at microwave frequencies**. The cause is the negative differential mobility above.
Suppose the field in the bar exceeds E_{th} and a small region happens to have a slightly higher field. Electrons there slow down (negative differential mobility), so electrons pile up behind it and a depleted layer forms in front of it. This makes the field in that region even higher: a **high-field domain** grows. The field outside the domain falls below E_{th}. The domain then drifts with the electrons towards the anode at about $v_d \approx 10^7$ cm/s. When it reaches the anode it disappears, the current rises for a moment, and a new domain forms at the cathode.

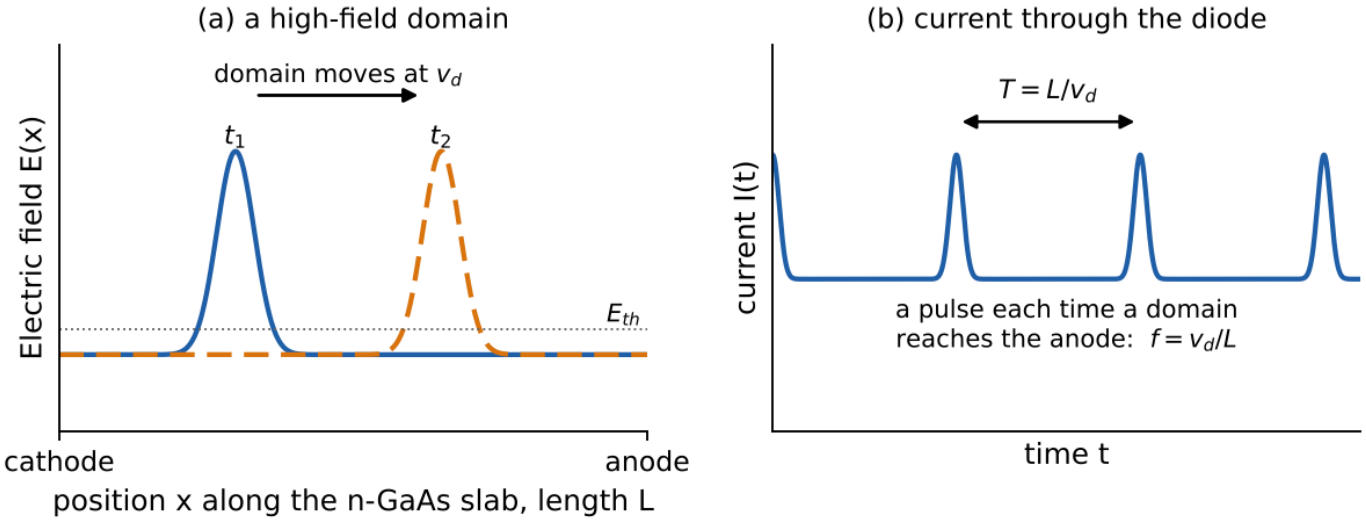


Figure 5: A high-field domain travelling along a Gunn diode, and the current pulses it produces

Reading the figure. Panel (a): the field along the bar at time t_1 (solid blue) and a little later at t_2 (dashed orange); the domain is a narrow peak of field, far above E_{th} (dotted line), that moves from the cathode to the anode. Panel (b): the current (solid blue) is low while a domain crosses and shows a pulse each time one reaches the anode. The pulses repeat with period $T = L/v_d$.
The oscillation frequency is therefore set by the length L of the sample:

$$f = \frac{1}{T} = \frac{v_d}{L}$$

For example, $L = 10\ \mu\text{m} = 10^{-3}$ cm and $v_d = 10^7$ cm/s give $f = 10^7/10^{-3} = 10^{10}$ Hz $= 10$ GHz. Domains form only in a long enough, well enough doped sample, $n_0 L \gtrsim 10^{12}$ cm^{-2} (Kroemer criterion, Sze).

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2. Answer

Velocity saturation

At low fields the drift velocity of carriers rises in proportion to the field, $v_d = \mu E$. At high fields the carriers gain energy from the field faster than they can give it to the lattice; these **hot carriers** lose energy by emitting optical phonons, so their drift velocity stops increasing and approaches a constant, the **saturation velocity**.

- **Si:** v_d saturates at $v_{sat} \approx 10^7$ cm/s for $E \gtrsim 3 \times 10^4$ V/cm (graph in the Theory).
- **GaAs:** v_d rises fast (high mobility), reaches a peak of about 2×10^7 cm/s at $E_{th} \approx 3.2$ kV/cm, then decreases and settles near 10^7 cm/s.

Consequence: in short-channel devices the current is limited by v_{sat}, not by the mobility: $I \approx e\, n\, v_{sat}\, A$. The transit time across a device of length L is about L/v_{sat}.

Gunn effect

Definition: the Gunn effect is the generation of microwave current oscillations in a uniformly doped n-type sample of GaAs (or InP) when the d.c. field exceeds a threshold of about 3.2 kV/cm.

Cause, the transferred-electron mechanism (two-valley picture in the Theory):

- GaAs has a lower Γ valley ($m^* = 0.067\, m_0$, high mobility) and upper L valleys 0.31 eV higher ($m^* \approx 0.55\, m_0$, low mobility, more states).
- Above E_{th} the hot electrons transfer from Γ to L, so the average velocity falls as the field rises: **negative differential mobility (resistance)**, $dv_d/dE < 0$.

Domains and oscillation:

- Any small field fluctuation grows into a **high-field domain** (an accumulation layer of electrons followed by a depletion layer), because the electrons inside it are slowed.
- The domain drifts from cathode to anode at $v_d \approx 10^7$ cm/s; when it reaches the anode, the current rises and a new domain forms at the cathode.
- The current is therefore a series of pulses with frequency

$$f = \frac{v_d}{L}$$

For $L = 10$ μm: $f = 10^7/(10 \times 10^{-4}) = 10$ GHz.

Application: Gunn diodes are simple, cheap microwave oscillators (1 to 100 GHz) used in radar speed guns, motion sensors and local oscillators.

3. Keyword Explanation

1. Saturation velocity

The highest steady drift velocity carriers reach at high field, about 10^7 cm/s for electrons in Si and GaAs.

2. Negative differential mobility

A range of field where the drift velocity falls as the field rises ($dv_d/dE < 0$); it gives a negative differential resistance.

3. Hot electrons

Electrons whose average energy is well above the lattice thermal energy, because the field heats them faster than collisions cool them.

4. Transit time

The time a carrier or a domain takes to cross a device, L/v; it sets the Gunn frequency.

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4. Notes and Remarks

- **Si has no Gunn effect:** its lowest valleys are the X valleys themselves, and there is no higher valley with heavier electrons that is reachable before impact ionization. Its velocity just saturates.
- **Velocity overshoot:** in very short GaAs devices (below about 0.1 µm) electrons can cross before they have time to transfer to L, and their velocity briefly exceeds the steady values in the graph.
- **InP** shows the same effect (valley separation about 0.53 eV) and is used for higher-frequency transferred-electron devices.

1.(c) Prove that along the [100] and [111] direction in k-space, for Si, the variation of electron energy with respect to wave number is higher for light holes than for heavy holes. Marks: 6

1. Theory

1. What is the wave number k? (Core Idea)

In quantum mechanics an electron is also a wave. A wave has a wavelength λ, and the **wave number** is

$$k = \frac{2\pi}{\lambda}$$

The electron's momentum is $p = \hbar k$, where $\hbar = h/2\pi = 1.055 \times 10^{-34}$ J s. So k is simply momentum measured in different units, and a big k means a fast-moving electron.
In a crystal, k is a vector: the electron can travel along any direction. The space of all possible k vectors is called **k-space**. Directions in it are named with Miller indices, just like directions in the crystal:

- **[100]:** along a cube edge (the k_x axis). Going out along it from the centre Γ ($k = 0$), one reaches the zone-edge point **X**.
- **[111]:** along the cube body diagonal. Going out along it, one reaches the point **L**.

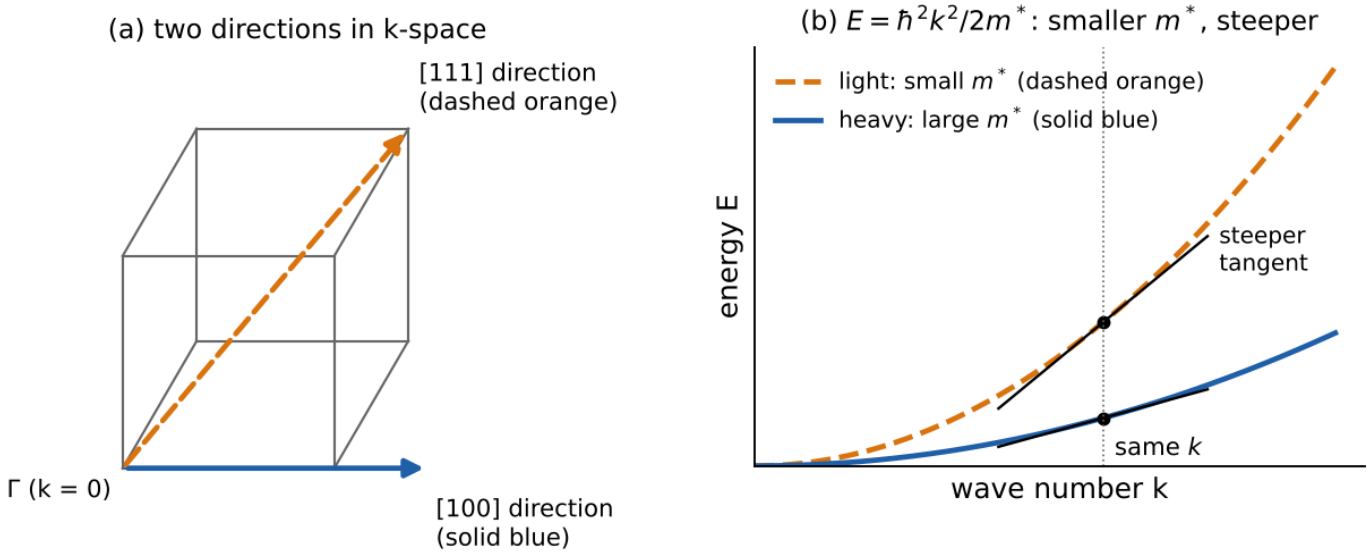


Figure 6: The [100] and [111] directions in k-space, and why a smaller effective mass gives a steeper E-k curve

Reading the figure. Panel (a): from the centre Γ, the solid blue arrow runs along a cube edge ([100]) and the dashed orange arrow along the body diagonal ([111]). Panel (b): two parabolas $E = \hbar^2 k^2/2m^*$; the dashed orange one has a small mass, the solid blue one a large mass. At the same k (dotted vertical line) the tangent (thin black line) on the dashed orange curve is steeper: its energy changes faster with k.

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2. The E–k relation and the effective mass

For a free electron all the energy is kinetic:

$$E = \frac{p^2}{2m_0} = \frac{\hbar^2 k^2}{2m_0}$$

which is a parabola in k. Inside a crystal, near the bottom or top of a band, the E–k curve is still very nearly a parabola, but with a different curvature. We keep the same form and write the change in curvature as an **effective mass** m^*:

$$E - E_0 = \pm\frac{\hbar^2 k^2}{2m^*}$$

Differentiating twice gives the definition used by Neamen:

$$\frac{d^2E}{dk^2} = \frac{\hbar^2}{m^*} \quad \Rightarrow \quad \frac{1}{m^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$$

So the **curvature** of the E–k curve gives the mass: a strongly curved band means a light particle, a flat band a heavy one. Also, the **slope** gives the velocity of the electron wave packet, its **group velocity**:

$$v_g = \frac{1}{\hbar}\frac{dE}{dk}$$

3. Holes and the valence band of Si

The valence band is almost full. An empty state in it behaves like a positive particle, a **hole**. At the top of the valence band the E–k curve bends **downward**: the electron energy is highest at $k = 0$ and falls as $|k|$ grows,

$$E = E_v - \frac{\hbar^2 k^2}{2m_h^*}$$

Since a hole is a *missing* electron, its energy is measured downward, and it has a positive mass m_h^*.
In Si (and all diamond and zinc-blende crystals) three valence bands meet near the top:

- **Heavy-hole band (hh):** the flatter curve, large m^*.
- **Light-hole band (lh):** the more curved one, small m^*. It touches the hh band at $k = 0$.
- **Split-off band:** pushed 0.044 eV lower by spin-orbit coupling in Si; it plays a minor role here.

4. Why do the masses depend on direction?

The hh and lh bands are not perfect spheres in k-space; they are **warped**. So the curvature, and the mass, is different along [100] and [111]. The masses along these two directions follow from three numbers measured for each crystal, the **Luttinger parameters** $\gamma_1, \gamma_2, \gamma_3$ (Sze and Ng):

$$\begin{aligned} [100]: &\quad \frac{m_0}{m_{hh}} = \gamma_1 - 2\gamma_2, &\quad \frac{m_0}{m_{lh}} = \gamma_1 + 2\gamma_2 \\ [111]: &\quad \frac{m_0}{m_{hh}} = \gamma_1 - 2\gamma_3, &\quad \frac{m_0}{m_{lh}} = \gamma_1 + 2\gamma_3 \end{aligned}$$

For Si, $\gamma_1 = 4.22$, $\gamma_2 = 0.39$, $\gamma_3 = 1.44$. Since all the γ are positive, the "+" sign always gives the larger $1/m$: **the light hole is lighter than the heavy hole in every direction**. Neamen does not list these direction-resolved masses (it gives only the density-of-states mass $m_p^* = 0.56\,m_0$), so they are taken from Sze and Ng.

2. Answer

Step 1: The shape of a hole band near its top

For a free electron, $E = p^2/2m_0$ and the momentum is $p = \hbar k$, so $E = \hbar^2 k^2/2m_0$. Near the top of a valence band the crystal changes only the mass: each band is a parabola with its own effective mass m^*, curving down from E_v (energy measured as electron energy):

$$E(k) = E_v - \frac{\hbar^2 k^2}{2m^*}$$

where m^* is m_{hh} or m_{lh} for the chosen direction.

Step 2 (Basic): Slope and curvature of the parabola

Differentiate once and twice with respect to k (E_v, $\hbar$ and m^* are constants):

$$\frac{dE}{dk} = -\frac{\hbar^2 k}{m^*}, \qquad \frac{d^2E}{dk^2} = -\frac{\hbar^2}{m^*}$$

The variation of energy with wave number is the slope dE/dk; its size is inversely proportional to m^*.

Step 3: Compare the two bands at the same k

Divide the light-hole slope by the heavy-hole slope; $\hbar^2 k$ cancels:

$$\frac{|dE/dk|_{lh}}{|dE/dk|_{hh}} = \frac{\hbar^2 k/m_{lh}}{\hbar^2 k/m_{hh}} = \frac{m_{hh}}{m_{lh}}$$

So the light-hole energy varies faster with k whenever $m_{lh} < m_{hh}$. The same ratio holds for the curvature and for the energy drop $|E - E_v|$. What remains is to show $m_{lh} < m_{hh}$ along both directions in Si.

Step 4: Effective masses of Si along [100] and [111]

The Si valence band is described by the Luttinger parameters $\gamma_1 = 4.22$, $\gamma_2 = 0.39$, $\gamma_3 = 1.44$. Along [100] the masses use γ_2, along [111] they use γ_3:

$$[100]: \quad m_{hh} = \frac{m_0}{\gamma_1 - 2\gamma_2} = \frac{m_0}{3.44} = 0.291\, m_0, \qquad m_{lh} = \frac{m_0}{\gamma_1 + 2\gamma_2} = \frac{m_0}{5.00} = 0.200\, m_0$$

$$[111]: \quad m_{hh} = \frac{m_0}{\gamma_1 - 2\gamma_3} = \frac{m_0}{1.34} = 0.746\, m_0, \qquad m_{lh} = \frac{m_0}{\gamma_1 + 2\gamma_3} = \frac{m_0}{7.10} = 0.141\, m_0$$

In both directions $m_{lh} < m_{hh}$, because the light hole adds 2γ in the denominator and the heavy hole subtracts it.

Step 5: Ratio of the slopes

Put the masses of Step 4 into Step 3:

$$[100]: \quad \frac{|dE/dk|_{lh}}{|dE/dk|_{hh}} = \frac{0.291}{0.200} = 1.45, \qquad [111]: \quad \frac{|dE/dk|_{lh}}{|dE/dk|_{hh}} = \frac{0.746}{0.141} = 5.30$$

$$\boxed{\left|\frac{dE}{dk}\right|_{lh} = 1.45 \left|\frac{dE}{dk}\right|_{hh} [100], \qquad \left|\frac{dE}{dk}\right|_{lh} = 5.30 \left|\frac{dE}{dk}\right|_{hh} [111]}$$

Both ratios are greater than 1, which proves the statement.

Step 6: Numbers at $k = 10^9$ m^{-1} ($= 1$ nm^{-1})

With $\hbar^2/m_0 = (1.055 \times 10^{-34})^2/(9.11 \times 10^{-31}) = 1.222 \times 10^{-38}$ J m^2 $= 0.0763$ eV nm^2, the slope is $|dE/dk| = 0.0763\, k/(m^*/m_0)$ and the energy drop is $|E - E_v| = 0.0763\, k^2/(2m^*/m_0)$:

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Direction	Band	m^*/m_0	$\lvert dE/dk \rvert$ (eV nm)	$\lvert E - E_v \rvert$ (meV)
[100]	heavy hole	0.291	0.262	131.2
[100]	light hole	0.200	0.381	190.7
[111]	heavy hole	0.746	0.102	51.1
[111]	light hole	0.141	0.541	270.7

Table 1: Slope and energy of the Si hole bands at k = 10^9~\text{m}^{-1}

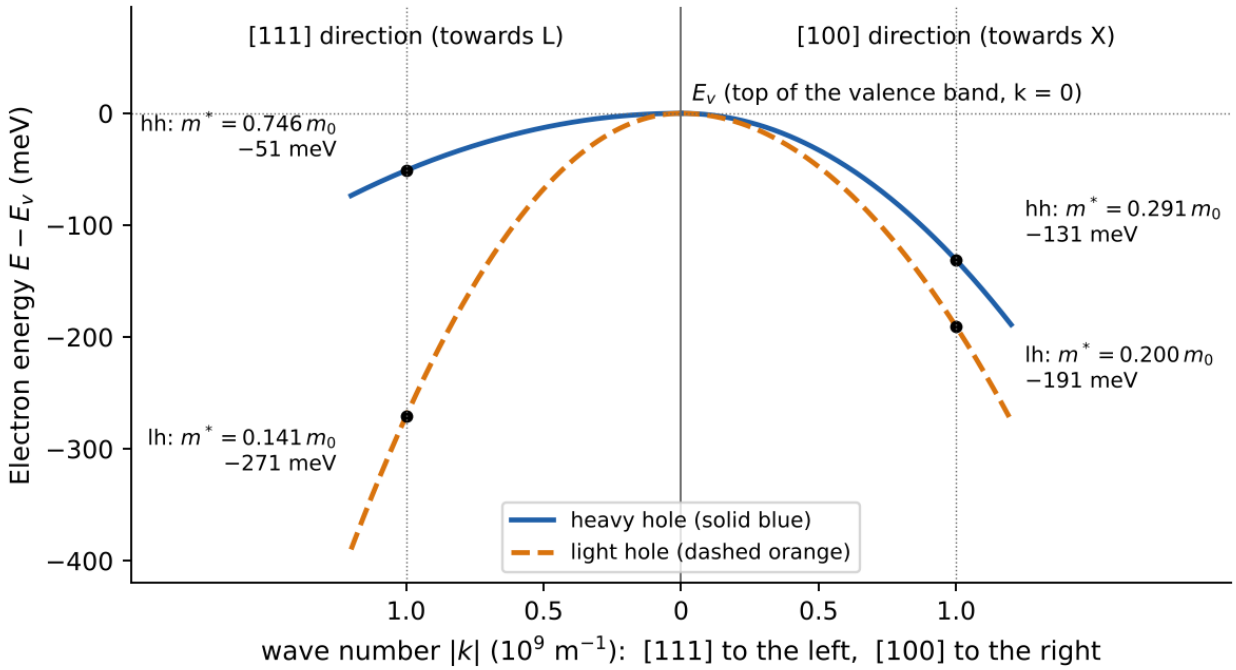


Figure 7: E-k diagram of the Si valence band along [111] (left) and [100] (right): heavy holes and light holes

Reading the figure. Energy is that of an electron, so both bands curve downward from E_v at $k = 0$. On both sides the dashed orange light-hole curve falls faster than the solid blue heavy-hole curve: at $|k| = 10^9$ m^{-1} (dotted lines, black dots) the light hole is 191 meV below E_v against 131 meV along [100], and 271 meV against 51 meV along [111]. The difference is largest along [111], where the heavy-hole band is very flat.

Conclusion: along both [100] and [111] the light-hole band has a smaller effective mass, so its energy changes faster with wave number, by factors of 1.45 and 5.30. Physically, light holes have a larger group velocity $v_g = (1/\hbar)\,dE/dk$ and a higher mobility than heavy holes.

Steps to remember

Parabolic band $E = E_v - \hbar^2k^2/2m^* \rightarrow$ slope $\hbar^2k/m^* \rightarrow$ ratio of slopes $= m_{hh}/m_{lh} \rightarrow$ Luttinger masses along [100] (γ_2) and [111] (γ_3) $\rightarrow$ ratios 1.45 and 5.30, both above 1.

3. Keyword Explanation

1. Wave number

$k = 2\pi/\lambda$; the electron momentum divided by $\hbar$.

2. Heavy and light holes

The two valence bands that meet at $k = 0$; the heavy-hole band is flatter (larger m^*), the light-hole band more curved (smaller m^*).

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3. Warped bands

Bands whose curvature depends on the direction of k, so the effective mass differs along [100], [110] and [111].

4. Group velocity

The velocity of an electron (or hole) wave packet, $v_g = (1/\hbar)\, dE/dk$.

4. Notes and Remarks

- **Average masses:** many tables (Sze and Ng, Appendix) give direction-averaged masses for Si, $m_{hh} \approx 0.49\, m_0$ and $m_{lh} \approx 0.16\, m_0$. With them the ratio is $0.49/0.16 = 3.1$; the conclusion is the same.
- **Sign of the slope:** for the electron energy, dE/dk is negative for $k > 0$ in a valence band. "Higher variation" compares magnitudes, $|dE/dk|$. Measured as hole energy (downward), both are positive and the comparison is the same.
- **Why it matters:** most holes are heavy holes (the flat band has more states), so the hole density of states is set mainly by m_{hh}, while the light holes contribute more than their share to conduction.

1.(d) For a $GaAs - Al_xGa_{1-x}As$ heterojunction, determine the following: (i) Energy-Band diagram: Case-1: Before contact, Case-2: After contact (ii) Show that the built-in potential (using hole concentration) $eV_{bi} = -\Delta E_v + kT \ln\left(\frac{P_P\, N_{VN}}{P_N\, N_{VP}}\right)$. Given, GaAs is doped P type, $Al_xGa_{1-x}As$ is doped N^+ type. Assume: $x = 0.25$ & $\Delta E_c = \frac{1}{2}\Delta E_g$ P_p = hole concentration in P type region, P_N = hole concentration in N type region, N_{VN} and N_{VP} are density of states of electron and hole respectively. Marks: 8

1. Theory

1. Energy bands, Fermi level and vacuum level (Core Idea)

An **energy-band diagram** shows the allowed energies of electrons against position:

- E_c is the bottom of the **conduction band** (free electrons live just above it).
- E_v is the top of the **valence band** (holes live just below it).
- $E_g = E_c - E_v$ is the **band gap**: no electron states exist inside it.
- E_F is the **Fermi level**: the energy at which a state has a 50 % chance of being filled. In n-type material it is close to E_c; in p-type material close to E_v.
- E_0 is the **vacuum level**: the energy an electron needs to just escape from the material and be at rest outside it.

Two material constants connect the bands to the vacuum level:

- **Electron affinity** χ: the energy from E_c up to E_0, $\chi = E_0 - E_c$. It is fixed for a material (GaAs: 4.07 eV).
- **Work function** $e\phi$: the energy from E_F up to E_0, $e\phi = E_0 - E_F$. It depends on the doping, because E_F does.

2. How many carriers? (Effective density of states)

The hole concentration is linked to the position of E_F above E_v (Neamen, Boltzmann approximation):

$$p_0 = N_v \exp\left[\frac{-(E_F - E_v)}{kT}\right], \qquad n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$$

where

- N_v and N_c are the **effective densities of states** of the valence and conduction bands (cm^{-3}); they depend on the hole and electron effective masses as $m^{*3/2}$ (GaAs: $N_v = 7.0 \times 10^{18}$, $N_c = 4.7 \times 10^{17}$ cm^{-3})

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- $kT = 0.0259$ eV at 300 K

Inverted, $E_F - E_v = kT \ln(N_v/p_0)$: the more holes, the closer E_F is to E_v.

3. What is a heterojunction?

A **homojunction** joins two regions of the *same* semiconductor (a Si p-n junction). A **heterojunction** joins two *different* semiconductors, with different band gaps. GaAs and $Al_xGa_{1-x}As$ are the classic pair: Al atoms replace a fraction x of the Ga atoms. Their lattice constants are almost equal, so one grows on the other without defects. The band gap of the alloy rises with x (for $x < 0.45$):

$$E_g(Al_xGa_{1-x}As) = 1.424 + 1.247x \ \text{ eV}$$

For $x = 0.25$: $E_g = 1.424 + 1.247(0.25) = 1.736$ eV, against 1.424 eV for GaAs. The difference is $\Delta E_g = 0.312$ eV.

Notation: a capital letter marks the **wide-gap** side and a small letter the **narrow-gap** side. Our junction, p-type GaAs with N-type AlGaAs, is a **pN** heterojunction. The "+" in N^+ means heavily doped.

4. How do the bands line up? (Anderson's rule)

Before contact, both materials see the same vacuum (an electron at rest outside either has the same energy). So we draw them with a **common vacuum level**. Then E_c of each lies χ below it, and the band gaps are split between the conduction and valence bands:

$$\Delta E_c = \chi_1 - \chi_2, \qquad \Delta E_v = \Delta E_g - \Delta E_c$$

where material 1 is the narrow-gap GaAs and material 2 the wide-gap AlGaAs. This is **Anderson's electron-affinity rule**. Depending on how the gaps sit relative to each other, there are three types:

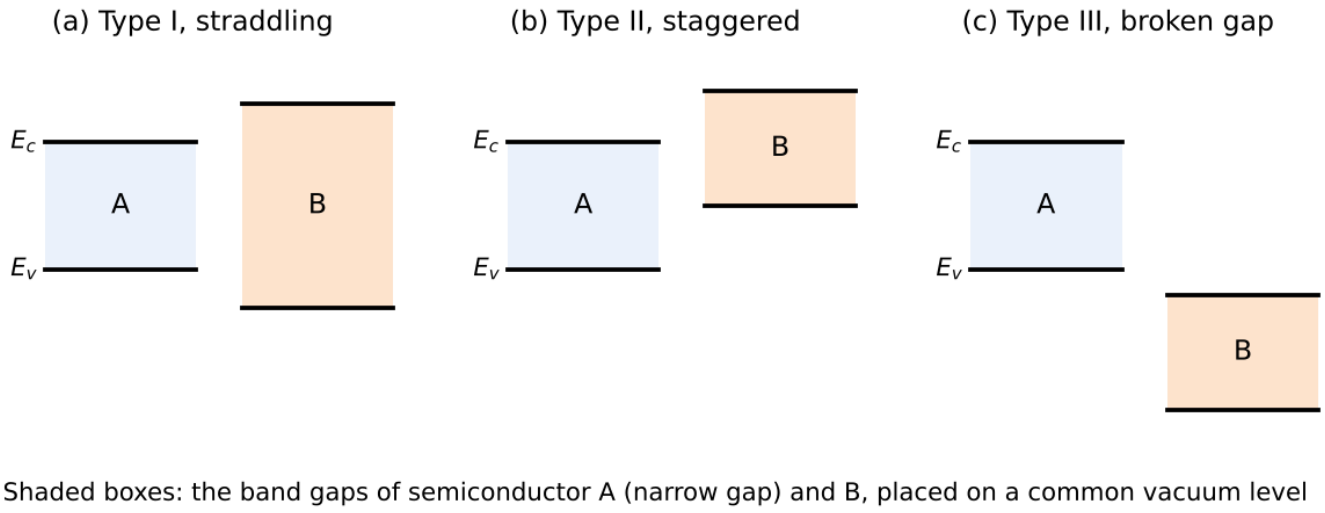


Figure 8: The three types of heterojunction band alignment: straddling, staggered and broken gap

Reading the figure. Each shaded box is a band gap, from E_v (lower thick line) to E_c (upper thick line). (a) **Straddling (type I):** the narrow gap of A lies completely inside the wide gap of B; E_c steps up and E_v steps down going from A to B. GaAs/AlGaAs is of this type. (b) **Staggered (type II):** both edges of B are higher than those of A, so the gaps overlap only partly. (c) **Broken gap (type III):** the conduction band of B is below the valence band of A; the gaps do not overlap at all (InAs/GaSb).

5. What happens on contact?

When the two materials touch, electrons flow from the side where E_F is higher (the N side) to the side where it is lower (the p side), and holes flow the other way. This continues until E_F **is the same everywhere**: in equilibrium the Fermi level is flat. The flow leaves fixed charges near the junction (positive donor ions on the N side, negative acceptor ions on the p side), a **depletion region**. Their field bends the bands. Three rules fix the shape:

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- **Far from the junction** each material is as it was before contact (same $E_c - E_F$, same $E_v - E_F$).
- **The vacuum level is continuous** (no infinite field), so it bends smoothly through the junction. Since χ and E_g are properties of each material, E_c and E_v follow E_0 in parallel on each side.
- **At the interface** E_c and E_v jump by ΔE_c and ΔE_v, as in Anderson's rule.

The total bending of the vacuum level is the **built-in potential energy** eV_{bi}, which equals the difference of the two work functions, that is, of the two Fermi levels before contact:

$$eV_{bi} = e\phi_P - e\phi_N = E_{FN} - E_{FP} \quad \text{(before contact)}$$

2. Answer

Data: $x = 0.25$, $\Delta E_c = \frac{1}{2}\Delta E_g$, GaAs p-type (narrow gap), $Al_{0.25}Ga_{0.75}As$ N^+-type (wide gap), $\chi_{GaAs} = 4.07$ eV (Neamen), $kT = 0.0259$ eV.

Band parameters:

$$E_{g1} = 1.424 \text{ eV (GaAs)}, \qquad E_{g2} = 1.424 + 1.247(0.25) = 1.736 \text{ eV}$$
$$\Delta E_g = 1.736 - 1.424 = 0.312 \text{ eV}$$
$$\Delta E_c = \tfrac{1}{2}\Delta E_g = 0.156 \text{ eV}, \qquad \Delta E_v = \Delta E_g - \Delta E_c = 0.156 \text{ eV}$$
$$\chi_2 = \chi_1 - \Delta E_c = 4.07 - 0.156 = 3.914 \text{ eV}$$

The AlGaAs gap straddles the GaAs gap: E_c of AlGaAs is 0.156 eV higher and E_v is 0.156 eV lower than those of GaAs (a type I junction).

(i) Energy-band diagrams

The positions of E_F need the doping, which the paper does not give. For the drawings we take $p_P = 10^{17}$ cm^{-3} in the GaAs and $N_D = 10^{17}$ cm^{-3} in the AlGaAs (an illustration only); then $E_{FP} - E_{vP} = kT\ln(7.0 \times 10^{18}/10^{17}) = 0.110$ eV and $E_{cN} - E_{FN} = 0.051$ eV.

Case 1: Before contact. The two materials are drawn side by side with a common vacuum level E_0; each has its own Fermi level.

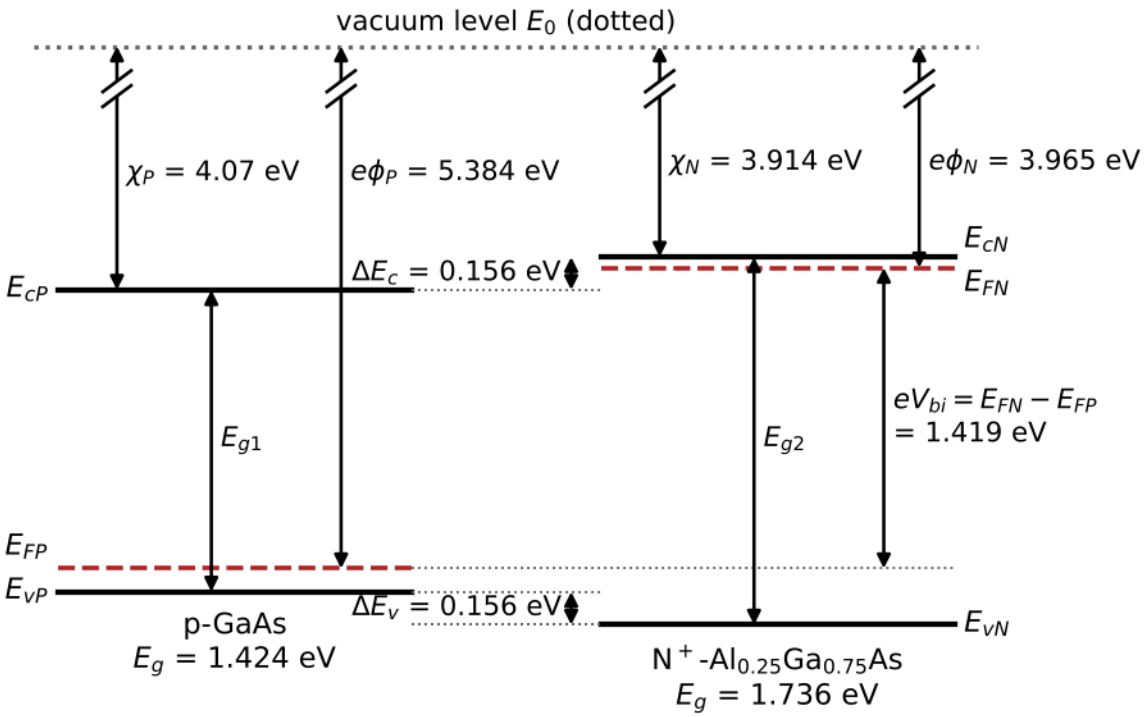


Figure 9: p-GaAs and N-$Al_{0.25}Ga_{0.75}As$ before contact, on a common vacuum level

Reading the figure. Solid lines are E_c and E_v, dashed red lines the Fermi levels, the dotted line the vacuum level (the long χ and $e\phi$ arrows are broken, not to scale). In the p-GaAs, E_{FP} is just above E_{vP};

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in the N-AlGaAs, E_{FN} is just below E_{cN}. E_{cN} is $\Delta E_c = 0.156$ eV above E_{cP} and E_{vN} is $\Delta E_v = 0.156$ eV below E_{vP}. The Fermi levels differ by $E_{FN} - E_{FP} = e\phi_P - e\phi_N = 5.384 - 3.965 = 1.419$ eV: this is eV_{bi}.

Case 2: After contact (thermal equilibrium). Electrons flow from the N-AlGaAs into the p-GaAs and holes the other way, until E_F is flat. A depletion region forms on both sides of the interface ($-x_p$ to x_n), and the bands bend by eV_{bi} in total.

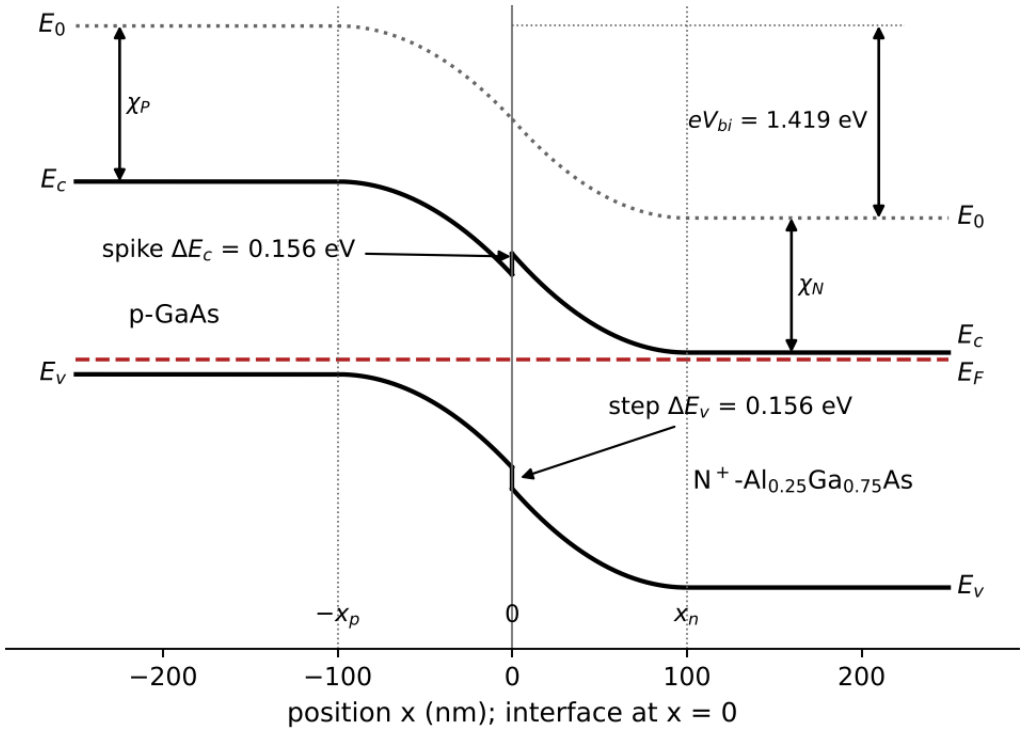


Figure 10: p-GaAs / N-$Al_{0.25}Ga_{0.75}As$ heterojunction after contact, in thermal equilibrium (illustration doping 10^{17} cm^{-3} on both sides)

Reading the figure. The dashed red E_F is now one flat line. The dotted vacuum level E_0 bends smoothly and drops by $eV_{bi} = 1.419$ eV from the p side to the N side; the solid E_c and E_v bend with it, parallel to it on each side. At the interface $x = 0$ the conduction band has a **spike**: it jumps up by $\Delta E_c = 0.156$ eV going into the AlGaAs, an extra barrier for electrons. The valence band **steps down** by $\Delta E_v = 0.156$ eV, which adds to the barrier for holes going from p-GaAs into the N side. Far from the junction each side has its original band positions relative to E_F. (With equal doping the band bending is shared almost equally; with a much heavier N^+ side most of it would lie in the p-GaAs.)

(ii) Built-in potential in terms of hole concentrations

Step 1: Hole concentration far from the junction, before contact

In the p-GaAs and in the N-AlGaAs, using the valence-band effective density of states of each material:

$$p_P = N_{VP} \exp\left[\frac{-(E_{FP} - E_{vP})}{kT}\right], \qquad p_N = N_{VN} \exp\left[\frac{-(E_{FN} - E_{vN})}{kT}\right]$$

Step 2: Fermi levels

Taking logarithms:

$$E_{FP} = E_{vP} + kT \ln\left(\frac{N_{VP}}{p_P}\right), \qquad E_{FN} = E_{vN} + kT \ln\left(\frac{N_{VN}}{p_N}\right)$$

Step 3: Built-in potential

On contact the Fermi levels line up; the energy by which the bands of the N side move down relative to the p side is the difference of the Fermi levels before contact:

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$$\begin{aligned} eV_{bi} &= E_{FN} - E_{FP} \\ &= (E_{vN} - E_{vP}) + kT\ln\left(\frac{N_{VN}}{p_N}\right) - kT\ln\left(\frac{N_{VP}}{p_P}\right) \\ &= (E_{vN} - E_{vP}) + kT\ln\left(\frac{p_P\,N_{VN}}{p_N\,N_{VP}}\right) \end{aligned}$$

Step 4: Valence-band offset

On the common vacuum level (Case 1), the valence band of the wide-gap AlGaAs lies ΔE_v *below* that of GaAs:

$$E_{vN} - E_{vP} = -\Delta E_v$$

Substituting:

$$\boxed{eV_{bi} = -\Delta E_v + kT\ln\left(\frac{P_P\,N_{VN}}{P_N\,N_{VP}}\right)}$$

which is the required result. Here $\Delta E_v = 0.156$ eV for $x = 0.25$.

Steps to remember

Hole concentration on each side, $p = N_V e^{-(E_F - E_v)/kT}$ → solve each for E_F → $eV_{bi} = E_{FN} - E_{FP}$ (Fermi levels line up) → combine the logarithms → put $E_{vN} - E_{vP} = -\Delta E_v$.

Numerical illustration (doping not given in the paper; assumed $p_P = 10^{17}$ cm^{-3}, $N_D = 10^{17}$ cm^{-3}):

- $N_{VP} = 7.0 \times 10^{18}$ cm^{-3} (GaAs, Neamen); $N_{VN} = 8.576 \times 10^{18}$ cm^{-3} and $N_{CN} = 7.045 \times 10^{17}$ cm^{-3} for $Al_{0.25}Ga_{0.75}As$, scaled from GaAs with the masses $m_h^* = (0.5 + 0.29x)\,m_0$ and $m_e^* = (0.067 + 0.083x)\,m_0$
- $n_i(\text{AlGaAs}) = \sqrt{N_{CN}N_{VN}}\,\exp[-E_{g2}/(2kT)] = 6.886 \times 10^3$ cm^{-3}, so $p_N = n_i^2/N_D = 4.741 \times 10^{-10}$ cm^{-3}

$$eV_{bi} = -0.156 + 0.0259\ln\left(\frac{(10^{17})(8.576\times10^{18})}{(4.741\times10^{-10})(7.0\times10^{18})}\right) = -0.156 + 0.0259 \times 60.82 = 1.419 \text{ eV}$$

$$\boxed{V_{bi} = 1.419 \text{ V} \quad \text{(illustration)}}$$

This agrees with the Fermi-level difference read from Case 1 ($5.384 - 3.965 = 1.419$ eV), which checks the formula.

3. Keyword Explanation

1. Electron affinity

$\chi = E_0 - E_c$: the energy to lift an electron from the conduction-band edge to the vacuum level; a material constant.

2. Band offsets

ΔE_c and ΔE_v: the steps in the conduction- and valence-band edges at a heterojunction; $\Delta E_c + \Delta E_v = \Delta E_g$ for a straddling junction.

3. Anderson's rule

Aligning the vacuum levels of the two materials before contact, so that $\Delta E_c = \chi_1 - \chi_2$.

4. Built-in potential

The total band bending in equilibrium, equal to the difference of the Fermi levels (work functions) of the two sides before contact.

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5. Effective density of states

N_v (or N_c): the number of states per cm^3 that, placed at the band edge, give the same carrier count as the real band; proportional to $(m^*T)^{3/2}$.

4. Notes and Remarks

- **About the paper's wording:** the paper says "N_{VN} and N_{VP} are density of states of electron and hole respectively". In the formula both are **valence-band (hole)** effective densities of states: N_{VN} of the N-type AlGaAs and N_{VP} of the p-type GaAs. They differ only because the hole effective mass of AlGaAs is a little larger than that of GaAs; if they were taken equal, the formula would become $eV_{bi} = -\Delta E_v + kT\ln(p_P/p_N)$.
- **The same result with electrons:** working with $n = N_c \exp[-(E_c - E_F)/kT]$ instead gives $eV_{bi} = \Delta E_c + kT\ln(n_N N_{CP}/(n_P N_{CN}))$; with $\Delta E_c = \Delta E_v$ here, both forms give 1.419 eV.
- **The real offset ratio:** the paper's $\Delta E_c = \frac{1}{2}\Delta E_g$ is a simplifying assumption. Measurements for GaAs/AlGaAs give $\Delta E_c \approx 0.60$ to $0.65\,\Delta E_g$ (Sze and Ng); Anderson's rule with tabulated χ values is also known to be only approximate.
- **Spike and notch:** for this pN junction the conduction-band spike is a barrier that electrons injected from the N side must cross. When N-AlGaAs is grown on undoped or lightly doped n-GaAs (an Nn junction), the bending forms a **notch** (potential well) in the GaAs conduction band at the interface, which holds the two-dimensional electron gas of a HEMT.

5. Final Concise Answer

(a) GaAs has the higher electron mobility: $\mu_n = 8500$ cm^2/V s against 1350 cm^2/V s for Si (6.3 times). Since $\mu_n = e\tau/m^*$, the main reason is the much lighter electron of GaAs: a single, sharply curved direct Γ valley with $m^* = 0.067\,m_0$, against the six flat X valleys of indirect Si with conductivity mass $0.26\,m_0$ (3.9 times lighter). GaAs also has little intervalley scattering at low field (L valley 0.31 eV above), so its mean free time is slightly longer (0.32 ps against 0.20 ps), despite polar optical phonon scattering.

(b) **Velocity saturation:** at high fields hot carriers lose their extra energy to optical phonons, so v_d stops rising and saturates at about 10^7 cm/s (Si, for $E \gtrsim 3\times10^4$ V/cm). **Gunn effect:** in n-GaAs above $E_{th} \approx 3.2$ kV/cm (peak $v_d \approx 2\times10^7$ cm/s) electrons transfer from the light Γ valley to the heavy L valleys 0.31 eV higher, giving negative differential mobility; high-field domains form, drift to the anode, and produce microwave current oscillations at $f = v_d/L$ (10 GHz for $L = 10$ μm).

(c) With $E = E_v - \hbar^2k^2/2m^*$, $|dE/dk| = \hbar^2k/m^*$, so at the same k the slope ratio is m_{hh}/m_{lh}. From the Luttinger parameters of Si: [100] $m_{hh} = 0.291\,m_0$, $m_{lh} = 0.200\,m_0$, ratio 1.45; [111] $m_{hh} = 0.746\,m_0$, $m_{lh} = 0.141\,m_0$, ratio 5.30. Both exceed 1, so the light-hole energy varies faster with k in both directions (at $k = 10^9$ m^{-1}: 191 against 131 meV along [100], 271 against 51 meV along [111]).

(d) For $x = 0.25$: E_g(AlGaAs) $= 1.736$ eV, $\Delta E_g = 0.312$ eV, $\Delta E_c = \Delta E_v = 0.156$ eV, $\chi_{AlGaAs} = 3.914$ eV; a straddling (type I) junction. (i) Before contact: common vacuum level, separate Fermi levels, E_c of AlGaAs 0.156 eV higher and E_v 0.156 eV lower. After contact: flat E_F, bands bent by eV_{bi}, a conduction-band spike ΔE_c and a valence-band step ΔE_v at the interface. (ii) From $p = N_V \exp[-(E_F - E_v)/kT]$ on each side, $eV_{bi} = E_{FN} - E_{FP} = (E_{vN} - E_{vP}) + kT\ln(p_P N_{VN}/p_N N_{VP})$ with $E_{vN} - E_{vP} = -\Delta E_v$, giving $eV_{bi} = -\Delta E_v + kT\ln(P_P N_{VN}/(P_N N_{VP}))$; for an illustrative 10^{17} cm^{-3} on both sides, $V_{bi} = 1.419$ V.

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2. (a) Which semiconductors are most extensively used in near infra-red region? Explain the reasons behind it. Marks: 5

(b) Why is lattice matching of semiconductor important for hetero-structured devices? Discuss the effect of strain and critical thickness in epitaxial growth. Marks: 9

(c) Discuss the effect of isoelectric trap in the ternary compound semiconductor $GaAs_{1-x}P_x$. Marks: 4

(d) Fig. 1 shows the Bandgap Vs. lattice constant for different binary compounds. Assume that a quaternary compound $In_xGa_{1-x}As_ySb_{1-y}$ is grown on InP substrate. Find the composition if the bandgap is 1 eV and lattice parameter is 5.93 Å. Marks: 7

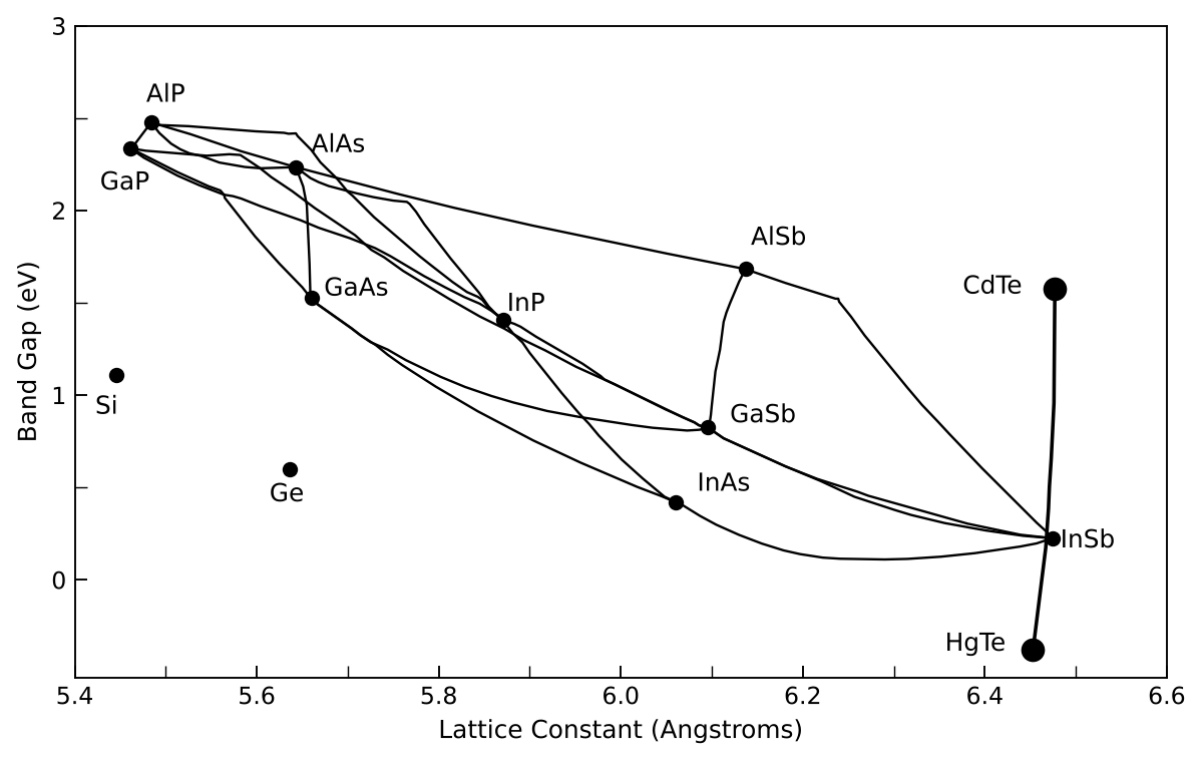


Figure 11: Fig. 1: Band gap vs. lattice constant of binary compounds (redrawn from the paper)

2.(a) Which semiconductors are most extensively used in near infra-red region? Explain the reasons behind it. Marks: 5

1. Theory

1. What is the near-infrared region? (Core Idea)

Light is an electromagnetic wave. Our eyes see wavelengths from about 0.4 µm (violet) to 0.75 µm (red). Just beyond red, from about **0.75 µm to 2 µm**, lies the **near-infrared** (near-IR). We cannot see it, but it is the most important band for optical-fibre communication, remote controls, laser printers, night cameras and sensing.

2. How does a semiconductor emit or absorb light?

A semiconductor has an almost full **valence band** and an almost empty **conduction band**, separated by the **band gap** E_g (energy bands are built in 1(d)).

- **Emission (LED, laser):** an electron in the conduction band falls into an empty state (a hole) in the valence band. The energy it loses, about E_g, leaves as one photon.
- **Absorption (photodetector):** a photon with energy at least E_g lifts an electron from the valence band to the conduction band. A photon with less energy passes straight through.

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The photon energy and its wavelength are linked by $E = hc/\lambda$. With E in eV and λ in µm:

$$\lambda\ (\mu\text{m}) = \frac{hc}{E_g} = \frac{1.24}{E_g\ (\text{eV})}$$

So an emitter gives light at $\lambda \approx 1.24/E_g$, and a detector responds to every wavelength **shorter** than this **cut-off wavelength**. For near-IR we need E_g between about $1.24/2 = 0.62$ eV and $1.24/0.75 = 1.65$ eV.

3. Direct and indirect band gaps

In an E-k diagram (built in 1(c)) the bottom of the conduction band and the top of the valence band may sit at the same crystal momentum k or at different k.

- **Direct gap** (GaAs, InP, InGaAs, InGaAsP): both at $k = 0$. An electron can drop straight down and give a photon. Emission is fast and efficient.
- **Indirect gap** (Si, Ge, GaP, AlAs): the minimum is at another k. A photon carries almost no momentum, so a **phonon** (a lattice vibration) must also take part. This three-body event is rare, so Si and Ge are poor light emitters. They still **absorb** well enough to be good detectors.

4. Why do fibres pick the wavelengths?

Silica optical fibre loses the least light in three **windows**: about 0.85 µm, 1.3 µm (least **dispersion**, so pulses do not spread) and 1.55 µm (least **loss**, about 0.2 dB/km). Sources and detectors are therefore made for these three wavelengths.

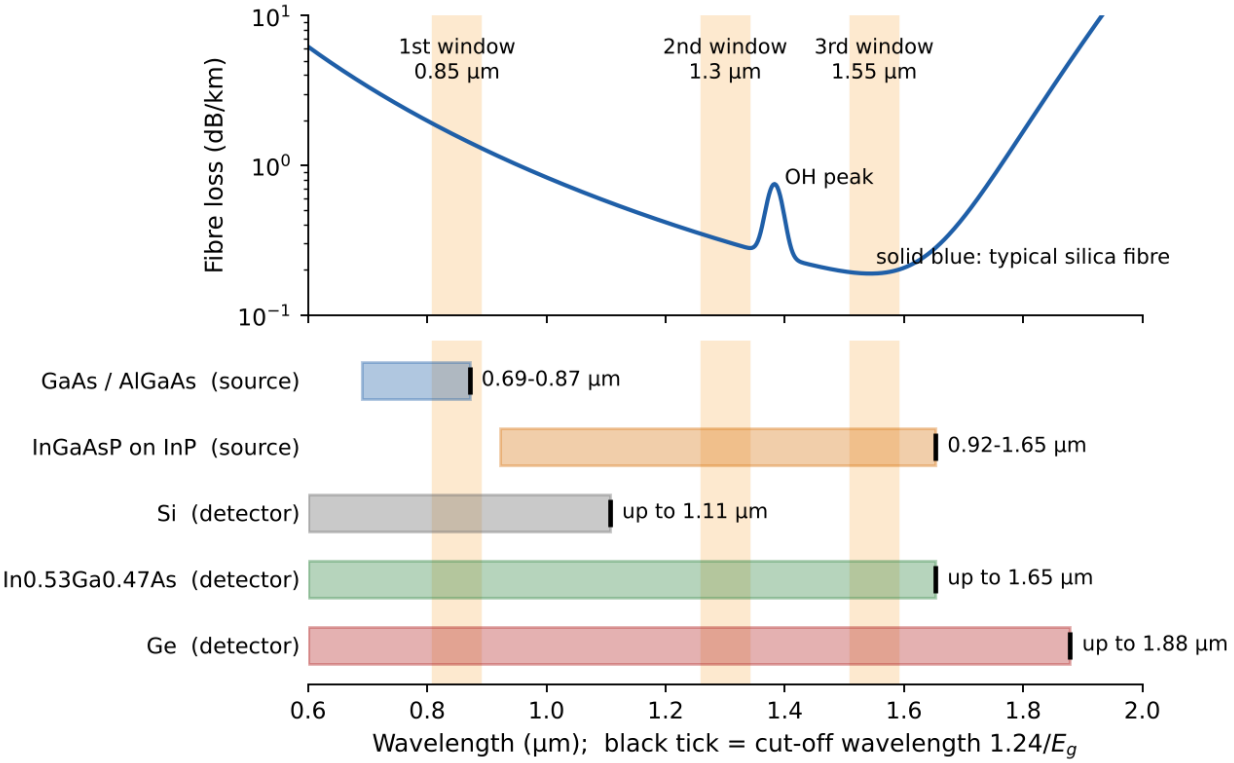


Figure 12: Fibre loss windows (top, solid blue curve) and the wavelength range each semiconductor covers (bottom, shaded bars ending at the cut-off tick)

Reading the figure. The top panel shows the typical loss of a silica fibre (solid blue curve) on a log scale: it falls with wavelength (Rayleigh scattering) until infrared absorption of the glass makes it rise after 1.6 µm; the bump at 1.38 µm is water (OH) absorption. The shaded vertical stripes are the three windows. In the bottom panel each shaded bar is a material; its black tick is the cut-off wavelength $1.24/E_g$. GaAs/AlGaAs covers the first window; InGaAsP on InP can be tuned across both the second and the third. Among detectors, Si stops at 1.11 µm, so it is useless at 1.3 and 1.55 µm, while $In_{0.53}Ga_{0.47}As$ (to 1.65 µm) and Ge (to 1.88 µm) cover all three windows.

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5. Why alloys? (Band-gap engineering)

Mixing two compounds gives an **alloy** whose band gap lies between theirs. In $Al_xGa_{1-x}As$, raising x raises E_g; in $In_{1-x}Ga_xAs_yP_{1-y}$ two fractions can be chosen, so the band gap and the lattice constant can be set independently (lattice constants and Vegard's law are built in 2(b) and 2(d)).

2. Answer

The near-IR region (about 0.75-2 µm) is served mainly by the **III-V compound semiconductors** GaAs, AlGaAs, InP, InGaAs and InGaAsP, plus **Ge** (and Si up to 1.1 µm) for detectors.

Material	E_g (eV)	$\lambda = 1.24/E_g$ (µm)	Main use
GaAs	1.424	0.871	LEDs and lasers, 0.8-0.9 µm (first window)
$Al_{0.3}Ga_{0.7}As$	1.798	0.690	barrier/cladding layers of GaAs lasers
InP	1.344	0.923	substrate for 1.3 and 1.55 µm devices
InGaAsP (on InP)	0.954 / 0.800	1.30 / 1.55	lasers and LEDs for the 2nd and 3rd fibre windows
$In_{0.53}Ga_{0.47}As$ (on InP)	0.75	1.65	photodetectors 1.0-1.65 µm
Ge	0.66	1.88	photodetectors up to about 1.8 µm
Si	1.12	1.107	detectors below 1.1 µm only

Table 2: Near-IR semiconductors (Sze & Ng band gaps, 300 K)

Reasons:

1. **Right band gap:** E_g between about 0.62 and 1.65 eV puts $\lambda = 1.24/E_g$ in the near-IR. GaAs (0.87 µm) and InGaAsP (1.3, 1.55 µm) fall exactly on the fibre windows of low dispersion and low loss.
2. **Direct band gap:** GaAs, InP, InGaAs and InGaAsP are direct, so electron-hole recombination emits a photon without a phonon. Emission and absorption are strong, giving efficient LEDs and lasers (Si and Ge are indirect, so they are used only as detectors).
3. **Lattice matching:** AlGaAs matches GaAs at every composition, and $In_{0.53}Ga_{0.47}As$ and InGaAsP match InP. Defect-free **heterostructures** can therefore be grown (double-heterostructure lasers, quantum wells), which confine carriers and light.
4. **Tunable band gap:** in a ternary or quaternary alloy the composition sets E_g, so the emission wavelength can be chosen (InGaAsP covers 0.92-1.65 µm while staying matched to InP).
5. **High electron mobility** of GaAs and InGaAs gives fast photodetectors and lets the optical devices share a chip with high-speed transistors.

3. Keyword Explanation

1. Cut-off wavelength

The longest wavelength a semiconductor can absorb, $\lambda_c = 1.24/E_g$ (µm, with E_g in eV).

2. Fibre window

A wavelength range where silica fibre has low loss: 0.85 µm, 1.3 µm (zero dispersion) and 1.55 µm (minimum loss, about 0.2 dB/km).

3. Lattice matching

Two crystals have the same lattice constant, so one can be grown on the other without defects (see 2(b)).

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4. Notes and Remarks

- The 1.24 comes from $hc/e = (6.626 \times 10^{-34})(3 \times 10^{8})/(1.6 \times 10^{-19}) = 1.24 \times 10^{-6}$ V · m.
- The fibre-loss curve in the figure is a typical shape for illustration, not the data of a particular fibre.
- For the mid-IR (3-5 µm, 8-12 µm) narrower gaps are used instead: InSb (0.17 eV), InAs (0.354 eV) and HgCdTe.

2.(b) Why is lattice matching of semiconductor important for hetero-structured devices? Discuss the effect of strain and critical thickness in epitaxial growth. Marks: 9

1. Theory

1. What is a lattice constant? (Core Idea)

A crystal is a regular, repeating pattern of atoms. The smallest repeating box of the pattern is the **unit cell**, and the length of its side is the **lattice constant** a. GaAs, InP, Si and the other materials of Fig. 1 all have cubic cells, with a between about 5.4 and 6.5 Å (1 Å = 0.1 nm).

2. What is epitaxy and a heterostructure?

Epitaxy means growing a thin crystal layer on top of a crystal **substrate** so that the layer continues the substrate's atomic pattern (methods: MBE, MOCVD, LPE). If the layer is a different material, the result is a **heterostructure**, and the plane where they meet is a **heterojunction** (its band diagram is built in 1(d)). Lasers, HEMTs and HBTs are all heterostructures.

3. Lattice mismatch

The layer's atoms must sit on top of the substrate's atoms. This is easy only if both have the same lattice constant. The mismatch is measured by the **misfit**

$$f = \frac{a_L - a_S}{a_S}$$

where a_L is the natural lattice constant of the layer and a_S that of the substrate. f is usually given in percent.

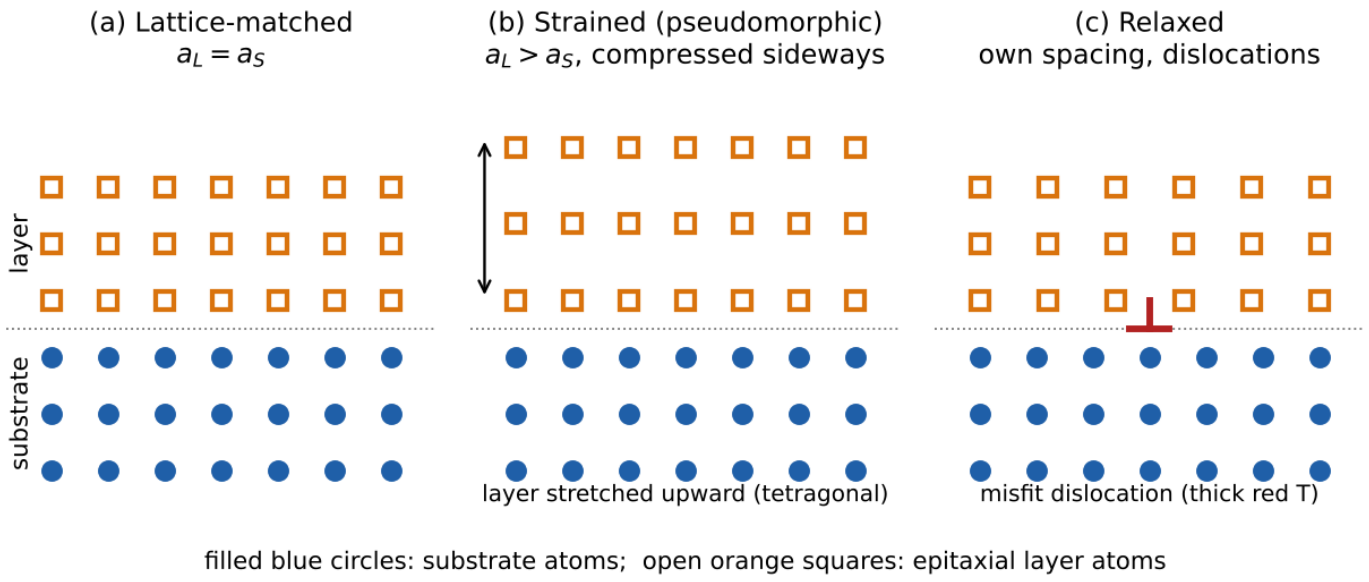


Figure 13: Three ways a layer can sit on a substrate: matched, strained and relaxed with a misfit dislocation

Reading the figure. The filled blue circles are substrate atoms and the open orange squares are layer atoms; the dotted line is the interface. **(a)** With $a_L = a_S$ every layer atom sits above a substrate atom, and nothing is disturbed. **(b)** If $a_L > a_S$ but the layer is thin, the layer is squeezed sideways to the

substrate spacing and, to keep its volume, stretches upward (the double arrow). This is a **strained** or **pseudomorphic** layer. **(c)** If the layer is too thick, it relaxes to its own spacing. Then there is one column of atoms fewer in the layer than in the substrate, and an extra half-plane ends at the interface: a **misfit dislocation** (the thick red T symbol).

4. Why are dislocations harmful?

At a dislocation, atoms have broken (dangling) bonds. These create energy levels inside the band gap, which act as:

- **recombination centres:** electrons and holes meet there and recombine without light, so the carrier **lifetime** and the efficiency of LEDs and lasers fall;
- **scattering centres:** they lower the **mobility** of carriers (mobility is built in 1(a));
- **leakage paths:** junctions leak current, and the defects grow during operation (dark lines in lasers), so **reliability** falls.

That is why lattice matching matters so much.

5. Strain

Strain is the fractional change of a length. In the pseudomorphic layer of panel (b):

- **In-plane strain:** $\varepsilon_{\parallel} = (a_S - a_L)/a_L \approx -f$. If $a_L > a_S$ it is negative: **compressive** strain (InGaAs on GaAs, SiGe on Si). If $a_L < a_S$ it is positive: **tensile** strain (GaAsP on GaAs).
- **Out-of-plane strain:** the cell grows (or shrinks) in the growth direction, $\varepsilon_{\perp} = -(2C_{12}/C_{11})\,\varepsilon_{\parallel}$, with the elastic constants C_{11}, C_{12} ($2C_{12}/C_{11} \approx 0.9$ for GaAs). The cubic cell becomes a slightly tall box: a **tetragonal distortion**.

Strain is not only a nuisance. It changes the band structure:

- the band gap shifts (compressive strain raises it, tensile lowers it, for the same alloy);
- the heavy-hole and light-hole bands, which touch at $k = 0$ in a cubic crystal (see 1(c)), **split** apart, and the top valence band gets a lighter mass.

Engineers use this on purpose: **strained-layer quantum-well lasers** (InGaAs on GaAs, lower threshold current), **strained SiGe** HBTs and strained-Si MOSFETs (higher mobility), and **pseudomorphic InGaAs HEMTs** (pHEMTs, larger band offset and mobility).

6. Critical thickness

A strained layer stores elastic energy, and this energy grows with its thickness h. Making a misfit dislocation also costs energy, but it releases part of the strain. Below a certain thickness the strained layer is cheaper; above it dislocations form. That thickness is the **critical thickness** h_c. Its classic model is by **Matthews and Blakeslee** (force balance on a threading dislocation):

$$h_c = \frac{b\,(1 - \nu\cos^2\alpha)}{2\pi f\,(1+\nu)\cos\lambda}\left[\ln\left(\frac{h_c}{b}\right) + 1\right]$$

where

- b is the length of the dislocation's Burgers vector ($a/\sqrt{2} \approx 4$ Å in GaAs)
- ν is Poisson's ratio (about 0.31)
- α, λ are angles of the dislocation (60° for the usual dislocation in zinc-blende crystals)

A rough order-of-magnitude estimate, useful in an exam, is

$$h_c \approx \frac{a}{2f}$$

It says the same thing: h_c **is roughly inversely proportional to the misfit.** It is only an estimate (the model forms differ by factors of 2-4).

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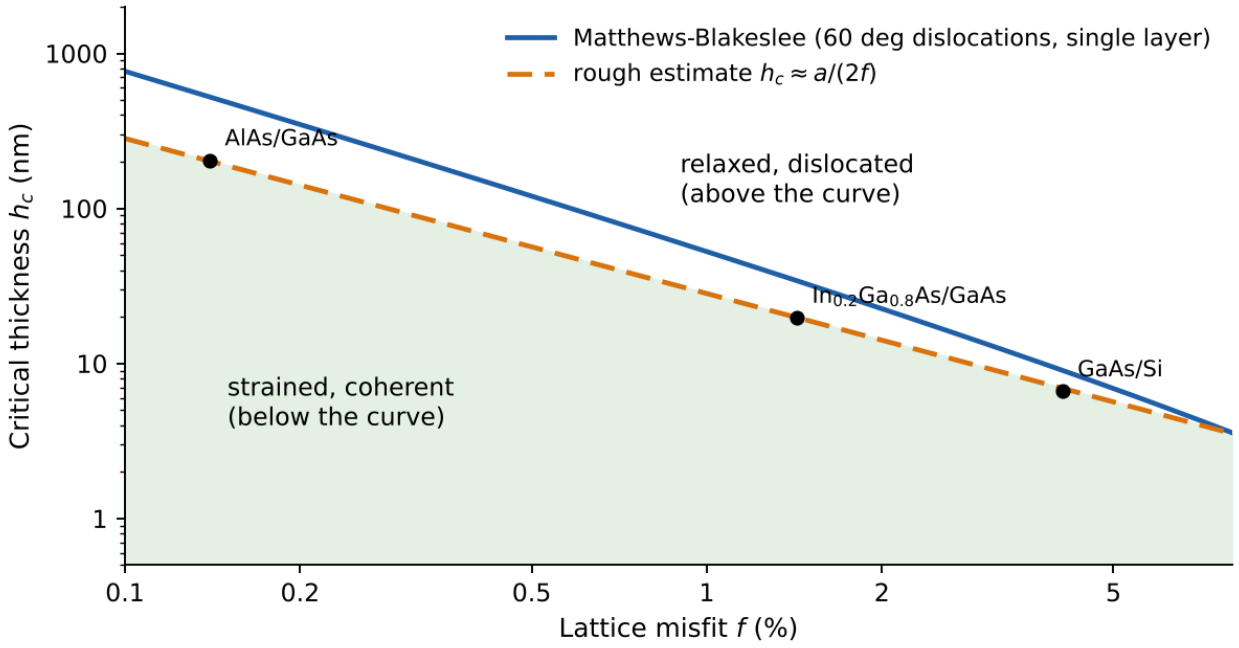


Figure 14: Critical thickness against misfit on log-log axes: Matthews-Blakeslee (solid blue) and the rough estimate a/(2f) (dashed orange)

Reading the figure. Both curves fall as the misfit grows; on log-log axes they are nearly straight with slope -1. A layer whose thickness is below the curve (the shaded region under the dashed orange line) stays **strained and coherent**; above the curve it **relaxes** with misfit dislocations. The marked points use the rough estimate: AlAs on GaAs (0.14 %) can be about 200 nm thick, $In_{0.2}Ga_{0.8}As$ on GaAs (1.43 %) only about 20 nm, and GaAs on Si (4.1 %) only a few nanometres.

2. Answer

Why lattice matching is important:

- In a heterostructure the layer must continue the crystal of the substrate. The misfit is $f = (a_L - a_S)/a_S$.
- If $f \neq 0$ and the layer is thick, **misfit dislocations** form at the interface and **threading dislocations** run up through the active layer.
- Dislocations give energy levels in the gap. They act as **non-radiative recombination centres** (lower lifetime, lower efficiency of LEDs and lasers, higher threshold current), **scattering centres** (lower mobility), and **leakage paths** (higher dark current); they also grow during operation, so reliability falls.
- Interface states at a defective heterojunction also pin the bands and spoil the designed band offsets.
- Hence near-matched pairs are used: AlGaAs/GaAs ($f = 0.14$ % for AlAs), $In_{0.53}Ga_{0.47}As$/InP and InGaAsP/InP ($f \approx 0$).

Effect of strain:

- A thin mismatched layer can grow **pseudomorphically**: it is forced to the substrate's in-plane lattice constant, so it is under biaxial **compressive** strain ($a_L > a_S$) or **tensile** strain ($a_L < a_S$), with in-plane strain $\varepsilon_{\parallel} \approx -f$.
- Its cell then distorts **tetragonally**: it lengthens (compressive case) or shortens (tensile) along the growth direction, $\varepsilon_{\perp} = -(2C_{12}/C_{11})\varepsilon_{\parallel}$.
- The strain shifts the band gap and **splits the heavy- and light-hole bands** at $k = 0$, lowering the hole effective mass. This is used in strained quantum-well lasers (lower threshold), SiGe HBTs and strained-Si MOSFETs, and InGaAs pHEMTs (larger ΔE_c, higher mobility).

Critical thickness:

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- The elastic energy grows with layer thickness. Above the **critical thickness** h_c it becomes cheaper to form misfit dislocations, and the layer relaxes (Matthews-Blakeslee model).
- h_c falls roughly as $1/f$; a rough estimate is $h_c \approx a/(2f)$ (order of magnitude only).
- **Examples**

  (rough estimate): $In_{0.2}Ga_{0.8}As$ on GaAs, $f = 1.43$ % gives $h_c \approx 20$ nm (measured about 10-20 nm); GaAs on Si, $f = 4.1$ %, gives only about 7 nm, so thick GaAs on Si is always full of dislocations.
- **Design rule:** a strained layer must be thinner than h_c (quantum wells of a few nm are fine); a thick layer must be lattice-matched.

(Figures of the three growth cases and of h_c against f are in the Theory.)

3. Keyword Explanation

1. Pseudomorphic layer

A strained epitaxial layer that has taken the substrate's in-plane lattice constant without dislocations.

2. Misfit dislocation

A line defect at the interface where an extra half-plane of atoms ends; it relieves strain but creates traps.

3. Critical thickness h_c

The largest thickness a mismatched layer can have and still stay strained and dislocation-free.

4. Tetragonal distortion

The change of the cubic unit cell into a box with one side longer (or shorter) than the other two, caused by biaxial strain.

4. Notes and Remarks

- **Matthews-Blakeslee numbers:** with $b = 4.0$ Å, $\nu = 0.31$ and 60° dislocations, the formula gives $h_c \approx 34$ nm for $f = 1.43$ % and 9 nm for $f = 4.1$ % (solved by iteration). The rough $a/(2f)$ gives 19.7 nm and 6.6 nm. Measured values for InGaAs on GaAs are 10-20 nm; real growth can exceed the equilibrium value (metastable layers).
- **Misfit examples** (Sze & Ng lattice constants): AlAs/GaAs 0.14 %, GaAs/Si 4.09 %, Ge/Si 4.18 %, InAs/GaAs 7.16 %.
- **Graded buffers:** when a mismatched layer is unavoidable (GaAs on Si), the composition is graded slowly, or a thick buffer is grown, so that the dislocations stay below the active region.

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2.(c) Discuss the effect of isoelectric trap in the ternary compound semiconductor $GaAs_{1-x}P_x$. Marks: 4

1. Theory

1. What is $GaAs_{1-x}P_x$? (Core Idea)

$GaAs_{1-x}P_x$ is a **ternary** alloy: gallium with a mixture of arsenic and phosphorus. A fraction x of the group-V sites hold P and $1-x$ hold As. At $x = 0$ it is GaAs ($E_g = 1.424$ eV, direct); at $x = 1$ it is GaP ($E_g = 2.26$ eV, indirect). It was the material of the first red, orange and yellow LEDs.

2. The direct-indirect crossover

The conduction band has more than one minimum (valley): the Γ valley at $k = 0$ and the X valleys at the zone edge (E-k diagrams are built in 1(c)). As x rises, both move up, but the Γ valley rises faster.

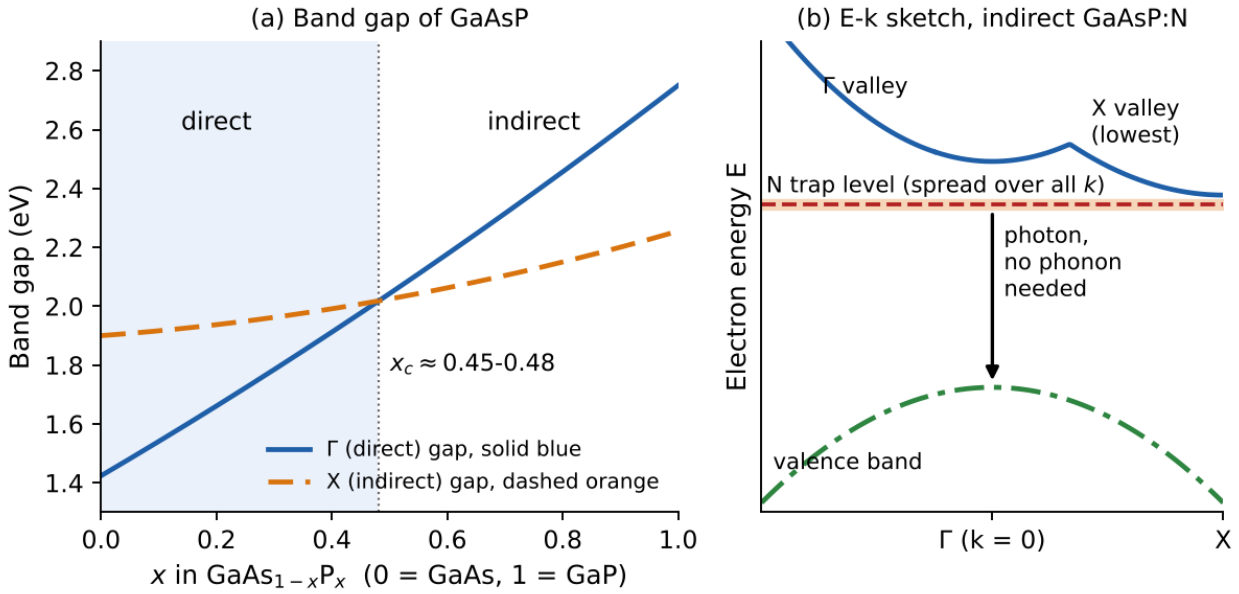


Figure 15: Band gap of GaAsP against x, with the direct gap (solid blue) and the indirect gap (dashed orange); E-k sketch of indirect GaAsP:N with the nitrogen level (dashed red line)

Reading the figure. In panel (a) the solid blue curve is the direct (Γ) gap and the dashed orange curve the indirect (X) gap. The band gap is whichever is lower. Below $x_c \approx 0.45$ (the shaded region) the Γ valley is lowest, so the alloy is **direct** and emits light well. Above x_c the X valley is lowest, so it is **indirect**: electrons collect at the zone edge, where they cannot drop to the valence-band top (at $k = 0$) without a phonon, and light emission becomes very weak. Panel (b) shows such an indirect alloy: the solid blue conduction band has its lowest valley at X, the dash-dot green valence band peaks at $k = 0$, and the dashed red line is the nitrogen level explained below.

So, without help, GaAsP cannot give bright light above $E_g \approx 2$ eV (wavelengths shorter than about 0.63 µm).

3. What is an isoelectronic trap?

An **isoelectronic** impurity is an atom from the **same group** of the periodic table as the atom it replaces. Nitrogen (N) replacing P (or As) is the standard example: all three are group V, so N brings no extra electron and is neither a donor nor an acceptor. (The paper says "isoelectric"; the standard term is **isoelectronic**.)

Still, N is much smaller and pulls electrons much more strongly (higher electronegativity) than P. Near an N atom there is a short-range potential well that can capture an electron. This is an **isoelectronic trap**. Its level lies just below the conduction band (about 0.01 eV below E_c in GaP; N-N pairs give deeper levels).

4. Why does the trap give light?

- The trapped electron is held within a very small distance Δx of the N atom.

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- By the **uncertainty principle**, $\Delta x\,\Delta p \gtrsim \hbar$, a small Δx means a large spread $\Delta p = \hbar\,\Delta k$. The electron's wavefunction is spread over a wide range of k, including $k = 0$.
- So the trapped electron has a part at $k = 0$, right above the valence-band top. It can recombine with a hole **directly**, giving a photon **without a phonon** (the vertical arrow in panel (b)).
- Once the electron is trapped (negative), it attracts a hole by Coulomb force, forming a bound **exciton**, which makes recombination even more likely.

The indirect material now emits nearly as efficiently as a direct one, at an energy a little below E_g.

2. Answer

Isoelectronic ("isoelectric") trap: an impurity of the same group as the host atom it replaces, so it adds no free carrier, but its different size and electronegativity make a short-range potential that traps an electron (or hole). In $GaAs_{1-x}P_x$ the impurity is **nitrogen** on a P/As site.

Effect in $GaAs_{1-x}P_x$:

1. $GaAs_{1-x}P_x$ is **direct for** $x < 0.45$ and **indirect for** $x > 0.45$ ($E_g \approx 1.98$ eV at the crossover). Above it, radiative recombination needs a phonon, so the efficiency of an LED falls sharply.
2. Adding N creates a trap level just below E_c (about 10 meV in GaP).
3. The electron is **localised in space** at the N atom, so by the uncertainty principle ($\Delta x\,\Delta k \gtrsim 1$) its wavefunction is **spread in k-space** and includes $k = 0$.
4. The trapped electron then captures a hole (bound exciton) and recombines **vertically, without a phonon**. The indirect alloy becomes an efficient light emitter.
5. Photon energy is slightly below E_g, so the colour shifts a little to longer wavelength.

Result: bright LEDs from indirect compositions: $GaAs_{0.35}P_{0.65}$:N (red-orange, about 632 nm), $GaAs_{0.15}P_{0.85}$:N (yellow), and GaP:N (green, about 565 nm). Without N these compositions would be very dim.

3. Keyword Explanation

1. Isoelectronic impurity

An impurity from the same column of the periodic table as the host atom it replaces (N for P, Bi for P); it is neither a donor nor an acceptor.

2. Direct-indirect crossover

The composition at which the lowest conduction-band valley changes from Γ ($k = 0$) to X; $x \approx 0.45$ in $GaAs_{1-x}P_x$.

3. Bound exciton

An electron and a hole held together by their Coulomb attraction at an impurity.

4. Notes and Remarks

- The crossover value depends on the fit used: Neamen quotes $x \approx 0.45$; the commonly used fits $E_\Gamma = 1.424 + 1.150x + 0.176x^2$ and $E_X = 1.900 + 0.144x + 0.211x^2$ (eV), drawn in the figure, cross at $x = 0.48$, $E_g = 2.02$ eV.
- GaP itself has $E_g = 2.26$ eV (549 nm); the GaP:N emission (about 565 nm) is at a slightly lower energy because the trap lies below E_c and phonon replicas are present.
- Today bright red, yellow and green LEDs are made from AlGaInP and InGaN instead, but GaAsP:N and GaP:N are still used in cheap indicator LEDs.

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2.(d) Fig. 1 shows the Bandgap Vs. lattice constant for different binary compounds. Assume that a quaternary compound $In_xGa_{1-x}As_ySb_{1-y}$ is grown on InP substrate. Find the composition if the bandgap is 1 eV and lattice parameter is 5.93 Å. Marks: 7

1. Theory

1. What is a quaternary alloy? (Core Idea)

$In_xGa_{1-x}As_ySb_{1-y}$ is a III-V crystal with **four** elements. Half of the atoms are group III (a fraction x indium and $1-x$ gallium), and half are group V (a fraction y arsenic and $1-y$ antimony). It is a mixture of the four **binary** compounds at its corners: InAs, InSb, GaAs and GaSb.

A ternary alloy (one fraction) moves along a line in Fig. 1. A quaternary alloy has *two* fractions, x and y, so it covers an area. That is its advantage: we can choose the band gap *and* the lattice constant separately, for example to match a substrate (why matching matters is built in 2(b)).

2. Vegard's law

Vegard's law says that an alloy's lattice constant is the weighted average of the lattice constants of its parts. For a ternary $A_xB_{1-x}C$:

$$a(x) = x\,a_{AC} + (1-x)\,a_{BC}$$

For a quaternary with two fractions, each binary is weighted by the product of its two fractions (bilinear interpolation):

$$Q(x,y) = xy\,Q_{\text{InAs}} + x(1-y)\,Q_{\text{InSb}} + (1-x)y\,Q_{\text{GaAs}} + (1-x)(1-y)\,Q_{\text{GaSb}}$$

The four weights add up to 1. Q can be the lattice constant. The same rule is used for the band gap as a first estimate. Real band gaps sag below the straight line (**bowing**), and Fig. 1 itself draws the ternary lines bowed. The question gives no bowing parameters, so the method it expects is the linear (Vegard) interpolation between the four corner compounds read from the chart; we use it, and the Notes say what the printed curves give.

3. How to read Fig. 1

The four binaries are corners of a four-sided region in Fig. 1. Every $In_xGa_{1-x}As_ySb_{1-y}$ alloy is a point inside it. The two given numbers, a and E_g, are one point on the chart; the composition is found by solving two equations, one for a and one for E_g.

Read the corners from the figure itself. The question says "Fig. 1 shows…", so the four corner values are taken from the chart. The lattice constants read from it agree with the textbook to within 0.01 Å, but its band gaps are a little higher than the usual room-temperature values (GaAs 1.515 eV on the chart, 1.424 eV at 300 K): the chart plots the gaps measured near 0 K. Using the chart's own corner numbers, with straight edges between them, is what makes the point (5.93 Å, 1 eV) fall inside the region.

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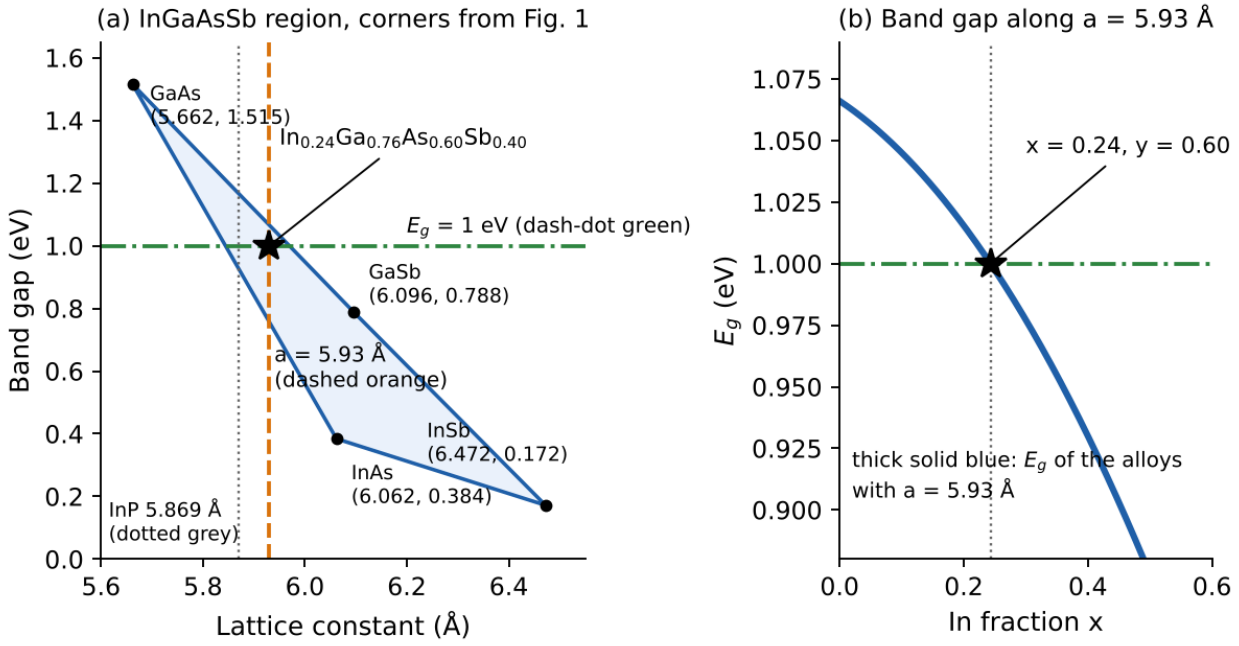


Figure 16: The InGaAsSb region of Fig. 1 (corners read from the chart) with the line a = 5.93 Å (dashed orange), the level 1 eV (dash-dot green) and the solution marked

Reading the figure. Panel (a): the shaded region with the solid blue border holds every InGaAsSb alloy, with the corners at the chart's values. The dashed orange vertical line is $a = 5.93$ Å, the dash-dot green line is $E_g = 1$ eV, and the dotted grey line is InP. The two lines cross inside the region, at the star. Panel (b) zooms in: the thick solid curve is the band gap of every alloy with $a = 5.93$ Å, and it crosses 1 eV at $x = 0.24$.

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2. Answer

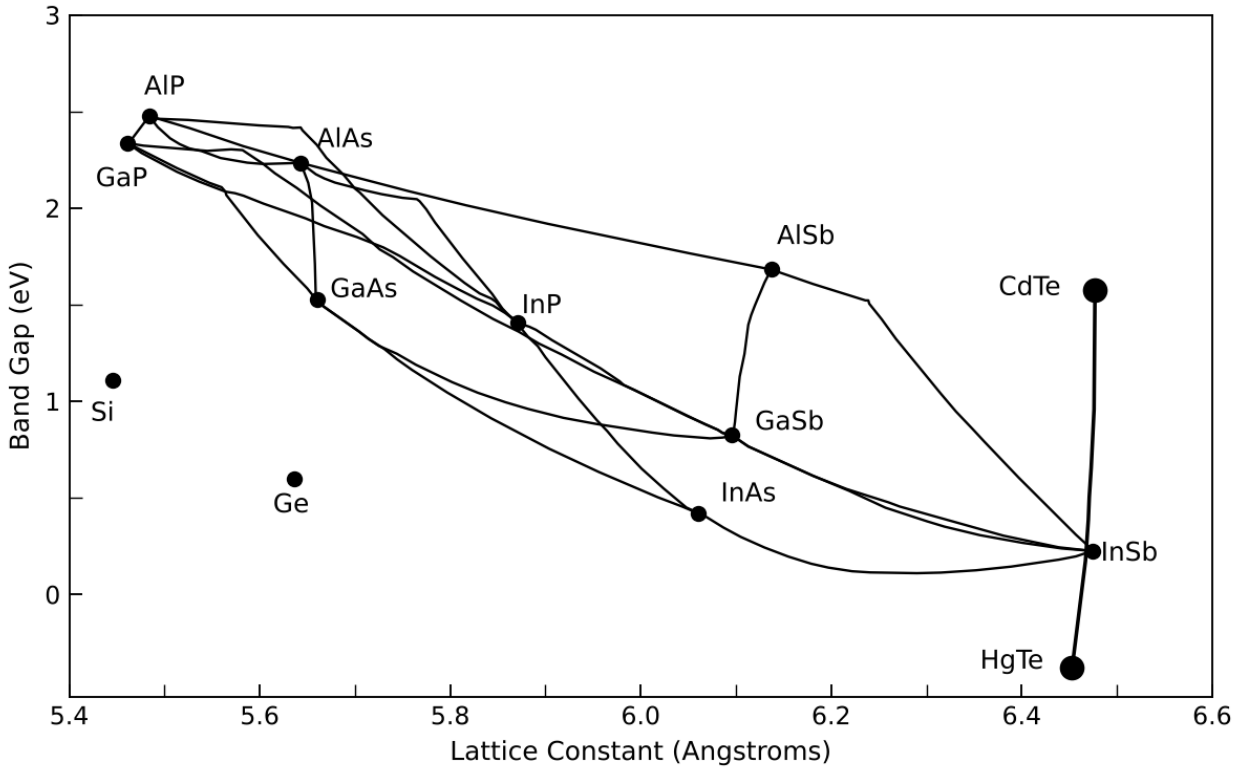


Figure 17: Repeated from the question: Fig. 1: Band gap vs. lattice constant of binary compounds (redrawn from the paper)

Given:

- Alloy $\mathrm{In}_x\mathrm{Ga}_{1-x}\mathrm{As}_y\mathrm{Sb}_{1-y}$ on InP, $E_g = 1$ eV, $a = 5.93$ Å.
- Corner binaries, read from Fig. 1:

Binary	a (Å)	E_g (eV)
InAs	6.062	0.384
InSb	6.472	0.172
GaAs	5.662	1.515
GaSb	6.096	0.788

Table 3: Corner binaries of InGaAsSb, read from Fig. 1

Assumption: Vegard's law (linear interpolation between the four corners, no bowing) for both a and E_g. This is the standard exam method; Fig. 1 itself draws the ternary lines bowed, and the Notes give what those printed curves imply.

Step 1: Interpolation formula

$$Q(x,y) = xy\,Q_{\mathrm{InAs}} + x(1-y)\,Q_{\mathrm{InSb}} + (1-x)y\,Q_{\mathrm{GaAs}} + (1-x)(1-y)\,Q_{\mathrm{GaSb}}$$

Expanding around GaSb ($x = y = 0$):

$$Q = Q_{\mathrm{GaSb}} + x\,(Q_{\mathrm{InSb}} - Q_{\mathrm{GaSb}}) + y\,(Q_{\mathrm{GaAs}} - Q_{\mathrm{GaSb}}) + xy\,(Q_{\mathrm{InAs}} - Q_{\mathrm{InSb}} - Q_{\mathrm{GaAs}} + Q_{\mathrm{GaSb}})$$

Step 2: Lattice constant

$$\begin{aligned} a &= 6.096 + x(6.472 - 6.096) + y(5.662 - 6.096) + xy(6.062 - 6.472 - 5.662 + 6.096) \\ &= 6.096 + 0.376x - 0.434y + 0.024xy \quad \text{Å} \end{aligned}$$

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Step 3: Band gap

$$E_g = 0.788 + x(0.172 - 0.788) + y(1.515 - 0.788) + xy(0.384 - 0.172 - 1.515 + 0.788)$$
$$= 0.788 - 0.616x + 0.727y - 0.515xy \ \text{eV}$$

Step 4: Apply $a = 5.93$ Å

$$6.096 + 0.376x - 0.434y + 0.024xy = 5.93 \quad \Rightarrow \quad y = \frac{0.166 + 0.376x}{0.434 - 0.024x}$$

Putting this y into E_g gives the band gap of every alloy with $a = 5.93$ Å:

x (In)	y (As)	E_g (eV)
0.00	0.382	1.066
0.10	0.472	1.045
0.20	0.562	1.015
0.24	0.601	1.000
0.30	0.653	0.977
0.40	0.746	0.930

Table 4: Alloys with a = 5.93 Å (Vegard's law, corners from Fig. 1)

Step 5: Apply $E_g = 1$ eV

The band gap falls through 1 eV between $x = 0.20$ and $x = 0.30$. Solving the two equations together (substitute the y of Step 4 into Step 3 and solve for x):

$$x = 0.243, \qquad y = \frac{0.166 + 0.376(0.243)}{0.434 - 0.024(0.243)} = 0.601$$

Check: $a = 6.096 + 0.376(0.243) - 0.434(0.601) + 0.024(0.243)(0.601) = 5.930$ Å and $E_g = 0.788 - 0.616(0.243) + 0.727(0.601) - 0.515(0.243)(0.601) = 1.000$ eV.

$$\boxed{x = 0.24, \quad y = 0.60: \quad \mathrm{In_{0.24}Ga_{0.76}As_{0.60}Sb_{0.40}}}$$

The alloy is about a quarter indium on the group-III sites and 60 % arsenic on the group-V sites.

Final Answer: $x \approx 0.24$, $y \approx 0.60$: $\mathrm{In_{0.24}Ga_{0.76}As_{0.60}Sb_{0.40}}$ has $a = 5.93$ Å and $E_g = 1$ eV.

Check with InP: $a(\mathrm{InP}) = 5.869$ Å, so a 5.93 Å layer has misfit $f = (5.93 - 5.869)/5.869 = 1.04$ %; by 2(b), it stays strain-free only below roughly $h_c \approx a/(2f) \approx 28$ nm.

3. Keyword Explanation

1. Vegard's law

An alloy's lattice constant is the composition-weighted average of those of its components.

2. Quaternary alloy

A compound of four elements, with two independent composition fractions; it lets E_g and a be chosen separately.

3. Bowing

The downward sag of an alloy's band gap below the straight line between its end compounds.

4. Notes and Remarks

- **How sure is the reading?** If each band gap read from the chart is off by ±0.02 eV, x stays between 0.19 and 0.29; the answer "about a quarter indium, 60 % arsenic" does not change.
- **With the textbook's room-temperature values** (Sze & Ng, 300 K: GaAs 1.424, GaSb 0.726, InAs 0.354, InSb 0.170 eV) the band gap along the 5.93 Å line never reaches 1 eV: its largest value is 0.988 eV, at $x = 0$ ($\mathrm{GaAs_{0.375}Sb_{0.625}}$). So the question's numbers only work with the chart's own (low-temperature) band gaps, which is why the corners are read from Fig. 1.

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- **The given numbers do not fit InP exactly.** The question says "on InP" but gives 5.93 Å, a 1.04 % mismatch. An alloy exactly matched to InP (5.869 Å) with $E_g = 1$ eV would need more arsenic.
- **Bowing:** real InGaAsSb band gaps sag well below the straight-line estimate (GaAsSb bows by about 1.43 eV, Vurgaftman et al.), so a real 1 eV alloy has a different composition. The exam asks for the straight-line reading of Fig. 1. **What the printed curves of Fig. 1 give:** at $a = 5.93$ Å the printed GaAs–InAs curve is at about 0.70 eV and the printed GaAs–GaSb curve at about 0.90 eV (read from a 600 dpi scan, the paper's slightly skewed frame corrected; about 0.66 and 0.88 eV without the correction). So with the bowing drawn in Fig. 1 the InGaAsSb region spans only about 0.70 to 0.90 eV at this lattice constant, and no alloy has both $a = 5.93$ Å and $E_g = 1$ eV; the point lies about 0.1 eV above the region. The composition above is the straight-line (Vegard) answer the question expects. The corner values read against the left axis (1.515, 0.788, 0.384, 0.172 eV) are a few hundredths of an eV lower on the right side of the chart than a reading that corrects the skew; this does not change the conclusion.
- **Wavelength:** 1 eV corresponds to $1.24/1.0 = 1.24$ µm, in the near-IR (see 2(a)).

5. Final Concise Answer

(a) Near-IR (0.75-2 µm) devices use III-V compounds: GaAs/AlGaAs (0.8-0.9 µm), InGaAsP and InGaAs on InP (1.3 and 1.55 µm, the fibre windows of low dispersion and low loss), and Ge or InGaAs detectors. Reasons: band gaps of 0.6-1.65 eV give $\lambda = 1.24/E_g$ in the near-IR (GaAs 0.87 µm); direct gaps give efficient emission and absorption; lattice matching to GaAs or InP gives defect-free heterostructures; the alloy composition tunes E_g; high mobility gives fast devices.

(b) Lattice matching ($f = (a_L - a_S)/a_S \approx 0$) avoids misfit and threading dislocations, which act as non-radiative recombination centres and scattering centres and lower lifetime, efficiency, mobility and reliability. A thin mismatched layer grows pseudomorphically under compressive or tensile strain with a tetragonal distortion; strain shifts E_g and splits the heavy- and light-hole bands (used in strained lasers, SiGe, InGaAs pHEMTs). Above the critical thickness h_c (Matthews-Blakeslee; roughly $h_c \approx a/(2f)$) the layer relaxes by forming misfit dislocations: about 20 nm for $In_{0.2}Ga_{0.8}As$ on GaAs ($f = 1.43$ %).

(c) $GaAs_{1-x}P_x$ is direct for $x < 0.45$ and indirect above. Nitrogen on a P site is isoelectronic (same group V) and forms a trap just below E_c; the electron is localised in space, so it is spread in k-space and can recombine with a hole without a phonon. This makes indirect $GaAs_{0.35}P_{0.65}$:N (red-orange) and GaP:N (green) efficient LEDs.

(d) With Vegard's law and the corner values read from Fig. 1, $a = 6.096 + 0.376x - 0.434y + 0.024xy$ Å and $E_g = 0.788 - 0.616x + 0.727y - 0.515xy$ eV. Setting $a = 5.93$ Å and $E_g = 1$ eV gives $x = 0.24$, $y = 0.60$: $In_{0.24}Ga_{0.76}As_{0.60}Sb_{0.40}$ (1.04 % mismatched to InP).

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3. (a) Draw the ideal energy band diagram of a metal semiconductor junction under:

   (i) Reverse bias condition
   (ii) Forward bias condition Marks: 5

   (b) Explain how the density of states changes qualitatively as the dimensionality reduces from $3D \rightarrow 2D \rightarrow 1D$. Sketch $D(E)$ vs E for each case. Marks: 6

   (c) Consider a contact between tungsten and n-type silicon doped to $N_d = 1 \times 10^{16}$ cm^{-3} at $T = 300$ K. Determine the

   (i) theoretical barrier height
   (ii) built-in potential barrier
   (iii) maximum electric field in the metal-semiconductor diode for zero applied bias, and
   (iv) the junction capacitance. Given that, the metal work function for tungsten (W) is 4.55 V and the electron affinity for silicon is 4.01 V. Assume $N_c = 2.8 \times 10^{19}$ cm^{-3}. Marks: 8

   (d) Consider a gallium arsenide metal-semiconductor contact in which the electric field in the semiconductor is assumed to be $E = 6.8 \times 10^4$ V cm^{-1}. Calculate the Schottky barrier lowering (due to image force induced lowering) and the position of the maximum barrier height. Marks: 6

3.(a) Draw the ideal energy band diagram of a metal semiconductor junction under: (i) Reverse bias condition (ii) Forward bias condition Marks: 5

1. Theory

1. What is a metal-semiconductor junction? (Core Idea)

A **metal-semiconductor junction** is what you get when a metal film is put directly on a semiconductor crystal. Depending on the two materials it behaves in one of two ways:

- as a **rectifying contact**, also called a **Schottky barrier diode**: current flows easily in one direction and hardly at all in the other, like a pn junction diode;
- as an **ohmic contact**: current flows equally easily both ways, like a resistor. This is the kind used to connect wires to a chip (see 5(d)).

This question is about the rectifying kind, made from a metal and an **n-type** semiconductor (free electrons are the carriers), with the metal work function larger than that of the semiconductor ($\phi_m > \phi_s$).

2. The words used in a band diagram

An **energy band diagram** draws the energy of an electron (upwards) against position (sideways). The levels used here were built in 1(d); in one line each:

- E_c, the **conduction band edge**: the lowest energy a free electron can have in the semiconductor.
- E_v, the **valence band edge**: the top of the band of bound electrons. The gap $E_g = E_c - E_v$ between them has no states.
- E_F, the **Fermi level**: the "water level" of the electrons. In equilibrium (no applied voltage) it is flat, the same everywhere in the structure. In the metal the electrons fill the states up to E_{Fm}.
- **Work function** ϕ_m of the metal: the energy (in volts) needed to lift an electron from E_{Fm} out to the vacuum.
- **Electron affinity** χ of the semiconductor: the energy from E_c to the vacuum.

3. The junction at zero bias, in short

When the two materials touch, electrons flow from the n-type semiconductor (higher Fermi level) into the metal until the Fermi levels line up. This leaves a layer of positive, uncovered donor ions in the semiconductor next to the metal, the **depletion region** of width W, and bends the bands upwards there. Two barriers appear:

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- Seen **from the metal**, an electron must climb the **Schottky barrier height**

$$\phi_{B0} = \phi_m - \chi$$

  This barrier is fixed by the two materials, so it does *not* change with applied voltage.
- Seen **from the semiconductor**, an electron in the conduction band must climb the **built-in potential barrier**

$$V_{bi} = \phi_{B0} - \phi_n$$

  where $e\phi_n = E_c - E_F$ in the neutral semiconductor.

How these are found, with numbers for tungsten on silicon and the drawing to scale, is in 3(c).

4. What a bias voltage does

An applied voltage V across the junction drops almost entirely across the depletion region, because the rest of the semiconductor and the metal conduct well. The voltage shifts the semiconductor's bands up or down relative to the metal by eV. The two Fermi levels then no longer line up: they are separated by exactly eV.

- **Reverse bias** V_R (semiconductor positive with respect to the metal): the semiconductor's energies are **lowered** by eV_R. The band bending grows from eV_{bi} to $e(V_{bi} + V_R)$, so more donor ions must be uncovered and the depletion region **widens**.
- **Forward bias** V_a (metal positive): the semiconductor's energies are **raised** by eV_a. The band bending shrinks to $e(V_{bi} - V_a)$ and the depletion region **narrows**.
- In both cases the corner of the barrier at the metal stays at $e\phi_{B0}$ above E_{Fm}.

The width follows the same formula as at zero bias with V_{bi} replaced by the total band bending (derived in 3(c)):

$$W = \sqrt{\frac{2\varepsilon_s(V_{bi} + V_R)}{eN_d}}$$

with V_R replaced by $-V_a$ for forward bias.

5. Where does the current come from? (Thermionic emission)

The electrons in the conduction band have a spread of thermal energies. Only the few with enough energy to get over the top of the barrier can cross the junction. This is called **thermionic emission**: emission of electrons over a barrier by their heat energy. Two streams of electrons cross:

- **metal to semiconductor:** these electrons must climb $e\phi_{B0}$, which does not change with bias, so this stream is fixed;
- **semiconductor to metal:** these electrons must climb $e(V_{bi} - V_a)$ or $e(V_{bi} + V_R)$, which does change with bias. The number able to cross changes as $\exp(eV_a/kT)$.

At zero bias the two streams are equal and the net current is zero. Their difference gives the diode equation (Neamen):

$$J = J_{sT}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right], \qquad J_{sT} = A^*T^2\exp\left(-\frac{e\phi_{B0}}{kT}\right)$$

where

- J is the current density (A/cm^2); positive current flows from the metal into the semiconductor, which means electrons flow from the semiconductor into the metal
- A^* is the effective **Richardson constant** (A/cm^{2}K^2) and T the temperature
- J_{sT} is the reverse saturation current density

Current is carried by majority carriers (electrons) only, so there is no slow minority-carrier storage. This is why Schottky diodes switch fast.

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2. Answer

Ideal rectifying contact between a metal and an n-type semiconductor ($\phi_m > \phi_s$), drawn with the metal Fermi level E_{Fm} as the reference. Panel (a) is the zero-bias diagram for comparison.

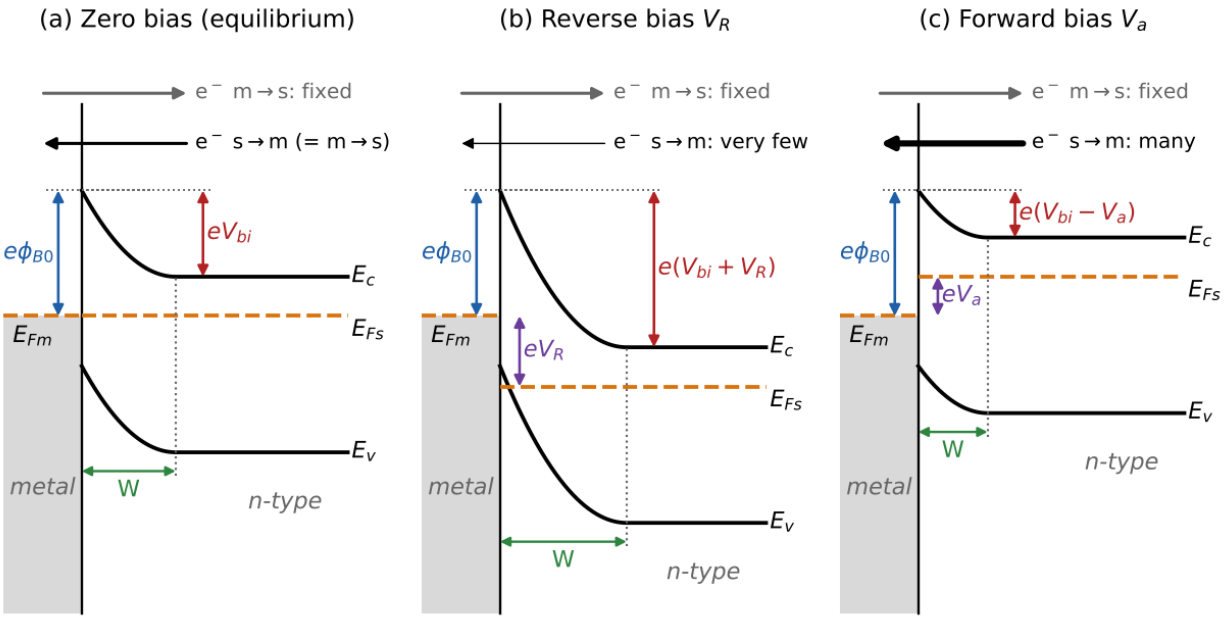


Figure 18: Ideal metal and n-type semiconductor band diagrams at zero bias, reverse bias and forward bias; the arrows at the top show the two thermionic electron streams

Reading the figure. The grey shaded block on the left is the metal, filled with electrons up to the dashed orange line E_{Fm}. The solid black curves are E_c and E_v of the semiconductor; the dashed orange line on the right is its Fermi level E_{Fs}. The blue double arrow $e\phi_{B0}$ is the barrier seen from the metal, the red double arrow the barrier seen from the semiconductor, the purple double arrow the applied voltage eV, and the green arrow the depletion width W. The two horizontal arrows above each diagram are the electron streams over the barrier: the solid grey arrow (metal to semiconductor) is the same in every panel; the solid black arrow (semiconductor to metal) is thin in reverse bias and thick in forward bias.

(i) Reverse bias (V_R applied, semiconductor positive):

- E_{Fs} lies eV_R **below** E_{Fm}; all the semiconductor energies are lowered by eV_R.
- Barrier from the semiconductor side: $e(V_{bi}+V_R)$, larger than at equilibrium.
- Depletion width grows: $W = \sqrt{2\varepsilon_s(V_{bi}+V_R)/eN_d}$.
- Barrier from the metal side stays $e\phi_{B0} = e(\phi_m - \chi)$.
- Almost no electrons from the semiconductor can cross; only the small fixed flow from the metal remains, so a small reverse current $-J_{sT}$ flows.

(ii) Forward bias (V_a applied, metal positive):

- E_{Fs} lies eV_a **above** E_{Fm}; the semiconductor energies are raised by eV_a.
- Barrier from the semiconductor side: $e(V_{bi}-V_a)$, smaller than at equilibrium.
- Depletion width shrinks: $W = \sqrt{2\varepsilon_s(V_{bi}-V_a)/eN_d}$.
- Barrier from the metal side is still $e\phi_{B0}$.
- Many more electrons now cross from the semiconductor into the metal by thermionic emission, so a large forward current flows: $J = J_{sT}[\exp(eV_a/kT) - 1]$.

Key point: the bias changes only the semiconductor-side barrier $V_{bi} \pm V$ and the width W; the Schottky barrier height ϕ_{B0} seen from the metal is unchanged (ideal case, no image-force lowering).

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3. Keyword Explanation

1. Schottky barrier height ϕ_{B0}

The energy step (in volts) an electron in the metal must climb to enter the semiconductor's conduction band; ideally $\phi_m - \chi$.

2. Built-in potential barrier V_{bi}

The band bending inside the semiconductor at equilibrium; the step an electron in the semiconductor's conduction band must climb to reach the metal.

3. Thermionic emission

Transport of carriers over a potential barrier by their thermal energy. It is the main current mechanism in a Schottky diode on moderately doped silicon or GaAs.

4. Notes and Remarks

- **Sign convention:** "reverse bias" means the n-type semiconductor is made positive with respect to the metal, just as the n side is made positive in a reverse-biased pn junction.
- **Real diodes:** the barrier seen from the metal does change slightly with bias, because the image force lowers it by an amount that grows with the field (see 3(d)). This is why the reverse current of a real Schottky diode rises slowly with V_R instead of staying flat.
- **p-type semiconductor:** with a p-type semiconductor and $\phi_m < \phi_s$, the diagrams are mirror images: the bands bend down and the barrier is for holes.

3.(b) Explain how the density of states changes qualitatively as the dimensionality reduces from $3D \to 2D \to 1D$. Sketch $D(E)$ vs E for each case. Marks: 6

1. Theory

1. What is the density of states? (Core Idea)

In a crystal an electron cannot have any energy it likes; only certain **allowed states** exist. The **density of states** $D(E)$ (Neamen writes $g(E)$) tells how many allowed states there are per unit energy, per unit size of the sample, at energy E:

$$\text{number of states between } E \text{ and } E + dE = D(E)\,dE \quad \text{(per volume, per area or per length)}$$

It matters because the number of electrons at an energy is (number of states) $\times$ (probability that a state is filled). So $D(E)$ decides the carrier density, the optical absorption and gain, and much more.

2. Why does dimensionality matter?

A free electron in a semiconductor moves like a free particle with an **effective mass** m^* (see 1(c) for E-k diagrams and effective mass). Its energy depends on its wave vector k as

$$E = E_c + \frac{\hbar^2 k^2}{2m^*}$$

where $\hbar = h/2\pi$ is the reduced Planck constant. In a box of size L, the allowed values of each component of k are spaced $2\pi/L$ apart, so the allowed states form a regular grid of points in "k-space". To find $D(E)$ we count how many grid points lie between k and $k + dk$, and how that number grows as E grows.

- If the electron can move in all three directions (**3D**, bulk crystal), the grid is three-dimensional.
- If one direction is squeezed to a few nanometres (**2D**, a **quantum well**), motion in that direction is frozen into discrete levels; the electron is free only in a plane.
- If two directions are squeezed (**1D**, a **quantum wire**), it is free only along a line.
- If all three are squeezed (**0D**, a **quantum dot**), it is not free at all: only discrete levels remain, like an atom.

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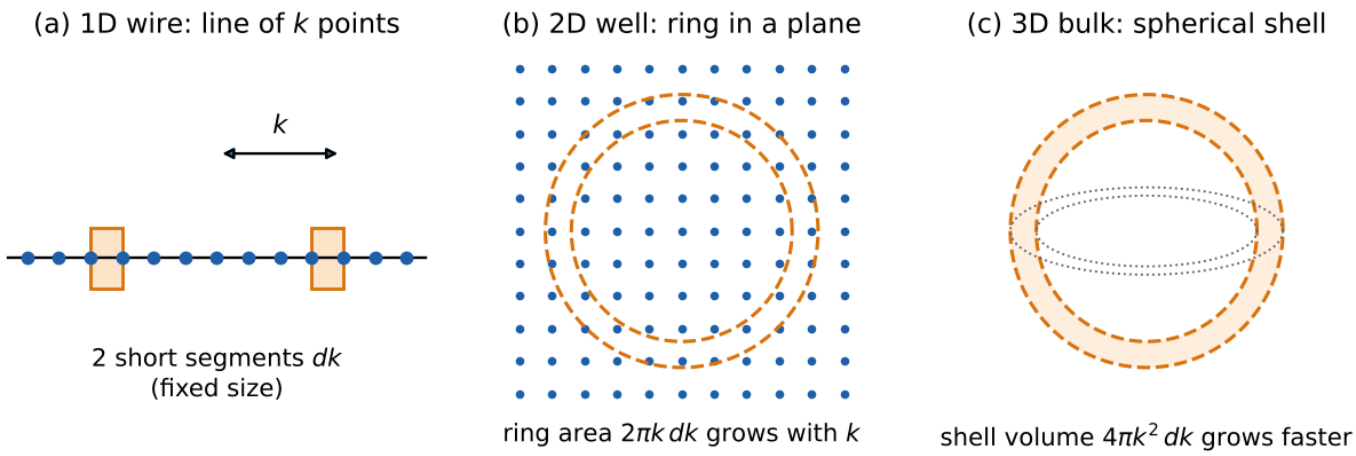


Figure 19: Counting allowed k points: two short segments on a line (1D), a ring in a plane (2D), a spherical shell (3D)

Reading the figure. The marked dots are the allowed k values. The shaded orange region between the dashed orange outlines holds the states between k and $k + dk$. In 1D it is two short segments of fixed length, whatever k is. In 2D it is a ring of area $2\pi k\,dk$, which grows in proportion to k. In 3D it is a shell of volume $4\pi k^2 dk$, which grows as k^2. This growth, divided by how fast E grows with k, gives the shape of $D(E)$.

3. The three formulas

Each k point holds two electrons (spin up and down). Using $dE/dk = \hbar^2 k/m^*$:

3D (bulk), per unit volume (Neamen):

$$D_{3D}(E) = \frac{4\pi(2m^*)^{3/2}}{h^3}\sqrt{E - E_c}$$

The shell grows as k^2 and $dk/dE \propto 1/k$, so $D \propto k \propto \sqrt{E - E_c}$: zero at the band edge, then rising smoothly.

2D (quantum well), per unit area, for each subband n:

$$D_{2D}(E) = \frac{m^*}{\pi\hbar^2}\sum_n H(E - E_n)$$

where E_n is the bottom of the n-th subband (the frozen energy of motion across the well) and H is the step function (0 below E_n, 1 above). The ring grows as k and $dk/dE \propto 1/k$, so the two cancel: each subband adds a **constant** $m^*/\pi\hbar^2$, independent of energy. The total is a **staircase**.

1D (quantum wire), per unit length, for each subband n:

$$D_{1D}(E) = \frac{1}{\pi\hbar}\sum_n \sqrt{\frac{2m^*}{E - E_n}}\,H(E - E_n)$$

The segments do not grow, and $dk/dE \propto 1/k$, so $D \propto 1/\sqrt{E - E_n}$: it is infinite at each subband edge and then falls. The total is a row of **spikes**.

0D (quantum dot): only discrete levels, $D(E) = \sum_i 2\,\delta(E - E_i)$, a set of sharp lines.

Units check: $m^*/\pi\hbar^2$ has units $\mathrm{kg}/(\mathrm{J^2 s^2}) = 1/(\mathrm{J\ m^2})$, that is states per energy per area, as a 2D density should.

Practical numbers (GaAs, $m^* = 0.067\,m_0$): each 2D step is $m^*/\pi\hbar^2 = 2.8 \times 10^{13}\ \mathrm{cm^{-2} eV^{-1}}$; in a 10 nm well the first subband sits $E_1 = 56$ meV above the bulk band edge.

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4. How the curves fit together

For a well of width L with infinitely high walls, $E_n = n^2\hbar^2\pi^2/(2m^*L^2)$. At each E_n the corner of the 2D staircase (divided by L to make it per volume) touches the 3D curve exactly. So the staircase follows the 3D $\sqrt{E}$ curve from below, as an envelope. As L grows the steps get finer and the staircase turns into the 3D curve. The same holds for the 1D spikes: averaged over energy they follow the 3D curve. Confinement does not create or destroy states; it **regroups** them into steps, spikes or lines.

2. Answer

Definition: The density of states $D(E)$ is the number of allowed electron states per unit energy per unit volume (3D), area (2D) or length (1D). It is found by counting the allowed k points between k and $k + dk$ and converting to energy with $E - E_c = \hbar^2k^2/2m^*$.

Change with dimension:

- **3D (bulk):** states between k and $k + dk$ lie in a spherical shell of volume $4\pi k^2 dk$. Hence

$$D_{3D}(E) = \frac{4\pi(2m^*)^{3/2}}{h^3}\sqrt{E - E_c}$$

  a smooth curve that starts at zero at E_c and rises as $\sqrt{E}$.
- **2D (quantum well):** confinement in one direction gives subbands $E_1, E_2, \ldots$; in each, the states lie in a ring of area $2\pi k\, dk$, which gives a constant

$$D_{2D}(E) = \frac{m^*}{\pi\hbar^2} \text{ per subband}$$

  so $D(E)$ is a **staircase**, rising by one step at each E_n. The step corners touch the 3D curve. There is a finite density of states right at the lowest edge E_1 (not zero as in 3D).
- **1D (quantum wire):** states lie on a line, two segments of length dk each, so

$$D_{1D}(E) \propto \frac{1}{\sqrt{E - E_n}}$$

  for each subband: a **spike** (van Hove singularity) at every subband edge, falling off above it.
- **0D (quantum dot), for completeness:** only discrete levels, $D(E) = \sum 2\delta(E - E_i)$, like an atom.

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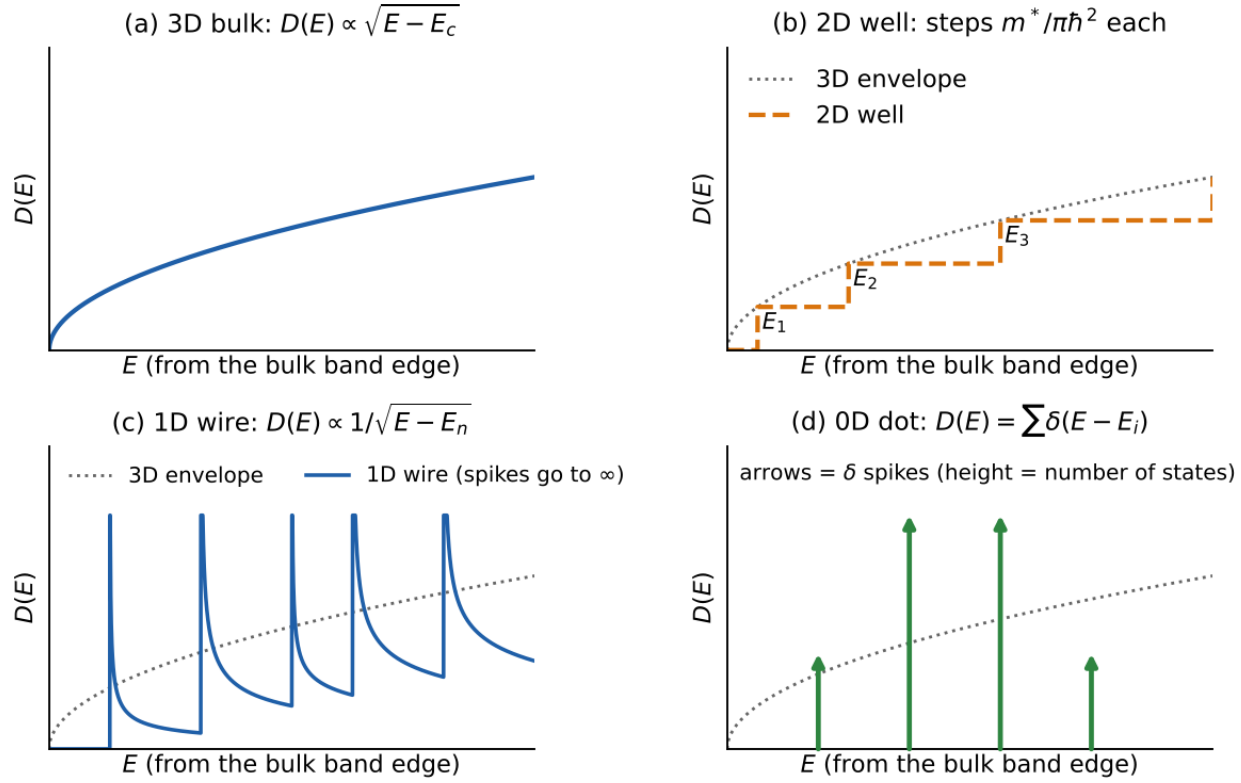


Figure 20: Density of states against energy in 3D, 2D, 1D and 0D, with the 3D curve dotted grey for comparison

Reading the figure. Panel (a) is the smooth solid blue $\sqrt{E}$ curve of bulk. In (b) the dashed orange staircase of a quantum well rises one step at each subband edge E_1, E_2, E_3, and each corner touches the dotted grey 3D curve. In (c) the solid blue curve of a quantum wire shoots up at each subband edge and then decays as $1/\sqrt{E - E_n}$ (the spikes are cut at the top; they really go to infinity). In (d) the thick green arrows are the delta functions of a quantum dot; a taller arrow means more states at that level.

Qualitative trend: as the dimension drops from 3D to 2D to 1D (to 0D), the states are squeezed into ever narrower energy ranges near the subband edges: smooth → steps → spikes → lines.

Practical Note: the sharp, concentrated density of states is what makes quantum-well and quantum-dot **lasers** have lower threshold current and narrower gain spectra, and it gives the two-dimensional electron gas (2DEG) of a **HEMT** its high sheet density at a well-defined energy.

3. Keyword Explanation

1. Quantum confinement

Squeezing a carrier into a region comparable to its de Broglie wavelength (a few nm to tens of nm), so that its motion in that direction becomes quantised into discrete levels.

2. Subband

The set of states that share one quantised level E_n of the confined direction(s) and are free in the remaining direction(s).

3. Van Hove singularity

A point where the density of states becomes infinite, here at each subband edge of a 1D wire.

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4. Notes and Remarks

- **Textbook:** Neamen derives only the 3D formula (with $g(E)$ for the density of states). The 2D and 1D results are the standard ones of low-dimensional physics (Sze & Ng, and any quantum-well text); they follow from the same counting.
- **Real wells:** with finite barriers the levels E_n are a little lower than the infinite-well values, and only a few subbands are bound, but the step shape stays.
- **Common confusion:** the 2D density is per area ($\mathrm{cm}^{-2}\mathrm{eV}^{-1}$) and the 3D density per volume ($\mathrm{cm}^{-3}\mathrm{eV}^{-1}$); compare them only after dividing the 2D value by the well width.

3.(c) Consider a contact between tungsten and n-type silicon doped to $N_d = 1 \times 10^{16}$ cm^{-3} at $T = 300$ K. Determine the (i) theoretical barrier height (ii) built-in potential barrier (iii) maximum electric field in the metal-semiconductor diode for zero applied bias, and (iv) the junction capacitance. Given that, the metal work function for tungsten (W) is 4.55 V and the electron affinity for silicon is 4.01 V. Assume $N_c = 2.8 \times 10^{19}$ cm^{-3}. Marks: 8

1. Theory

The basics

A semiconductor's electrons occupy energy bands. The conduction band edge is E_c, the valence band edge is E_v, and the Fermi level E_F marks how full they are. The **vacuum level** is the energy of a free electron at rest. The **work function** is the energy from E_F to the vacuum level, and the **electron affinity** χ is the energy from E_c to the vacuum level. A **Schottky diode** is a metal on a semiconductor, and it rectifies like a p-n junction.

1. Before and after contact (Core Idea)

The energy-band words (E_c, E_v, E_F, vacuum level, work function ϕ_m, electron affinity χ) are built in 1(d). Here we use them for a metal on n-type silicon.

- **Before contact**, both materials hang from the same vacuum level. The metal's Fermi level is $e\phi_m$ below it. The semiconductor's conduction band edge is $e\chi$ below it, and its Fermi level lies a further $e\phi_n$ lower.
- For tungsten on n-Si, $\phi_m = 4.55$ V is larger than $\phi_s = \chi + \phi_n$, so the semiconductor's Fermi level is **higher**. Electrons are at a higher "water level" in the semiconductor.
- **After contact**, electrons flow from the semiconductor into the metal until the Fermi levels are equal. The silicon near the interface loses its free electrons and keeps the positive donor ions: a **depletion region** of width x_n. The metal gets an equal and opposite thin sheet of negative charge on its surface.
- The vacuum level must be continuous, so the semiconductor bands **bend upwards** near the metal.

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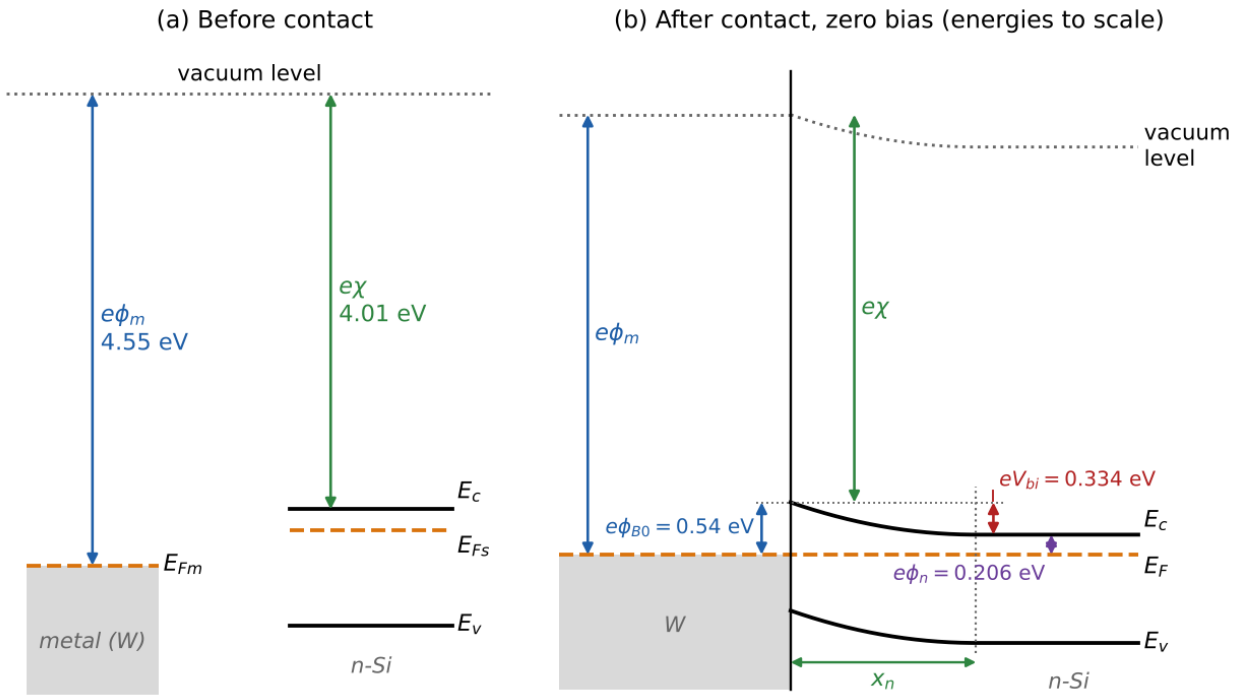


Figure 21: Tungsten and n-type silicon before contact and after contact at zero bias, drawn to scale

Reading the figure. In panel (a) the dotted grey vacuum level is common; the blue double arrow $e\phi_m$ hangs the metal's dashed orange Fermi level from it, the green double arrow $e\chi$ hangs the silicon's E_c. In panel (b) the dashed orange E_F is flat across the junction. At the interface, E_c (solid black) sits $e\phi_{B0} = 0.54$ eV above E_F (short blue double arrow); in the neutral silicon it sits $e\phi_n = 0.206$ eV above E_F (short purple double arrow). The difference, the red double arrow, is the band bending $eV_{bi} = 0.334$ eV across the depletion width x_n (green arrow).

2. The barrier height and built-in potential

Reading the barriers straight off the diagram (Neamen):

$$\phi_{B0} = \phi_m - \chi$$

$$V_{bi} = \phi_{B0} - \phi_n, \qquad \phi_n = \frac{kT}{e}\ln\left(\frac{N_c}{N_d}\right)$$

where

- ϕ_{B0} is the ideal **Schottky barrier height**, the barrier for electrons going from the metal into the semiconductor
- V_{bi} is the **built-in potential barrier**, the barrier for electrons going from the semiconductor into the metal
- ϕ_n is the depth of the Fermi level below E_c in the neutral n-type region; it comes from $n_0 = N_d = N_c \exp[-(E_c - E_F)/kT]$
- N_c is the effective density of states in the conduction band and $kT/e = 0.0259$ V at 300 K

3. Charge, field and width: Poisson's equation (depletion approximation)

To find the field we use the **depletion approximation**: inside $0 < x < x_n$ there are no free electrons at all, only donor ions of density N_d; beyond x_n the silicon is neutral. So the charge density is a simple rectangle, $\rho(x) = +eN_d$ for $0 < x < x_n$.

Poisson's equation links the field E to the charge (we write E for the electric field, as Neamen does, so that it is not confused with energy E):

$$\frac{dE}{dx} = \frac{\rho(x)}{\varepsilon_s} = \frac{eN_d}{\varepsilon_s}$$

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where $\varepsilon_s = \varepsilon_r \varepsilon_0$ is the permittivity of the semiconductor. A constant charge gives a field that changes **linearly** with x. With $E = 0$ at the edge $x = x_n$ (neutral region):

$$E(x) = -\frac{eN_d}{\varepsilon_s}(x_n - x), \qquad |E_{max}| = \frac{eN_d x_n}{\varepsilon_s} \text{ at } x = 0$$

The field is a triangle, largest at the metal. The voltage across the region is the area of this triangle:

$$V_{bi} = \frac{1}{2}|E_{max}|\, x_n = \frac{eN_d x_n^2}{2\varepsilon_s} \quad \Rightarrow \quad x_n = \sqrt{\frac{2\varepsilon_s(V_{bi} + V_R)}{eN_d}}$$

with $V_R = 0$ at zero bias. The same tools are used for the heterojunction in 4(a).

4. Junction capacitance

The depletion region is like a parallel-plate capacitor: the metal's surface sheet of charge on one side, the donor charge on the other, and a charge-free gap of width x_n between. A small change in voltage moves the edge x_n and changes the stored charge $Q' = eN_d x_n$, so the capacitance per unit area is (Neamen)

$$C' = \frac{dQ'}{dV_R} = \sqrt{\frac{e\varepsilon_s N_d}{2(V_{bi} + V_R)}} = \frac{\varepsilon_s}{x_n}$$

in F/cm^2. For a real diode multiply by its area A.

2. Answer

Given:

- Metal: tungsten, $\phi_m = 4.55$ V; semiconductor: n-Si, $\chi = 4.01$ V
- $N_d = 1 \times 10^{16}$ cm^{-3}, $N_c = 2.8 \times 10^{19}$ cm^{-3}, $T = 300$ K, $kT/e = 0.0259$ V
- $\varepsilon_s = 11.7\varepsilon_0 = 11.7 \times 8.85 \times 10^{-14} = 1.0355 \times 10^{-12}$ F/cm, $e = 1.602 \times 10^{-19}$ C
- Zero applied bias, $V_R = 0$

Step 1: (i) Theoretical barrier height

$$\phi_{B0} = \phi_m - \chi = 4.55 - 4.01$$

$$\boxed{\phi_{B0} = 0.54 \text{ V}}$$

An electron in the tungsten needs 0.54 eV to enter the silicon's conduction band.

Step 2: (ii) Built-in potential barrier

$$\phi_n = \frac{kT}{e}\ln\left(\frac{N_c}{N_d}\right) = 0.0259 \ln\left(\frac{2.8 \times 10^{19}}{1 \times 10^{16}}\right) = 0.0259 \times \ln(2800) = 0.0259 \times 7.937 = 0.2056 \text{ V}$$

$$V_{bi} = \phi_{B0} - \phi_n = 0.54 - 0.2056$$

$$\boxed{V_{bi} = 0.334 \text{ V}}$$

An electron in the silicon's conduction band sees a barrier of 0.334 eV towards the metal.

Step 3: (iii) Maximum electric field at zero bias

Depletion width:

$$x_n = \sqrt{\frac{2\varepsilon_s V_{bi}}{eN_d}} = \sqrt{\frac{2(1.0355 \times 10^{-12})(0.3344)}{(1.602 \times 10^{-19})(1 \times 10^{16})}} = 2.079 \times 10^{-5} \text{ cm} = 0.208\ \mu\text{m}$$

Maximum field (at the metal-semiconductor interface):

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$$|E_{max}| = \frac{eN_d x_n}{\varepsilon_s} = \frac{(1.602\times10^{-19})(1\times10^{16})(2.079\times10^{-5})}{1.0355\times10^{-12}}$$

$$\boxed{|E_{max}| = 3.22\times10^{4}\ \text{V/cm}}$$

Check: the field is a triangle of area V_{bi}, so $|E_{max}| = 2V_{bi}/x_n = 2(0.3344)/(2.079\times10^{-5}) = 3.22\times10^4$ V/cm ✔.

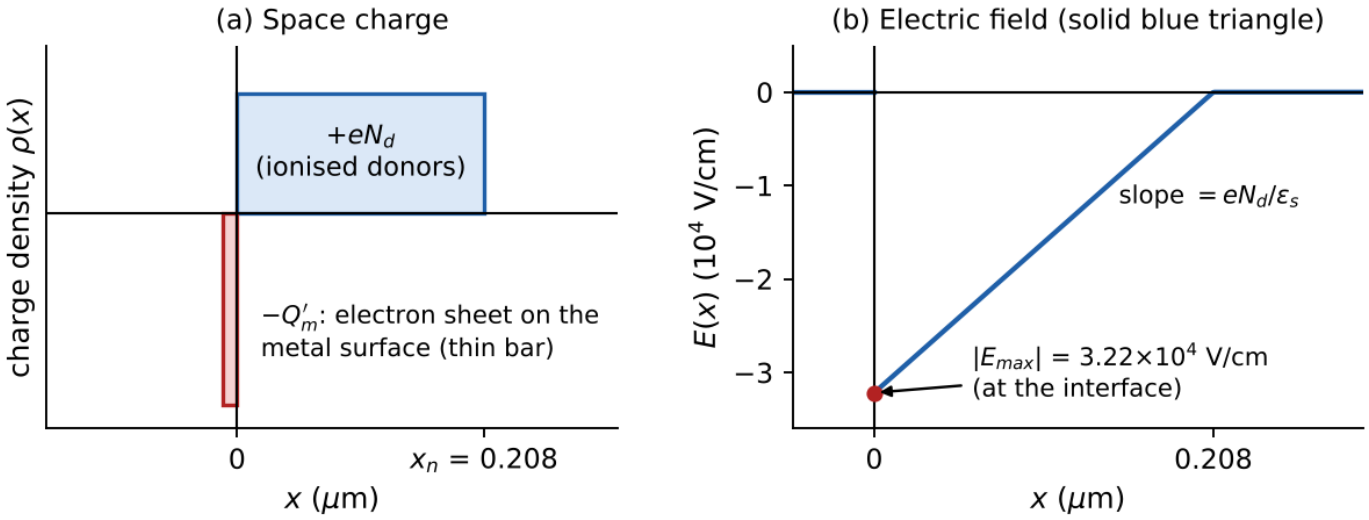


Figure 22: Space charge and electric field in the tungsten and n-Si depletion region at zero bias

Reading the figure. In panel (a) the shaded rectangle (blue outline) is the donor charge $+eN_d$ spread over 0.208 μm; the thin shaded bar (red outline) at $x = 0$ is the equal negative charge sheet on the metal surface. In panel (b) the solid blue line is the field: it falls linearly from 3.22×10^4 V/cm in magnitude at the interface (marked dot) to zero at x_n.

Step 4: (iv) Junction capacitance at zero bias

$$C' = \frac{\varepsilon_s}{x_n} = \frac{1.0355\times10^{-12}}{2.079\times10^{-5}}$$

$$\boxed{C' = 4.98\times10^{-8}\ \text{F/cm}^2 = 49.8\ \text{nF/cm}^2}$$

The area is not given, so the capacitance is stated per unit area. **Example:** a diode of area $A = 10^{-4}$ cm^2 (100 μm $\times$ 100 μm) has $C = C'A = 4.98$ pF.

Final Answer:

- (i) $\phi_{B0} = 0.54$ V
- (ii) $\phi_n = 0.206$ V, $V_{bi} = 0.334$ V
- (iii) $x_n = 0.208$ μm, $|E_{max}| = 3.22\times10^4$ V/cm
- (iv) $C' = 4.98\times10^{-8}$ F/cm^2 at zero bias (4.98 pF for 10^{-4} cm^2)

3. Keyword Explanation

1. Depletion approximation

The assumption that the depletion region is completely free of mobile carriers and ends sharply at x_n, so its charge is just eN_d.

2. Effective density of states N_c

The number that, multiplied by $\exp[-(E_c - E_F)/kT]$, gives the electron density in the conduction band; 2.8×10^{19} cm^{-3} for Si at 300 K.

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3. Depletion (junction) capacitance

The small-signal capacitance $C' = dQ'/dV$ of the depletion layer; it falls as the reverse bias grows, because the layer widens.

4. Notes and Remarks

- **Measured barrier:** the measured barrier of W on n-Si is about 0.67 V (Neamen, Sze), higher than the ideal 0.54 V. Real barriers depend only weakly on ϕ_m because of **interface states** (see 4(b)).
- **Rounding:** Neamen's own worked example on these numbers uses $e = 1.6 \times 10^{-19}$ C and gets $|E_{max}| = 3.21 \times 10^4$ V/cm; the small difference is only rounding.
- **With reverse bias:** replace V_{bi} by $V_{bi} + V_R$; for example $V_R = 5$ V would widen x_n and lower C'. A plot of $1/C'^2$ against V_R is a straight line whose intercept on the voltage axis is $-V_{bi}$, which is how V_{bi} and ϕ_{B0} are measured.

3.(d) Consider a gallium arsenide metal-semiconductor contact in which the electric field in the semiconductor is assumed to be $E = 6.8 \times 10^4$ V cm^{-1}. Calculate the Schottky barrier lowering (due to image force induced lowering) and the position of the maximum barrier height. Marks: 6

1. Theory

1. What is an image charge? (Core Idea)

A metal is a perfect conductor: its surface must stay at one potential. When an electron sits at a distance x from a metal surface, the metal's own electrons rearrange to keep it so. The result is a positive charge induced on the surface. The force this induced charge exerts on the electron is exactly the same as if a single charge $+e$ were sitting at the mirror position $-x$ inside the metal. This imaginary charge is called the **image charge**.

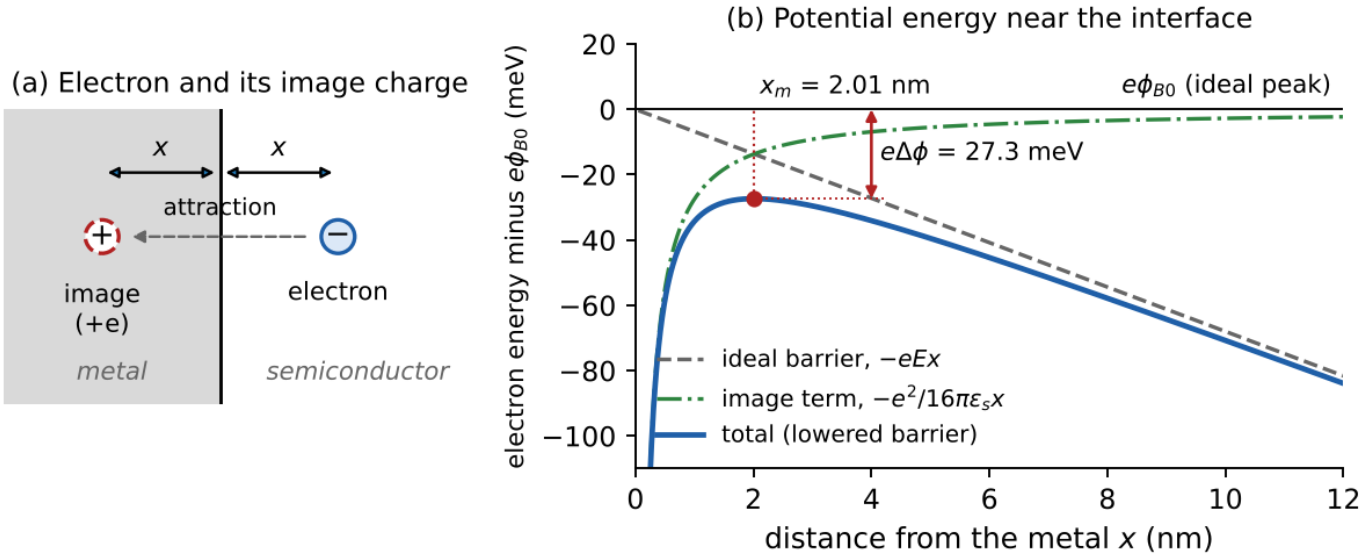


Figure 23: An electron at distance x from the metal and its image charge at the mirror position; the potential energy near the interface with and without the image force

Reading the figure. In panel (a) the electron (solid outline, minus sign) is at x in the semiconductor; the image (dashed outline, plus sign) is at $-x$ in the shaded metal. They are $2x$ apart and attract each other. Panel (b) is explained in the steps below.

2. The image force and its potential energy

The two charges are $2x$ apart, so by Coulomb's law the attraction is

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$$F = \frac{-e^2}{4\pi\varepsilon_s(2x)^2} = \frac{-e^2}{16\pi\varepsilon_s x^2}$$

where ε_s is the permittivity of the semiconductor. The work needed to bring the electron from far away to x against this force gives its potential energy:

$$PE_{image}(x) = -\frac{e^2}{16\pi\varepsilon_s x}$$

This is negative and grows large near the surface: the image force pulls the electron's energy *down* close to the metal.

3. Adding the field of the barrier

In the ideal picture, the electron energy just inside the semiconductor falls from the barrier top $e\phi_{B0}$ with the slope of the junction field E. Near the interface the field is close to its maximum and nearly constant, so the ideal energy is $e\phi_{B0} - eEx$. Adding the image term:

$$E(x) = e\phi_{B0} - eEx - \frac{e^2}{16\pi\varepsilon_s x}$$

The ideal barrier is a sharp corner at $x = 0$. The image term rounds the corner off: very close to the metal it pulls the energy steeply down, and far away it vanishes. In between there is a **maximum**, lower than $e\phi_{B0}$, a little way inside the semiconductor.

Reading panel (b). The dashed grey line is the ideal barrier $-eEx$ (energies measured from $e\phi_{B0}$); the dash-dot green curve is the image term; the solid blue curve is their sum. The sum peaks at the marked dot, at x_m, a height $e\Delta\phi$ below the ideal peak (red double arrow).

4. Finding the maximum (derivation)

At the maximum, $dE/dx = 0$:

$$\frac{dE}{dx} = -eE + \frac{e^2}{16\pi\varepsilon_s x^2} = 0 \quad \Rightarrow \quad x_m = \sqrt{\frac{e}{16\pi\varepsilon_s E}}$$

At x_m the two terms are equal, $e^2/(16\pi\varepsilon_s x_m) = eEx_m$, so the drop below the ideal peak is

$$e\Delta\phi = eEx_m + eEx_m = 2eE\sqrt{\frac{e}{16\pi\varepsilon_s E}} \quad \Rightarrow \quad \Delta\phi = \sqrt{\frac{eE}{4\pi\varepsilon_s}}$$

where $\Delta\phi$ is the **Schottky barrier lowering** in volts. It grows as $\sqrt{E}$, and E grows with reverse bias, so the barrier height seen from the metal drops a little as the reverse bias rises. These are Neamen's results.

2. Answer

Given:

- GaAs, $\varepsilon_s = 13.1\varepsilon_0 = 13.1 \times 8.85 \times 10^{-14} = 1.159 \times 10^{-12}$ F/cm
- Field in the semiconductor $E = 6.8 \times 10^4$ V/cm
- $e = 1.602 \times 10^{-19}$ C

Relationships. First find the formulas for the barrier lowering.

Image force. An electron at distance x from the metal induces a positive image charge at distance x behind the surface, so the two are $2x$ apart. They attract with the force

$$F = \frac{e^2}{4\pi\varepsilon_s(2x)^2} = \frac{e^2}{16\pi\varepsilon_s x^2}$$

Image energy. The potential energy is the work done against this force in bringing the electron from infinity to x:

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$$-\int_x^\infty F\,dx' = -\frac{e^2}{16\pi\varepsilon_s x}$$

Total barrier. Adding the barrier $e\phi_{B0}$ and the slope $-eEx$ made by the field:

$$U(x) = e\phi_{B0} - eEx - \frac{e^2}{16\pi\varepsilon_s x}$$

Position of the maximum. Set the derivative to zero:

$$\frac{dU}{dx} = -eE + \frac{e^2}{16\pi\varepsilon_s x^2} = 0 \quad \Rightarrow \quad x_m = \sqrt{\frac{e}{16\pi\varepsilon_s E}}$$

Size of the lowering. The top of the barrier is lower than $e\phi_{B0}$ by

$$e\Delta\phi = eEx_m + \frac{e^2}{16\pi\varepsilon_s x_m}$$

From the zero-derivative condition, $e^2/(16\pi\varepsilon_s x_m^2) = eE$, so the second term also equals eEx_m. Then $\Delta\phi = 2Ex_m$. Putting in x_m:

$$\Delta\phi = 2E\sqrt{\frac{e}{16\pi\varepsilon_s E}} = \sqrt{\frac{4E^2 e}{16\pi\varepsilon_s E}} = \sqrt{\frac{eE}{4\pi\varepsilon_s}}$$

Here x_m is the position of the barrier maximum and $\Delta\phi$ the barrier lowering.

Step 1: Schottky barrier lowering

$$\Delta\phi = \sqrt{\frac{(1.602\times10^{-19})(6.8\times10^{4})}{4\pi(1.159\times10^{-12})}} = \sqrt{7.477\times10^{-4}}\ \text{V}$$

$$\boxed{\Delta\phi = 0.0273\ \text{V} = 27.3\ \text{mV}}$$

The barrier is lowered by 27.3 meV, a few per cent of a typical GaAs barrier height of about 0.8 V.

Step 2: Position of the maximum barrier

$$x_m = \sqrt{\frac{1.602\times10^{-19}}{16\pi(1.159\times10^{-12})(6.8\times10^{4})}} = \sqrt{4.043\times10^{-14}}\ \text{cm} = 2.01\times10^{-7}\ \text{cm}$$

$$\boxed{x_m = 2.01\ \text{nm}}$$

That is 20.1 Å. The peak of the barrier sits only about 2 nm inside the semiconductor.

Check: $\Delta\phi = 2Ex_m = 2(6.8\times10^{4})(2.01\times10^{-7}) = 0.0273$ V ✔.

Final Answer: $\Delta\phi = 27.3$ mV, at $x_m = 2.01$ nm (20.1 Å) from the metal.

3. Keyword Explanation

1. Image-force lowering (Schottky effect)

The reduction of a metal-semiconductor barrier height caused by the attraction between an electron and its image charge in the metal, in the presence of an electric field.

2. Image charge

The imaginary charge, at the mirror position inside a conductor, that reproduces the field of the real charges induced on the conductor's surface.

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4. Notes and Remarks

- **Why it matters:** the reverse current $J_{sT} \propto \exp[-e(\phi_{B0} - \Delta\phi)/kT]$ grows with reverse bias because $\Delta\phi$ grows with the field. This is one of the non-ideal effects discussed in 4(b).
- **Which permittivity:** strictly the image force should use the high-frequency permittivity if the electron crosses the region faster than the lattice can polarise (Sze & Ng). For GaAs the difference is small; Neamen and this answer use the static value 13.1.
- **Field at zero field:** with $E \to 0$, $x_m \to \infty$ and $\Delta\phi \to 0$: the lowering exists only because a field is present.

5. Final Concise Answer

(a) For an ideal metal/n-type rectifying contact, the barrier seen from the metal is always $e\phi_{B0} = e(\phi_m - \chi)$. **(i) Reverse bias** V_R (semiconductor positive) lowers the semiconductor bands by eV_R: the barrier from the semiconductor becomes $e(V_{bi} + V_R)$, the depletion region widens, and only a small reverse current J_{sT} flows. **(ii) Forward bias** V_a raises them by eV_a: the barrier becomes $e(V_{bi} - V_a)$, the depletion region narrows, and the thermionic emission current rises as $\exp(eV_a/kT)$.

(b) $D(E)$ counts states per unit energy. 3D: $D \propto \sqrt{E - E_c}$ (spherical shell in k-space). 2D quantum well: constant $m^*/\pi\hbar^2$ per subband, a staircase touching the 3D curve. 1D quantum wire: $D \propto 1/\sqrt{E - E_n}$ spikes at each subband edge. (0D dot: delta functions.) States are squeezed into narrower energy ranges as the dimension drops.

(c) W on n-Si, $N_d = 10^{16}$ cm^{-3}: (i) $\phi_{B0} = 0.54$ V; (ii) $\phi_n = 0.206$ V, $V_{bi} = 0.334$ V; (iii) $x_n = 0.208$ μm, $|E_{max}| = 3.22 \times 10^4$ V/cm; (iv) $C' = 4.98 \times 10^{-8}$ F/cm^2 (49.8 nF/cm^2) at zero bias.

(d) GaAs, $E = 6.8 \times 10^4$ V/cm: $\Delta\phi = \sqrt{eE/4\pi\varepsilon_s} = 27.3$ mV, at $x_m = \sqrt{e/16\pi\varepsilon_s E} = 2.01$ nm (20.1 Å) from the metal.

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4. (a) A narrow bandgap N-type and a wide bandgap P-type material are used to form a heterojunction. If they are straddling type, then draw the energy band diagram of the heterojunction before contact and after contact at thermal equilibrium. With necessary assumption, derive expression for electric field and electrostatic potential in both regions. Marks: 12

(b) Discuss the following non-ideal effects on barrier height of a metal-semiconductor heterojunction:

(i) Image-force-induced lowering

(ii) Interface states Marks: 8

(c) Describe how a linearly graded junction is formed. Marks: 5

4.(a) A narrow bandgap N-type and a wide bandgap P-type material are used to form a heterojunction. If they are straddling type, then draw the energy band diagram of the heterojunction before contact and after contact at thermal equilibrium. With necessary assumption, derive expression for electric field and electrostatic potential in both regions. Marks: 12

1. Theory

1. What is this junction? (Core Idea)

A **heterojunction** is a junction between two *different* semiconductors (for example GaAs and AlGaAs). The basic ideas (energy bands, the vacuum level E_0, the electron affinity χ, Anderson's rule and the three types of band alignment) are built in 1(d). In one line: each material keeps its own band gap and its own electron affinity, so where they meet the band edges jump.

In this question:

- **Material 1 (left):** narrow band gap E_{g1}, **N-type** (donor doping N_{d1}), permittivity ϵ_1, electron affinity χ_1.
- **Material 2 (right):** wide band gap E_{g2}, **P-type** (acceptor doping N_{a2}), permittivity ϵ_2, electron affinity χ_2.

A capital letter (N, P) is often used for the wide-gap side and a small letter for the narrow-gap side, so this junction is an **nP heterojunction** (Neamen §9.3).

Straddling (type I) means the narrow gap sits completely inside the wide gap: the conduction band of material 1 is *below* that of material 2, and the valence band of material 1 is *above* that of material 2. This needs $\chi_1 > \chi_2$. The two jumps are

$$\Delta E_c = e(\chi_1 - \chi_2), \qquad \Delta E_v = (E_{g2} - E_{g1}) - \Delta E_c$$

both positive for a straddling pair, and $\Delta E_c + \Delta E_v = \Delta E_g$.

2. Before contact

Before the two pieces touch, we draw them side by side with a **common vacuum level** E_0 (the energy of an electron just outside the material, at rest). Each Fermi level sits where its doping puts it: near E_{c1} in the N-type material, near E_{v2} in the P-type material.

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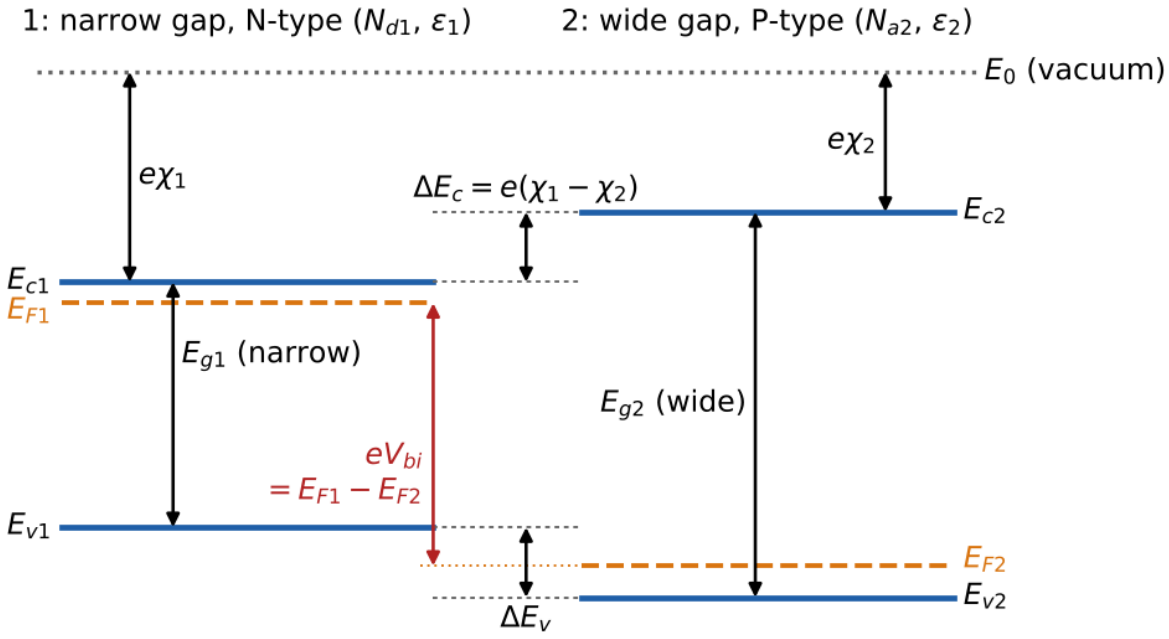


Figure 24: Band diagrams of the narrow-gap N and wide-gap P materials before contact, with a common vacuum level (energies not to scale)

Reading the figure. Solid blue lines are the band edges, dashed orange lines the Fermi levels and the dotted grey line the vacuum level E_0. The arrows measure $e\chi_1$ and $e\chi_2$ down from E_0, so E_{c1} is lower than E_{c2} by ΔE_c and E_{v1} is higher than E_{v2} by ΔE_v: the narrow gap "straddles" inside the wide one. The labelled red arrow shows that the Fermi level of the N side is higher than that of the P side; this difference is what becomes eV_{bi}.

3. What happens when they touch?

Electrons flow from where the Fermi level is high (N side) to where it is low (P side), and holes flow the other way, until one Fermi level runs flat through the whole structure. This is **thermal equilibrium**. The flow leaves behind:

- on the N side, a layer of positive donor ions ($+eN_{d1}$), width x_n;
- on the P side, a layer of negative acceptor ions ($-eN_{a2}$), width x_p.

Together these layers are the **space charge** or **depletion region**. Their charge makes an electric field, and the field makes a potential step, the **built-in potential** V_{bi}:

$$eV_{bi} = E_{F1} - E_{F2} \text{ (before contact)} = e\phi_{s2} - e\phi_{s1}$$

where $\phi_{s1} = \chi_1 + (E_{c1} - E_{F1})/e$ and $\phi_{s2} = \chi_2 + (E_{c2} - E_{F2})/e$ are the two **work functions**. With $E_{c1} - E_{F1} = kT\ln(N_{c1}/N_{d1})$ and $E_{F2} - E_{v2} = kT\ln(N_{v2}/N_{a2})$ this becomes

$$eV_{bi} = E_{g1} + \Delta E_v - kT\ln\left(\frac{N_{c1}\,N_{v2}}{N_{d1}\,N_{a2}}\right)$$

Part of V_{bi} drops in material 1 (V_{bi1}) and the rest in material 2 (V_{bi2}): $V_{bi} = V_{bi1} + V_{bi2}$.

4. The band diagram after contact

Three rules draw it (Anderson's model, 1(d)):

1. E_F is flat.
2. Far from the junction each material looks exactly as it did alone.
3. The vacuum level E_0 is continuous, and each band edge stays $e\chi$ (for E_c) or $e\chi + E_g$ (for E_v) below it. So the jumps ΔE_c and ΔE_v appear at the interface unchanged, while the bands bend smoothly in each depletion layer.

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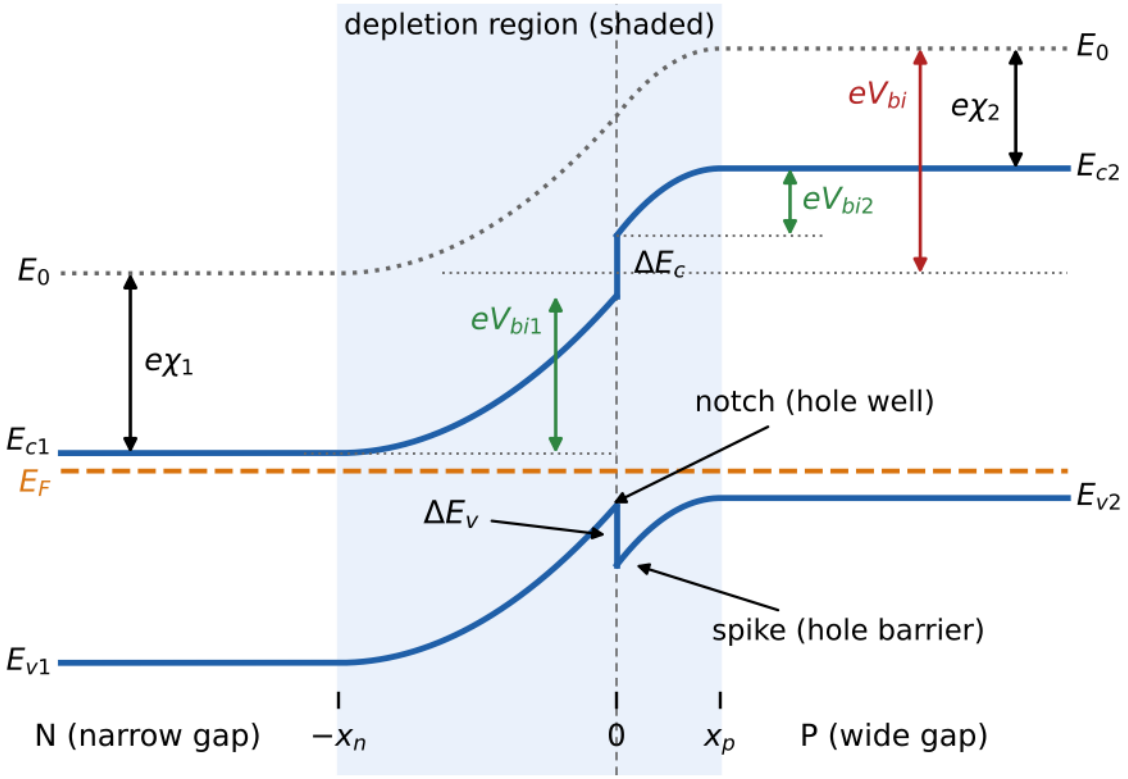


Figure 25: Equilibrium band diagram of the straddling nP heterojunction: flat Fermi level, band bending in both depletion layers, steps ΔEc and ΔEv at the interface

Reading the figure. The shaded strip from $-x_n$ to x_p is the depletion region. The dashed orange E_F is flat. The dotted grey vacuum level rises by eV_{bi} (labelled red arrow) from left to right, and E_c and E_v (solid blue) follow it. The labelled green arrows split this rise into eV_{bi1} (bending in the N layer) and eV_{bi2} (bending in the P layer). At $x = 0$, E_c jumps *up* by ΔE_c; since both bendings also go up, E_c rises in one smooth step with no spike. E_v instead jumps *down* by ΔE_v, so the valence band has a **notch** on the N side (a small well that can trap holes) and a **spike** on the P side (an extra barrier for holes). Electrons going from N to P must climb $eV_{bi} + \Delta E_c$; holes going from P to N climb only $eV_{bi} - \Delta E_v$.

5. From charge to field to potential: Poisson's equation

Poisson's equation (built for the Schottky diode in 3(c)) links the charge density ρ to the field E and the potential ϕ:

$$\frac{dE}{dx} = \frac{\rho(x)}{\epsilon}, \qquad E = -\frac{d\phi}{dx}$$

- Integrate the charge once and you get the field.
- Integrate the field once (with a minus sign) and you get the potential.
- Uniform charge gives a straight-line field and a parabolic potential.

The **depletion approximation** says: inside $-x_n < x < x_p$ there are no free carriers, only ionised dopants; outside, the material is neutral and the field is zero.

6. What is new at a heterojunction: the permittivity changes

At $x = 0$ the material, and so ϵ, changes. Gauss's law says the **electric displacement** $D = \epsilon E$ is continuous across an interface with no surface charge, not E itself:

$$\epsilon_1 E_1(0^-) = \epsilon_2 E_2(0^+)$$

So if $\epsilon_1 \neq \epsilon_2$, the field jumps at the interface. The potential is still continuous (a jump in ϕ would need an infinite field).

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2. Answer

Energy-band diagrams. Before contact: see the first figure in the Theory (common vacuum level; $\chi_1 > \chi_2$; E_{c1} below E_{c2} by $\Delta E_c = e(\chi_1 - \chi_2)$; E_{v1} above E_{v2} by $\Delta E_v = \Delta E_g - \Delta E_c$; E_{F1} near E_{c1}, E_{F2} near E_{v2}). After contact at equilibrium: see the second figure (E_F flat; E_0 continuous and bent by $eV_{bi} = E_{F1} - E_{F2}$; ΔE_c and ΔE_v kept at $x = 0$; bending eV_{bi1} in N and eV_{bi2} in P; notch and spike in E_v).

Assumptions:

- abrupt junction at $x = 0$: material 1 (N, narrow gap) for $x < 0$, material 2 (P, wide gap) for $x > 0$;
- uniform doping N_{d1} and N_{a2}, all dopants ionised;
- depletion approximation: depletion layer $-x_n < x < x_p$ with no free carriers, neutral outside ($E = 0$);
- no interface states or interface charge (Anderson model), so $D = \epsilon E$ is continuous at $x = 0$;
- one-dimensional problem; reference $\phi(-x_n) = 0$.

Charge density:

$$\rho(x) = \begin{cases} +eN_{d1}, & -x_n < x < 0 \\ -eN_{a2}, & 0 < x < x_p \end{cases}$$

Step 1: Electric field in region 1 (N side)

Poisson's equation in one dimension, $dE/dx = \rho/\epsilon$, says the field grows by the charge it passes, divided by the permittivity of that material. In region 1, $\rho = +eN_{d1}$ and the permittivity is ϵ_1; integrate once:

$$\frac{dE_1}{dx} = \frac{eN_{d1}}{\epsilon_1} \;\Rightarrow\; E_1(x) = \frac{eN_{d1}}{\epsilon_1}x + C_1$$

With $E_1(-x_n) = 0$, $C_1 = eN_{d1}x_n/\epsilon_1$:

$$\boxed{E_1(x) = \frac{eN_{d1}}{\epsilon_1}(x + x_n), \qquad -x_n \le x \le 0}$$

Step 2: Electric field in region 2 (P side)

Here $\rho = -eN_{a2}$ and the permittivity is ϵ_2; integrate once:

$$\frac{dE_2}{dx} = -\frac{eN_{a2}}{\epsilon_2} \;\Rightarrow\; E_2(x) = -\frac{eN_{a2}}{\epsilon_2}x + C_2$$

With $E_2(x_p) = 0$:

$$\boxed{E_2(x) = \frac{eN_{a2}}{\epsilon_2}(x_p - x), \qquad 0 \le x \le x_p}$$

Both are positive: the field points in $+x$, from the positive N side to the negative P side.

Step 3: Boundary condition at the interface

Continuity of D at $x = 0$:

$$\epsilon_1 E_1(0) = \epsilon_2 E_2(0) \;\Rightarrow\; eN_{d1}x_n = eN_{a2}x_p \;\Rightarrow\; \boxed{N_{d1}x_n = N_{a2}x_p}$$

The positive charge on the N side equals the negative charge on the P side. The field itself jumps from $E_1(0) = eN_{d1}x_n/\epsilon_1$ to $E_2(0) = eN_{a2}x_p/\epsilon_2$, so $E_2(0)/E_1(0) = \epsilon_1/\epsilon_2$.

Step 4: Potential in region 1

$$\phi_1(x) = -\int E_1\,dx = -\frac{eN_{d1}}{\epsilon_1}\left(\frac{x^2}{2} + x_n x\right) + C_3$$

With $\phi_1(-x_n) = 0$, $C_3 = -eN_{d1}x_n^2/(2\epsilon_1)$:

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$$\boxed{\phi_1(x) = -\frac{eN_{d1}}{2\epsilon_1}(x + x_n)^2, \qquad -x_n \le x \le 0}$$

At the interface, $\phi_1(0) = -V_{bi1}$, where

$$V_{bi1} = \frac{eN_{d1}x_n^2}{2\epsilon_1}$$

Step 5: Potential in region 2

$$\phi_2(x) = -\int E_2\,dx = -\frac{eN_{a2}}{\epsilon_2}\left(x_p x - \frac{x^2}{2}\right) + C_4$$

Continuity, $\phi_2(0) = \phi_1(0) = -V_{bi1}$, gives $C_4 = -V_{bi1}$:

$$\boxed{\phi_2(x) = -V_{bi1} - \frac{eN_{a2}}{\epsilon_2}\left(x_p x - \frac{x^2}{2}\right), \qquad 0 \le x \le x_p}$$

At $x = x_p$: $\phi_2(x_p) = -V_{bi1} - V_{bi2} = -V_{bi}$, with

$$V_{bi2} = \frac{eN_{a2}x_p^2}{2\epsilon_2}$$

Equivalently $\phi_2(x) = -V_{bi} + \dfrac{eN_{a2}}{2\epsilon_2}(x_p - x)^2$. The N side is at the higher potential, by V_{bi}.

Step 6: Built-in potential and depletion widths

$$V_{bi} = V_{bi1} + V_{bi2} = \frac{eN_{d1}x_n^2}{2\epsilon_1} + \frac{eN_{a2}x_p^2}{2\epsilon_2}$$

Put $x_p = N_{d1}x_n/N_{a2}$ (Step 3) into the second term, $\dfrac{eN_{a2}}{2\epsilon_2}\cdot\dfrac{N_{d1}^2x_n^2}{N_{a2}^2} = \dfrac{eN_{d1}^2x_n^2}{2\epsilon_2 N_{a2}}$, and take out the common factor $eN_{d1}x_n^2/2$:

$$V_{bi} = \frac{eN_{d1}x_n^2}{2}\left(\frac{1}{\epsilon_1} + \frac{N_{d1}}{\epsilon_2 N_{a2}}\right) = \frac{eN_{d1}x_n^2}{2}\cdot\frac{\epsilon_1 N_{d1} + \epsilon_2 N_{a2}}{\epsilon_1\epsilon_2 N_{a2}}$$

Solve for x_n; then $x_p = N_{d1}x_n/N_{a2}$:

$$\boxed{x_n = \left[\frac{2\epsilon_1\epsilon_2 N_{a2}\,V_{bi}}{eN_{d1}\,(\epsilon_1 N_{d1} + \epsilon_2 N_{a2})}\right]^{1/2}} \qquad \boxed{x_p = \left[\frac{2\epsilon_1\epsilon_2 N_{d1}\,V_{bi}}{eN_{a2}\,(\epsilon_1 N_{d1} + \epsilon_2 N_{a2})}\right]^{1/2}}$$

and the ratio of the two potential drops is

$$\frac{V_{bi1}}{V_{bi2}} = \frac{\epsilon_2 N_{a2}}{\epsilon_1 N_{d1}}$$

So the lighter-doped side takes most of the band bending, just as in a homojunction. Under a reverse bias V_R, replace V_{bi} by $V_{bi} + V_R$.

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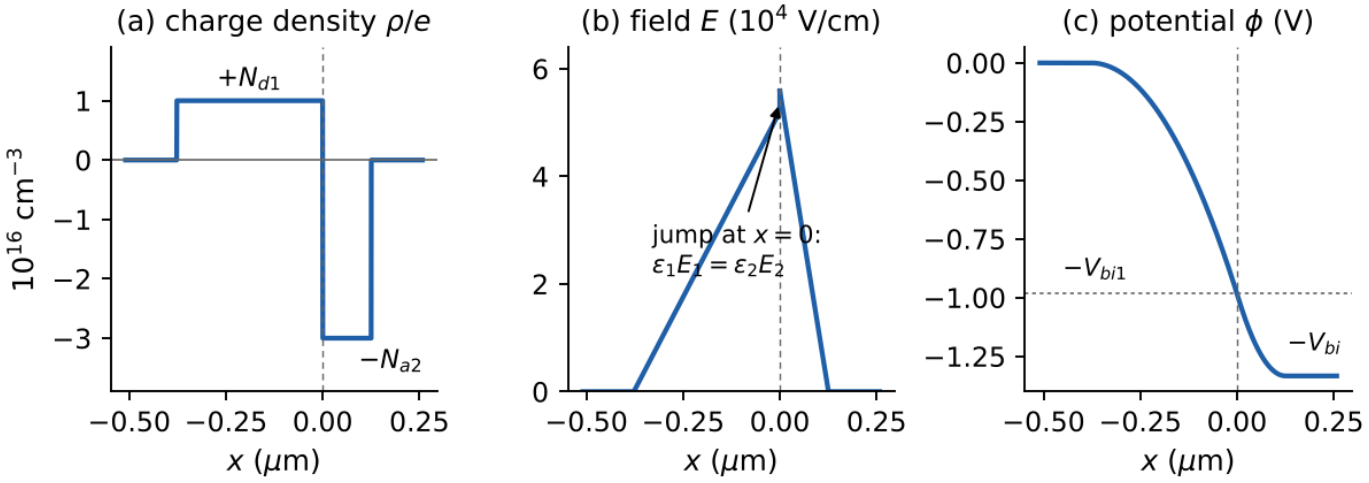


Figure 26: Charge density, electric field and potential in the nP heterojunction, drawn for the N-GaAs / P-$Al_{0.25}Ga_{0.75}As$ example of the Notes

Reading the figure. (a) The charge is a positive block on the N side and a taller, narrower negative block on the P side with the same area ($N_{d1}x_n = N_{a2}x_p$). (b) The field (solid blue) rises linearly across the N layer, falls linearly across the P layer, and jumps at $x = 0$ because $\epsilon_1 \neq \epsilon_2$; here $\epsilon_1 > \epsilon_2$, so the field is a little larger on the P side. (c) The potential falls along two parabolas, by V_{bi1} in the N layer and V_{bi2} in the P layer, with no jump.

Steps to remember

$\rho = +eN_{d1}, -eN_{a2} \to$ Poisson in each material with its own $\epsilon \to$ fields zero at the depletion edges $\to \epsilon_1 E_1(0) = \epsilon_2 E_2(0)$ gives $N_{d1}x_n = N_{a2}x_p \to$ integrate again with ϕ continuous $\to V_{bi} = V_{bi1} + V_{bi2}$ $\to$ solve for x_n, x_p.

Final Answer:

- $E_1(x) = \dfrac{eN_{d1}}{\epsilon_1}(x + x_n)$ and $E_2(x) = \dfrac{eN_{a2}}{\epsilon_2}(x_p - x)$, with $N_{d1}x_n = N_{a2}x_p$.
- $\phi_1(x) = -\dfrac{eN_{d1}}{2\epsilon_1}(x + x_n)^2$ and $\phi_2(x) = -V_{bi} + \dfrac{eN_{a2}}{2\epsilon_2}(x_p - x)^2$.
- $V_{bi} = V_{bi1} + V_{bi2}$, with x_n and x_p as boxed in Step 6.

3. Keyword Explanation

1. Straddling (type I) alignment

The narrow gap lies wholly inside the wide gap, so both ΔE_c and ΔE_v confine carriers in the narrow-gap material. GaAs/AlGaAs is the standard example.

2. Anderson model

The simplest rule for a heterojunction: the vacuum level is continuous and each material keeps its own χ and E_g, so $\Delta E_c = e(\chi_1 - \chi_2)$. It ignores interface states and interface dipoles.

3. Electric displacement D

$D = \epsilon E$. With no charge sheet at an interface, D is continuous across it; E is not, if ϵ changes.

4. Notch and spike

A notch is a narrow well in a band edge at the interface; a spike is a narrow barrier. Here both are in the valence band; in an nN or Np junction they appear in the conduction band and give the two-dimensional electron gas used in HEMTs.

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4. Notes and Remarks

- **Sign of the field.** The formulas above have the N side on the left, so the field points in $+x$ and is positive. Neamen draws homojunctions with the p side on the left, where the same formulas carry a minus sign ($E = -\frac{eN_a}{\epsilon_s}(x + x_p)$). Only the direction changes, not the size.
- **Homojunction check.** With $\epsilon_1 = \epsilon_2 = \epsilon_s$ the results reduce to the familiar $x_n = \left[\frac{2\epsilon_s V_{bi}}{e}\frac{N_a}{N_d}\frac{1}{N_a+N_d}\right]^{1/2}$ and the field becomes continuous.
- **Capacitance.** The junction capacitance per unit area follows from the same widths: $C' = \left[\frac{eN_{d1}N_{a2}\epsilon_1\epsilon_2}{2(\epsilon_1 N_{d1} + \epsilon_2 N_{a2})(V_{bi} + V_R)}\right]^{1/2}$.
- **A worked illustration (computed with a script).** N-GaAs ($N_{d1} = 10^{16}$ cm^{-3}, $\epsilon_1 = 13.1\epsilon_0$, $N_{c1} = 4.7 \times 10^{17}$ cm^{-3}) on P-$Al_{0.25}Ga_{0.75}As$ ($N_{a2} = 3 \times 10^{16}$ cm^{-3}, $N_{v2} = 8.576 \times 10^{18}$ cm^{-3}, $\epsilon_2 = (12.90 - 2.84x)\epsilon_0 = 12.19\epsilon_0$ from Sze & Ng), with $E_{g2} = 1.7357$ eV and $\Delta E_c = \Delta E_v = 0.1559$ eV as in 1(d):

  - ➢ $E_{c1} - E_{F1} = 0.0997$ eV, $E_{F2} - E_{v2} = 0.1465$ eV, so $eV_{bi} = 1.424 + 0.1559 - 0.0997 - 0.1465 = 1.3337$ eV;
  - ➢ $x_n = 0.3772$ μm, $x_p = 0.1257$ μm, $W = 0.5030$ μm;
  - ➢ $V_{bi1} = 0.9819$ V, $V_{bi2} = 0.3517$ V (ratio $2.79 = \epsilon_2 N_{a2}/\epsilon_1 N_{d1}$);
  - ➢ $E_1(0) = 5.206 \times 10^4$ V/cm, $E_2(0) = 5.595 \times 10^4$ V/cm: a 7.5 % jump, the ratio ϵ_1/ϵ_2;
  - ➢ $C'(0) = 22.63$ nF/cm^2; barrier for electrons $eV_{bi} + \Delta E_c = 1.4896$ eV, for holes $eV_{bi} - \Delta E_v = 1.1778$ eV.

  The 0.31 eV difference between the two barriers means that, under forward bias, hole injection into the N side is about $e^{0.31/0.0259} \approx 10^5$ times easier than electron injection into the P side (for similar prefactors): the idea behind the wide-gap emitter of an HBT.

4.(b) Discuss the following non-ideal effects on barrier height of a metal-semiconductor heterojunction: (i) Image-force-induced lowering (ii) Interface states Marks: 8

1. Theory

1. The ideal barrier height (Core Idea)

When a metal touches an n-type semiconductor, electrons see a barrier at the interface. In the ideal model of 3(c) its height is

$$\phi_{B0} = \phi_m - \chi$$

where ϕ_m is the metal work function and χ the semiconductor's electron affinity. This is the **Schottky limit**: change the metal and the barrier changes volt for volt. Real contacts differ for two main reasons, the two in the question.

2. Image-force lowering, in short

The full derivation is in 3(d). An electron at distance x from a metal surface induces a positive charge in the metal that attracts it as if a positive **image charge** sat at $-x$. The attraction adds a potential energy $-e^2/(16\pi\epsilon_s x)$ that pulls the top of the barrier down near the metal. Adding it to the straight-line energy $-eEx$ of the junction field and finding the maximum gives

$$\Delta\phi = \sqrt{\frac{eE}{4\pi\epsilon_s}}, \qquad x_m = \sqrt{\frac{e}{16\pi\epsilon_s E}}$$

where E is the field at the surface (the maximum field, $E = \sqrt{2eN_d(V_{bi} + V_R)/\epsilon_s}$) and x_m is where the barrier now peaks. For GaAs at $E = 6.8 \times 10^4$ V/cm, 3(d) found $\Delta\phi = 27.3$ mV at $x_m = 2.01$ nm.

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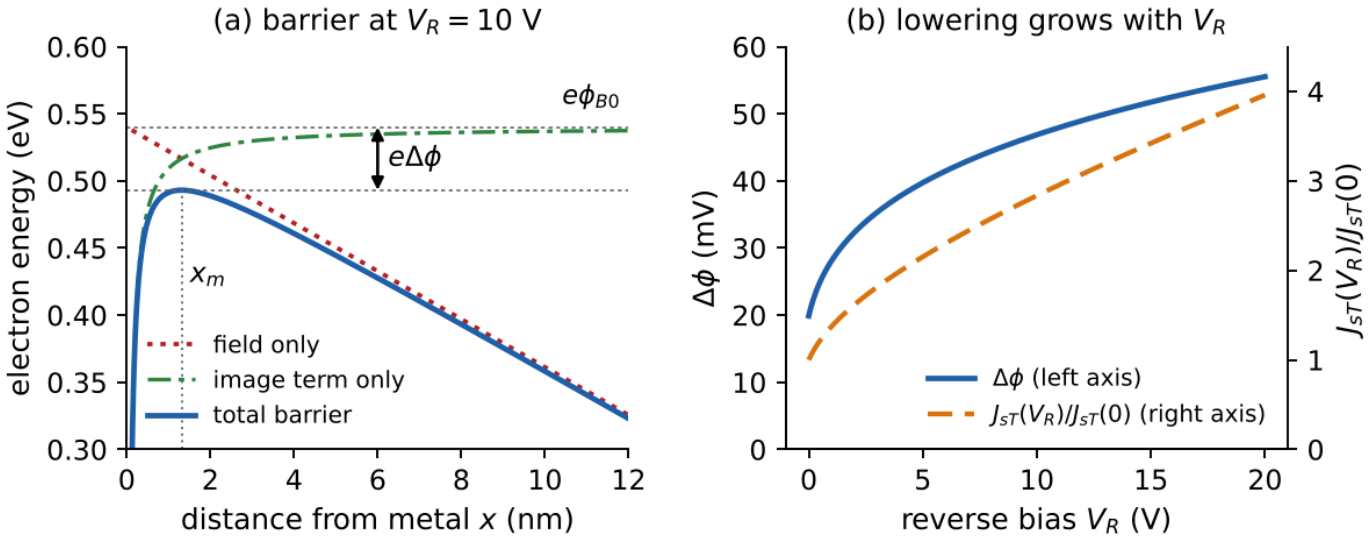


Figure 27: Image-force lowering for W on n-Si ($N_d = 10^{16}$ cm^{-3}): (a) barrier shape at $V_R = 10$ V; (b) lowering and saturation-current ratio versus reverse bias

Reading the figure. (a) The dotted red line is the barrier with the junction field alone; the dash-dot green curve is the image term alone; their sum (solid blue) is rounded off and peaks at x_m, $e\Delta\phi$ below the ideal $e\phi_{B0}$. (b) As the reverse bias grows, the surface field grows, so the lowering (solid blue, left axis) grows, from 19.9 mV at 0 V to 55.5 mV at 20 V, and the reverse saturation current (dashed orange, right axis) rises by a factor of about 4.

3. What are interface states?

A real surface is not a perfect end to the crystal. Broken (dangling) bonds, defects and a thin oxide leave allowed electron energy levels *inside the band gap* at the interface. These are **interface states** (or surface states), with a density D_{it} per unit area per unit energy. They are characterised by a **neutral level** $e\phi_0$, measured up from E_v at the surface:

- if the states up to E_F are filled exactly to $e\phi_0$, the surface is neutral;
- if E_F lies above $e\phi_0$, extra electrons fill states: a negative surface charge;
- if E_F lies below $e\phi_0$, states are emptied: a positive surface charge.

Between the metal and the semiconductor there is usually also a very thin **interfacial layer** (native oxide, thickness δ of a few ångström), which electrons tunnel through easily but which can hold a voltage Δ.

4. How interface states pin the Fermi level

Charge neutrality must hold: metal charge + interface-state charge + depletion charge = 0. With many interface states, a tiny shift of E_F at the surface changes the interface charge a lot. So the interface states absorb almost all the charge needed, and E_F at the surface stays almost at $e\phi_0$, whatever the metal. This is **Fermi-level pinning**. With E_F fixed at $e\phi_0$ above E_v, the barrier is fixed too:

$$\phi_{B0} \approx \frac{E_g}{e} - \phi_0 \qquad \text{(Bardeen limit, } D_{it} \to \infty\text{)}$$

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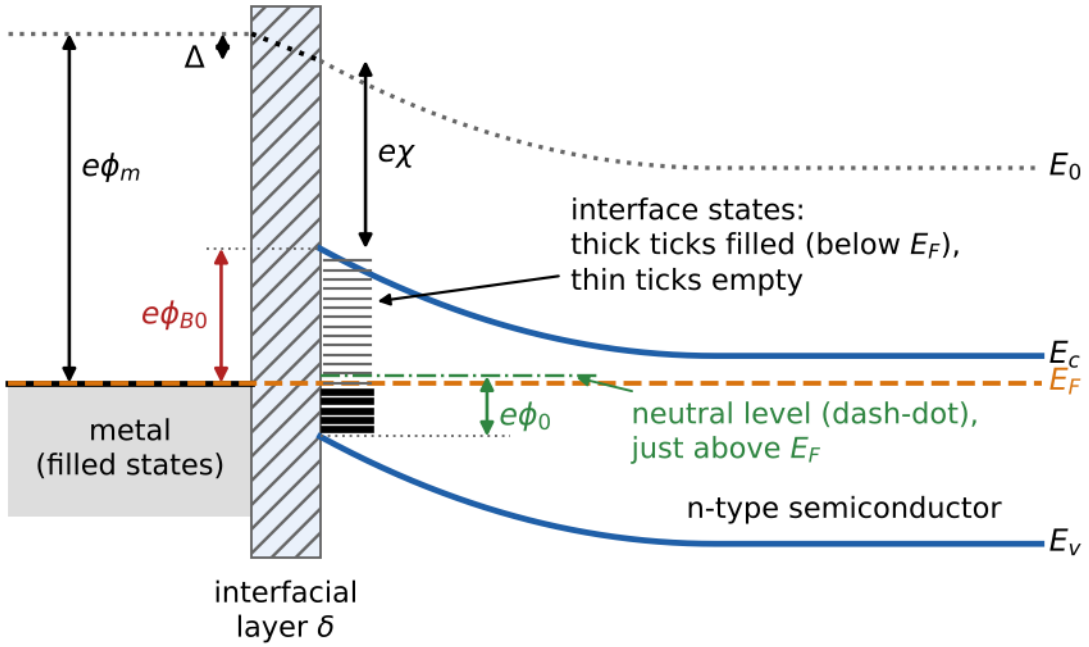


Figure 28: Band diagram of a metal on n-type semiconductor with an interfacial layer and interface states; the Fermi level at the surface is held near the neutral level

Reading the figure. The hatched strip is the interfacial layer of thickness δ, across which the vacuum level (dotted) drops by Δ. Short horizontal ticks at the surface are the interface states: thick black ticks below the dashed orange E_F are filled, thin grey ticks above it are empty. The dash-dot green line is the neutral level, $e\phi_0$ above the surface E_v; E_F sits just below it, so the barrier $e\phi_{B0}$ (labelled red arrow) is set mainly by $e\phi_0$ and E_g, not by $e\phi_m$.

5. The general result

Combining the charge balance with the voltage across the interfacial layer (Cowley and Sze; Sze & Ng, Neamen §9.1) gives, neglecting the small depletion-charge term,

$$\phi_{B0} = \gamma(\phi_m - \chi) + (1-\gamma)\left(\frac{E_g}{e} - \phi_0\right), \qquad \gamma = \frac{\epsilon_i}{\epsilon_i + e^2\delta D_{it}}$$

where ϵ_i is the permittivity of the interfacial layer and D_{it} the interface-state density (per unit area per unit energy).

- $D_{it} \to 0$: $\gamma \to 1$, the Schottky limit $\phi_{B0} = \phi_m - \chi$.
- $D_{it} \to \infty$: $\gamma \to 0$, the Bardeen limit $\phi_{B0} = E_g/e - \phi_0$.

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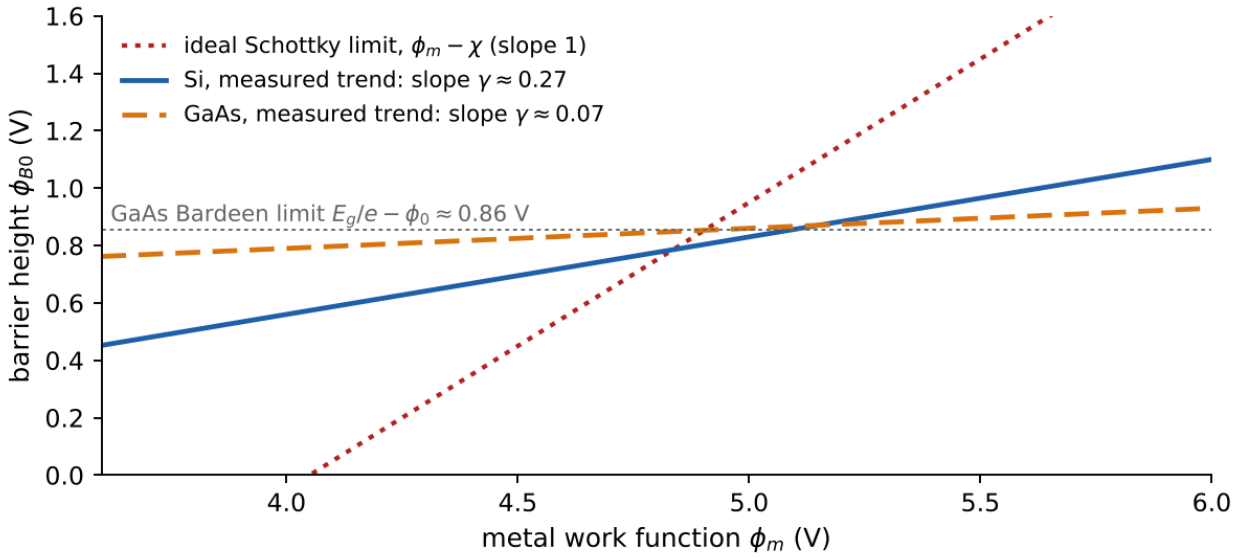


Figure 29: Barrier height versus metal work function: the ideal Schottky limit (dotted red, slope 1) against the measured trends for Si (solid blue, slope 0.27) and GaAs (dashed orange, slope 0.07)

Reading the figure. If the Schottky limit held, every point would sit on the dotted red line of slope 1. Measured barriers follow much flatter lines: slope 0.27 for Si (solid blue) and only 0.07 for GaAs (dashed orange). GaAs is almost fully pinned: whatever the metal, its barrier is close to 0.8–0.9 V, near its Bardeen limit (thin dashed grey line).

2. Answer

(i) Image-force-induced lowering

An electron in the semiconductor near the metal induces an equal positive image charge at the mirror point inside the metal. The attraction adds a potential energy $-e^2/(16\pi\epsilon_s x)$ to the barrier. Added to the linear potential of the junction field, it rounds the top of the barrier and lowers it by

$$\Delta\phi = \sqrt{\frac{eE}{4\pi\epsilon_s}}, \qquad \text{at } x_m = \sqrt{\frac{e}{16\pi\epsilon_s E}}$$

(derivation in 3(d)). The effective barrier is $\phi_B = \phi_{B0} - \Delta\phi$.

Bias dependence: $E = \sqrt{2eN_d(V_{bi}+V_R)/\epsilon_s}$, so $\Delta\phi \propto (V_{bi}+V_R)^{1/4}$. A larger reverse bias gives a larger field, a lower barrier and a larger current, since

$$J_{sT} = A^*T^2 \exp\left(-\frac{e\phi_{B0}}{kT}\right)\exp\left(\frac{e\Delta\phi}{kT}\right)$$

So the reverse current of a Schottky diode does not saturate; it keeps rising with V_R (a "soft" reverse characteristic). **Example** (W on n-Si, $N_d = 10^{16}$ cm^{-3}): $\Delta\phi = 19.9$ mV at $V_R = 0$ and 46.9 mV at $V_R = 10$ V, so the reverse current grows by a factor 2.84. The lowering is small (tens of mV) but matters because it enters an exponential.

(ii) Interface states

At a real metal-semiconductor contact a thin interfacial layer (thickness δ) and a large density D_{it} of interface states in the gap exist, characterised by a neutral level $e\phi_0$ above E_v. Charge neutrality, $Q_m + Q_{ss} + Q_{sc} = 0$, is met mostly by charge in these states, so the Fermi level at the surface is **pinned** near $e\phi_0$. Result (Cowley-Sze):

$$\phi_{B0} = \gamma(\phi_m - \chi) + (1-\gamma)\left(\frac{E_g}{e} - \phi_0\right), \qquad \gamma = \frac{\epsilon_i}{\epsilon_i + e^2\delta D_{it}}$$

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- Few states ($\gamma \to 1$): Schottky limit, $\phi_{B0} = \phi_m - \chi$.
- Many states ($\gamma \to 0$): Bardeen limit, $\phi_{B0} \approx E_g/e - \phi_0$, independent of the metal.

Experiment: ϕ_{B0} changes only weakly with ϕ_m: slope $\gamma \approx 0.27$ for Si and ≈ 0.07 for GaAs (Sze & Ng), so these contacts are close to pinned. This is why the measured barrier of a contact can differ from $\phi_m - \chi$ by a few tenths of a volt, and why choosing a metal by its work function alone does not set the barrier.
Comparison: image-force lowering is bias-dependent and small (tens of mV); interface states give a fixed shift, often of a few hundred mV, set at fabrication.

3. Keyword Explanation

1. Image charge

An imaginary charge, the mirror image of a real charge in a conducting surface, that gives the same field outside the conductor as the real induced surface charge.

2. Fermi-level pinning

The Fermi level at a surface or interface held at almost a fixed energy (near $e\phi_0$) by a large density of interface states, so that the barrier no longer follows the metal work function.

3. Neutral level $e\phi_0$

The energy, above E_v at the surface, up to which the interface states must be filled for the surface to be electrically neutral.

4. Notes and Remarks

- **Where the formula comes from.** For n-type material the interface-state charge is $Q_{ss} = -eD_{it}(E_g - e\phi_0 - e\phi_{B0})$; the voltage across the interfacial layer is $\Delta = -\delta Q_m/\epsilon_i = \phi_m - (\chi + \phi_{B0})$. With $Q_m \approx -Q_{ss}$ (the depletion charge is much smaller), eliminating Q_m and solving for ϕ_{B0} gives the formula above.
- **Numbers from the Si and GaAs fits (computed).** Sze & Ng fit $\phi_{B0} = c_2\phi_m + c_3$ with $c_2 = 0.27$, $c_3 = -0.52$ V for Si and $c_2 = 0.07$, $c_3 = 0.51$ V for GaAs. Taking $\delta = 5$ Å and $\epsilon_i = \epsilon_0$: $D_{it} \approx 3.0 \times 10^{13}$ cm^{-2}eV^{-1} and $\phi_0 \approx 0.35$ V for Si; $D_{it} \approx 1.5 \times 10^{14}$ cm^{-2}eV^{-1} and $\phi_0 \approx 0.57$ V for GaAs (Bardeen limit 0.86 V).
- **Link to 3(c).** For tungsten on Si the ideal barrier is $4.55 - 4.01 = 0.54$ V, but the Si fit gives $0.27(4.55) - 0.52 = 0.71$ V; measured values (Neamen's table, about 0.67 V) lie closer to the pinned estimate.
- **Other non-ideal effects:** an interfacial dipole, tunnelling through a thin barrier at high doping, and edge effects also change the effective barrier.

4.(c) Describe how a linearly graded junction is formed. Marks: 5

1. Theory

1. Abrupt versus graded (Core Idea)

A pn junction is where the material changes from p-type to n-type. What matters is the **net doping** $N_d - N_a$ (donors minus acceptors) along the depth x:

- In an **abrupt (step) junction** the net doping changes sign suddenly: uniform N_a on one side, uniform N_d on the other. Alloyed junctions, shallow ion-implanted junctions and epitaxial layers grown with a sudden change of dopant are close to this.
- In a **linearly graded junction** the net doping changes *gradually*. Near the junction it is a straight line through zero:

$$N_d(x) - N_a(x) = a\,x$$

where a (in cm^{-4}) is the **grading constant**, the slope of the net doping, and $x = 0$ is the metallurgical junction (where $N_a = N_d$).

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2. How it is formed: diffusion

The usual way is **diffusion** of acceptors into an n-type wafer:

1. Start with a wafer doped uniformly with donors N_d.
2. Heat it in a furnace (about 1000–1100 °C) with an acceptor source (boron) on the surface. The acceptor atoms diffuse into the crystal. Their concentration is highest at the surface and falls off with depth, following an erfc or Gaussian curve.
3. Near the surface $N_a > N_d$: p-type. Deep inside $N_a < N_d$: n-type. The **metallurgical junction** x_j is the depth where $N_a(x_j) = N_d$.

Close to x_j, the falling $N_a(x)$ curve is nearly a straight line, so $N_d - N_a$ is nearly $a(x - x_j)$: a linearly graded junction. A junction can also be graded on purpose by **epitaxial growth** in which the dopant flow is ramped gradually from acceptor to donor.

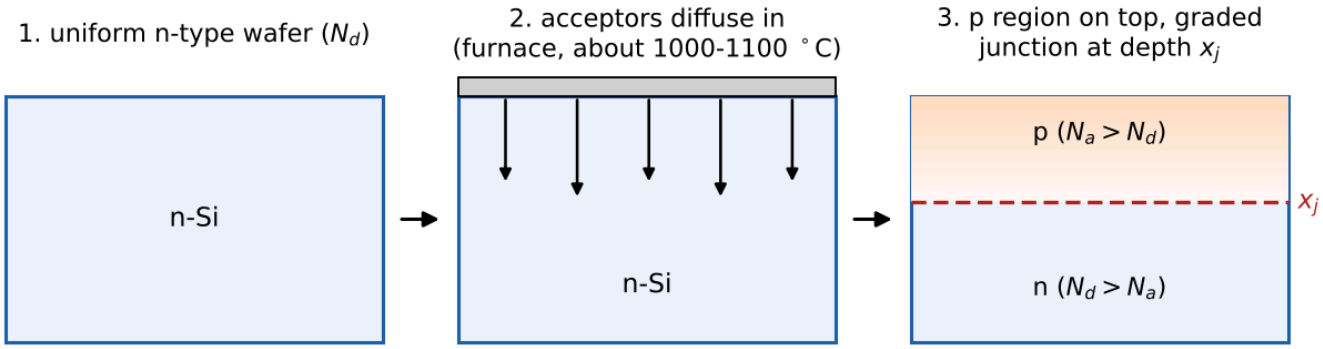


Figure 30: Forming a graded junction by diffusion: uniform n-type wafer, acceptors diffusing in from the surface, p region on top with the junction at depth xj

Reading the figure. Left to right: the uniform n-type wafer; the acceptor source on top (grey bar) with arrows showing the acceptors diffusing in; the result, a p region on top whose acceptor density fades with depth (shaded) and the junction (dashed red line, labelled x_j) where $N_a = N_d$.

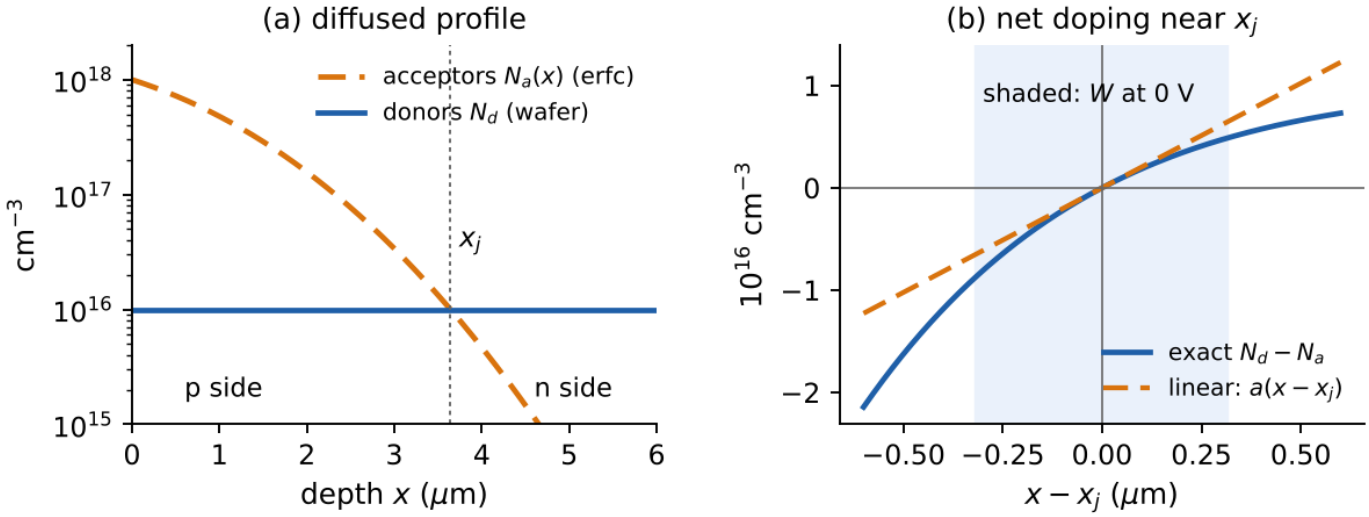


Figure 31: Diffused acceptor profile into a uniformly doped n-type wafer and the linear approximation of the net doping near the junction

Reading the figure. (a) On a log scale, the dashed orange erfc profile of acceptors falls from 10^{18} cm^{-3} at the surface and crosses the flat donor level (solid blue) at $x_j = 3.64$ μm. (b) Near x_j, the exact net doping (solid blue) is close to the straight line $a(x - x_j)$ (dashed orange) over the depletion width

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(shaded), which is what "linearly graded" means. Further away the real profile curves, so the linear model is an approximation valid only inside the depletion region.

3. Charge, field and potential

In the depletion region ($-x_0 < x < x_0$, with $W = 2x_0$) the charge is $\rho = eax$: negative on the p side, positive on the n side, growing linearly. Integrating once (Poisson, 4(a)) gives a parabolic field, and once more a cubic potential:

$$E(x) = \frac{ea}{2\epsilon_s}\left(x^2 - x_0^2\right), \qquad V_{bi} = \frac{2}{3}\frac{ea\,x_0^3}{\epsilon_s}$$

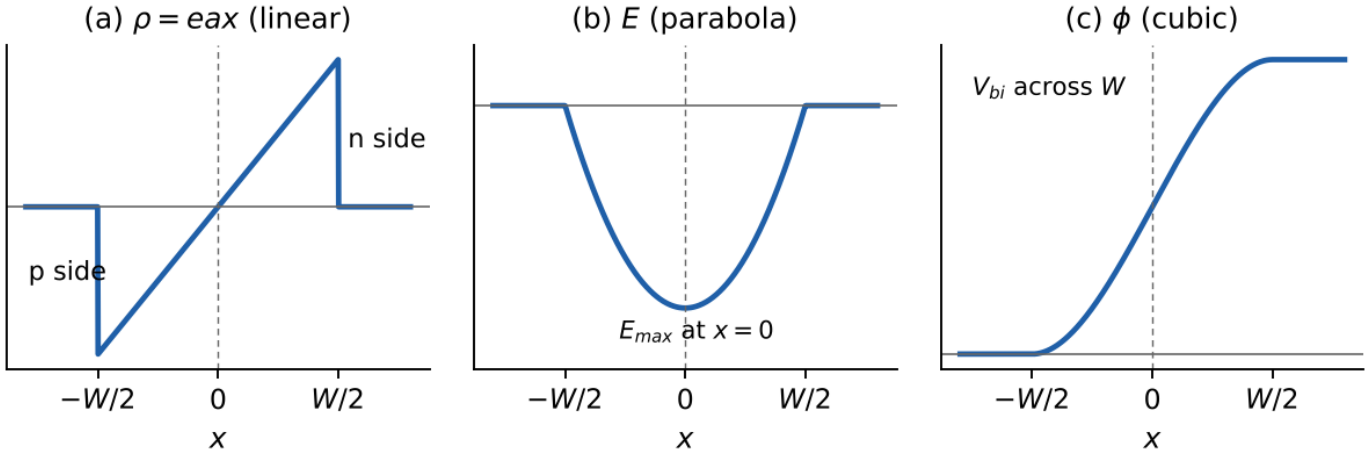


Figure 32: Linearly graded junction: (a) linear charge density, (b) parabolic electric field, (c) cubic potential

Reading the figure. (a) The charge (solid blue) is a straight line through zero, negative on the p side, positive on the n side. (b) The field is a parabola, zero at the edges and largest in size at the centre $x = 0$. (c) The potential rises along an S-shaped cubic by V_{bi} across the width W.

2. Answer

Definition: In a linearly graded junction the net impurity concentration changes linearly with distance through the junction: $N_d - N_a = ax$, where a is the grading constant (cm^{-4}), instead of changing in one step as in an abrupt junction.

Formation (diffusion):

1. Take an n-type wafer with uniform donor concentration N_d.
2. Diffuse acceptors (boron) into it from the surface at high temperature (about 1000–1100 °C). The acceptor concentration $N_a(x)$ is largest at the surface and decreases with depth (erfc or Gaussian profile).
3. Where $N_a > N_d$ the material becomes p-type; the junction forms at the depth x_j where $N_a(x_j) = N_d$.
4. Near x_j the acceptor profile is nearly straight, so the net doping $N_d - N_a \approx a(x - x_j)$: a linearly graded junction. (A graded junction can also be grown by epitaxy with gradually changed doping.)

(Process and profile figures in the Theory.)

Electrostatics: $\rho = eax$ in $-W/2 < x < W/2$; the field is parabolic, $E = \frac{ea}{2\epsilon_s}\left(x^2 - (W/2)^2\right)$, largest at $x = 0$. Integrating once more gives (Neamen §7.5)

$$W = \left[\frac{12\epsilon_s(V_{bi} + V_R)}{ea}\right]^{1/3}, \qquad C' = \left[\frac{ea\epsilon_s^2}{12(V_{bi} + V_R)}\right]^{1/3}$$

So $C' \propto (V_{bi} + V_R)^{-1/3}$, compared with $(V_{bi} + V_R)^{-1/2}$ for an abrupt junction; a $1/C'^3$ against V_R plot is a straight line and identifies a graded junction.

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3. Keyword Explanation

1. Grading constant a

The slope $d(N_d - N_a)/dx$ at the junction, in cm^{-4}. A large a (steep profile) behaves more like an abrupt junction.

2. Metallurgical junction

The plane where $N_a = N_d$ and the net doping is zero.

3. Diffusion

Movement of dopant atoms from high to low concentration at high temperature; it gives a smooth, falling profile from the surface inward.

4. Notes and Remarks

- **Example (computed)**

  Boron diffused into n-Si ($N_d = 10^{16}$ cm^{-3}) with surface concentration $N_s = 10^{18}$ cm^{-3} and $2\sqrt{Dt} = 2$ μm: $x_j = 3.64$ μm and $a = 2.04 \times 10^{20}$ cm^{-4}. Solving $V_{bi} = \frac{2kT}{e} \ln \frac{aW}{2n_i}$ together with W gives $V_{bi} = 0.672$ V, $W = 0.634$ μm and $C' = 16.3$ nF/cm^2 at zero bias; at $V_R = 5$ V, $W = 1.29$ μm and $C' = 8.0$ nF/cm^2.

- **Built-in potential.** For the graded junction, $V_{bi} = \frac{kT}{e} \ln\left[\frac{(aW/2)^2}{n_i^2}\right]$: the doping at the depletion edges is $aW/2$.
- **When is a diffused junction abrupt?** If the substrate is lightly doped and the diffusion is shallow and heavy (p^+n), the depletion region lies almost all in the n side, where the doping is uniform, so the one-sided abrupt model fits better. The linear model fits deep diffusions into a fairly heavily doped substrate.

5. Final Concise Answer

(a) In the straddling nP heterojunction ($\chi_1 > \chi_2$), before contact E_{c1} lies $\Delta E_c = e(\chi_1 - \chi_2)$ below E_{c2} and E_{v1} lies $\Delta E_v = \Delta E_g - \Delta E_c$ above E_{v2}, with $eV_{bi} = E_{F1} - E_{F2}$. After contact E_F is flat, the bands bend by eV_{bi1} in N and eV_{bi2} in P, and ΔE_c, ΔE_v stay at the interface (a notch and spike in E_v). With the depletion approximation and $\epsilon_1 E_1(0) = \epsilon_2 E_2(0)$: $E_1 = \frac{eN_{d1}}{\epsilon_1}(x + x_n)$, $E_2 = \frac{eN_{a2}}{\epsilon_2}(x_p - x)$, $N_{d1}x_n = N_{a2}x_p$; $\phi_1 = -\frac{eN_{d1}}{2\epsilon_1}(x + x_n)^2$, $\phi_2 = -V_{bi} + \frac{eN_{a2}}{2\epsilon_2}(x_p - x)^2$; $V_{bi1}/V_{bi2} = \epsilon_2 N_{a2}/(\epsilon_1 N_{d1})$ and $x_n = \left[\frac{2\epsilon_1\epsilon_2 N_{a2} V_{bi}}{eN_{d1}(\epsilon_1 N_{d1} + \epsilon_2 N_{a2})}\right]^{1/2}$. The field jumps at the interface; D and ϕ are continuous.

(b) (i) The image charge lowers the barrier by $\Delta\phi = \sqrt{eE/(4\pi\epsilon_s)}$ at $x_m = \sqrt{e/(16\pi\epsilon_s E)}$; since E grows with reverse bias, the reverse current rises with V_R (19.9 mV at 0 V to 46.9 mV at 10 V for W on n-Si). (ii) Interface states with neutral level $e\phi_0$ pin the surface Fermi level: $\phi_{B0} = \gamma(\phi_m - \chi) + (1-\gamma)(E_g/e - \phi_0)$, $\gamma = \epsilon_i/(\epsilon_i + e^2\delta D_{it})$; for large D_{it}, $\phi_{B0} \approx E_g/e - \phi_0$, almost independent of the metal (measured slopes 0.27 for Si, 0.07 for GaAs).

(c) A linearly graded junction is formed by diffusing acceptors into a uniformly doped n-type wafer (or by epitaxy with graded doping); near the depth x_j where $N_a = N_d$ the net doping is $N_d - N_a \approx a(x - x_j)$. Then $\rho = eax$, the field is parabolic, $W = [12\epsilon_s(V_{bi} + V_R)/(ea)]^{1/3}$ and $C' = [ea\epsilon_s^2/(12(V_{bi} + V_R))]^{1/3}$.

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5. (a) Draw the channel space charge region profiles of a single gate n-channel JFET for the following operating regions: (where V_p, V_g, and V_d are pinch-off, gate, and drain voltage respectively)

   (i) $V_g > V_p$ and $V_d < V_{d,sat}$
   (ii) $V_g > V_p$ and $V_d = V_{d,sat}$
   (iii) $V_g > V_p$ and $V_d > V_{d,sat}$ Marks: 6

   (b) Derive the formula of internal pinch-off voltage in an n-channel JFET. Marks: 8

   (c) Consider a GaAs n-channel pn JFET at $T = 300$ K with $N_a = 1 \times 10^{18}$ cm^{-3}, $N_d = 3 \times 10^{15}$ cm^{-3}, $\epsilon_s = 13.1$, and $a = 0.70$ μm. Determine the forward-bias gate voltage required to open a channel region that is 0.10 μm thick with zero drain voltage. Marks: 7

   (d) Sketch the ideal energy band diagram of a metal semiconductor junction in which $\Phi_m < \Phi_s$. Explain why this is an ohmic contact. Marks: 4

5.(a) Draw the channel space charge region profiles of a single gate n-channel JFET for the following operating regions: (where V_p, V_g, and V_d are pinch-off, gate, and drain voltage respectively) (i) $V_g > V_p$ and $V_d < V_{d,sat}$ (ii) $V_g > V_p$ and $V_d = V_{d,sat}$ (iii) $V_g > V_p$ and $V_d > V_{d,sat}$ Marks: 6

1. Theory

1. What is a JFET? (Core Idea)

JFET stands for **junction field-effect transistor**. It is a resistor whose thickness can be squeezed by a voltage.

- A thin slab of n-type semiconductor carries the current. This slab is the **channel**. Its electrons are the moving charges.
- The two ends of the channel have contacts: the **source** (S), where electrons enter, and the **drain** (D), where they leave. A positive drain voltage V_d pulls electrons from source to drain, so a drain current I_D flows.
- On top of the channel sits a heavily doped p-type region, the **gate** (G). Gate and channel form a pn junction.

In a **single-gate** (one-sided) JFET there is only one gate, on top; the channel of thickness a sits between the gate and the substrate. In a symmetric JFET there is a second gate below, and the same ideas hold for each half.

2. How does the gate control the current?

A pn junction always has a **space-charge region** (also called the **depletion region**) at its boundary: a layer from which the free carriers have been swept out, leaving only fixed ionised dopant atoms. No current can flow through it, because it has no free electrons.

- The gate is p$^+$ (very heavily doped) and the channel is lightly doped n. So the depletion layer lies almost entirely on the **channel side** (the light side must deplete a thicker layer to hold the same charge).
- The gate junction is normally **reverse biased**. A larger reverse bias makes the depletion layer thicker. Its thickness h obeys (built up in 5(b))

$$h = \left[\frac{2\epsilon_s\,(V_{bi} + V_R)}{eN_d}\right]^{1/2}$$

  where V_R is the reverse bias across the junction at that point.
- The part of the channel that is *not* depleted, of thickness $a - h$, is the only part that conducts. So a thicker depletion layer means a thinner conducting channel and a larger resistance.

When the depletion layer reaches all the way through the channel ($h = a$), the channel is **pinched off**.

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The gate voltage at which this happens at the source end (with $V_d = 0$) is the **pinch-off voltage** V_p. For an ordinary n-channel JFET V_p is negative.

Notation of the paper

$V_g > V_p$ means the gate is *less* negative than the pinch-off voltage. The channel is therefore still open at the source end. All three cases of the question have this, so the gate voltage is fixed and only the drain voltage changes.

3. Why is the depletion layer thicker at the drain end?

The gate is at V_g everywhere along its length, but the channel is not at one voltage. Current flowing through the channel's resistance makes the channel voltage $V(x)$ rise from 0 at the source to V_d at the drain. The reverse bias across the gate junction at a point x is therefore

$$V_R(x) = V(x) - V_g$$

It is smallest at the source ($-V_g$) and largest at the drain ($V_d - V_g$). So the depletion layer is thinnest at the source and thickest at the drain, and the open channel is a **wedge** that narrows towards the drain.

4. The three regions as V_d rises

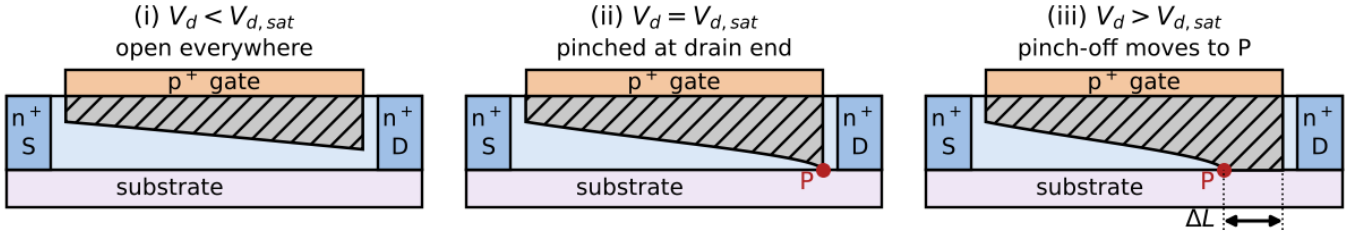


Figure 33: Channel space-charge profiles of a single-gate n-channel JFET at a fixed gate voltage above pinch-off: (i) below, (ii) at and (iii) above the saturation drain voltage

Reading the figure. Each panel is a cross-section: the p^+ gate on top, the n channel (light, unshaded) between the n^+ source S and drain D contacts, and the substrate below. The hatched grey area is the depletion region under the gate. Electrons flow from S to D through the open (unhatched) channel.

- **(i)** $V_d < V_{d,sat}$**:** the hatched depletion layer is thicker at the drain end, but a path of open channel remains all the way from S to D. For small V_d the channel acts like a resistor: I_D rises almost linearly with V_d. As V_d grows the wedge narrows and the curve bends over.
- **(ii)** $V_d = V_{d,sat}$**:** the depletion layer just touches the bottom of the channel at the drain end (point P). The channel is pinched off at one point. This drain voltage is the **saturation drain voltage**:

  $$V_{d,sat} = V_g - V_p$$

  (the reverse bias at the drain end, $V_d - V_g$, has reached the pinch-off value $-V_p$).
- **(iii)** $V_d > V_{d,sat}$**:** the extra drain voltage cannot pinch the channel "more" at the drain; instead the pinch-off point P moves towards the source by a short distance ΔL. The voltage at P stays at $V_{d,sat}$, so the open channel between S and P carries almost the same current as in (ii). Electrons that reach P are swept across the thin depleted region of length ΔL by its strong electric field (like carriers crossing the collector junction of a BJT). The current therefore **saturates**; it rises only slightly because the effective channel length $L - \Delta L$ is a little shorter (**channel length modulation**).

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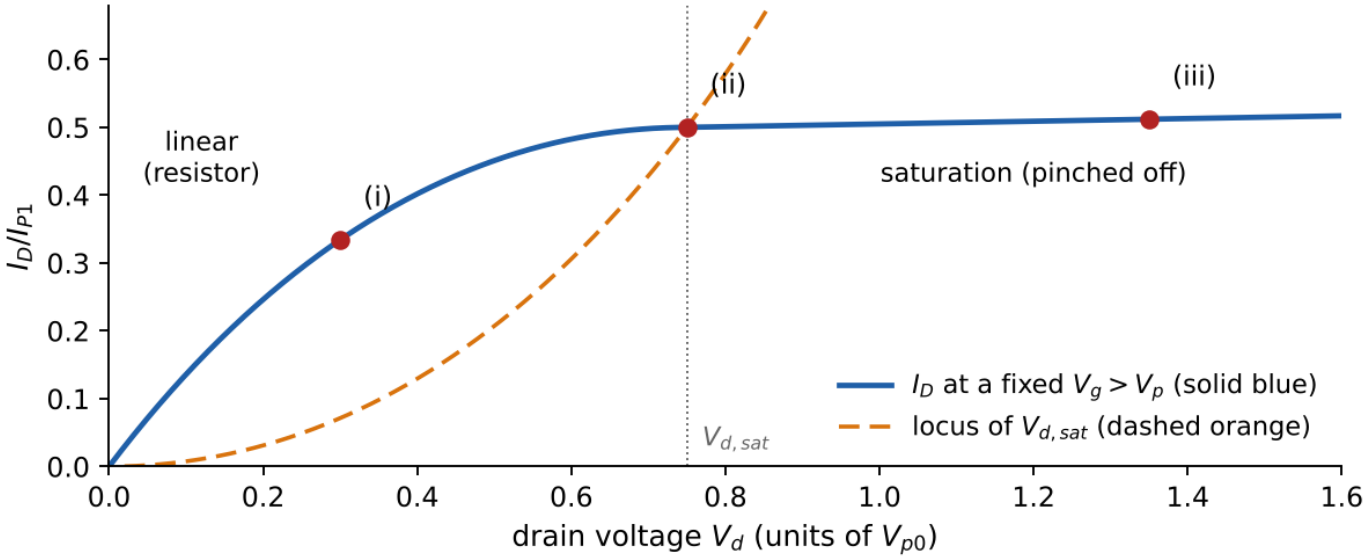


Figure 34: Drain current against drain voltage at a fixed gate voltage, with the three operating points of the question marked

Reading the figure. The solid blue curve is I_D at one gate voltage above pinch-off (Neamen's ideal JFET equation with a small channel-length-modulation slope added). The marked points (i), (ii) and (iii) are the three cases of the question. The dashed orange curve is the locus of $V_{d,sat}$ for different gate voltages: left of it the device is in the linear (non-saturated) region, right of it in saturation.

2. Answer

Device: single-gate n-channel JFET: p^+ gate on top of an n channel of thickness a; source at the left, drain at the right. The gate voltage is fixed with $V_g > V_p$, so the channel is open at the source in every case.

Key idea: the channel voltage rises from 0 (source) to V_d (drain), so the gate-channel reverse bias $V(x) - V_g$, and with it the depletion width $h(x)$, grows from source to drain:

$$h(x) = \left[\frac{2\epsilon_s\,(V_{bi} - V_g + V(x))}{eN_d}\right]^{1/2}$$

The three profiles are drawn in the first figure of the Theory.

(i) $V_g > V_p$, $V_d < V_{d,sat}$ (linear region): the depletion region is thinnest at the source and thicker at the drain, but $h < a$ everywhere: the channel is open along its whole length. The JFET behaves as a voltage-controlled resistor, I_D rising nearly linearly with V_d (bending over as V_d approaches $V_{d,sat}$).

(ii) $V_g > V_p$, $V_d = V_{d,sat}$ (onset of saturation): the depletion region just reaches the bottom of the channel at the drain end, $h(L) = a$. The channel is pinched off at a single point at the drain. Here $V_{d,sat} = V_g - V_p$ and $I_D = I_{D,sat}$.

(iii) $V_g > V_p$, $V_d > V_{d,sat}$ (saturation): the pinch-off point moves from the drain towards the source by ΔL; a short depleted region of length ΔL forms near the drain. The voltage at the pinch-off point stays at $V_{d,sat}$, so the current stays at about $I_{D,sat}$; electrons are carried across the depleted region by its high field. I_D increases only slightly with V_d (channel length modulation, effective length $L - \Delta L$).

3. Keyword Explanation

1. Pinch-off voltage V_p

The gate voltage at which the depletion region fills the whole channel thickness at the source end (with $V_d = 0$), so no open channel remains. Negative for an ordinary (depletion-mode) n-channel JFET.

2. Saturation drain voltage $V_{d,sat}$

The drain voltage at which the channel just pinches off at the drain end; $V_{d,sat} = V_g - V_p$. Above it the drain current stays nearly constant.

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3. Channel length modulation

The small rise of I_D above $V_{d,sat}$, caused by the pinch-off point moving towards the source, which shortens the conducting channel from L to $L - \Delta L$.

4. Notes and Remarks

- **Pinched off does not mean zero current.** In (ii) and (iii) the current is at its largest. Only the *open* channel stops at P; electrons arriving at P are pulled across the depleted gap by the field, just as electrons cross a reverse-biased collector junction.
- **If** $V_g < V_p$ (more negative than pinch-off), the channel is closed even at the source and $I_D \approx 0$: the device is **cut off**. This is the case the question excludes.
- **Two-sided JFET:** in the symmetric structure both gates deplete the channel from above and below, and the same three pictures hold for each half (with a the half-width).

5.(b) Derive the formula of internal pinch-off voltage in an n-channel JFET. Marks: 8

1. Theory

The basics

A **JFET** is a thin conducting channel with a gate junction on it. Reverse bias on the junction widens its **depletion layer**, a region with no free carriers, and this squeezes the channel. If the depletion layer becomes as thick as the channel, no current can flow. This is **pinch-off**.

1. What are we looking for? (Core Idea)

From 5(a): the channel is closed when the depletion layer under the gate becomes as thick as the channel, $h = a$. The **internal pinch-off voltage** V_{p0} is the **total** potential difference across the gate junction needed for this. "Total" means the built-in voltage of the junction plus the applied reverse bias. It depends only on the channel (its doping N_d and thickness a), not on how the voltage is applied; that is why it is called "internal".

To find it we need the thickness h of a depletion layer that holds a given voltage. This comes from **Poisson's equation**, which links charge, field and potential (also used in 4(a) and 3(c)).

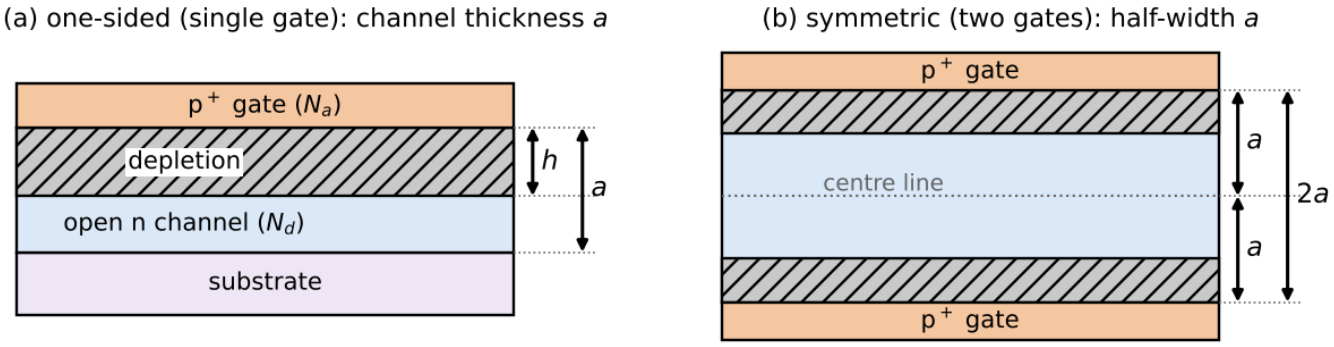


Figure 35: Geometry of the channel: (a) single-gate JFET with channel thickness a, (b) symmetric JFET with half-width a

Reading the figure. In panel (a) the hatched grey depletion layer of thickness h hangs down from the p^+ gate into the n channel of thickness a; the conducting part is $a - h$. In panel (b) two gates deplete the channel from both sides; by symmetry each gate only has to deplete half of the total thickness $2a$, so a is the half-width. In both cases pinch-off means $h = a$.

2. The tools: depletion approximation and Poisson's equation

- **Depletion approximation:** inside the depletion layer there are no free carriers at all, only the fixed ionised donors; outside it the semiconductor is neutral. On the n side, each ionised donor is a fixed positive charge $+e$, so the charge density is $\rho = +eN_d$ for $0 < y < h$ and $\rho = 0$ beyond h.

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- **Poisson's equation** (one dimension, y measured from the junction into the channel):

$$\frac{d^2\phi}{dy^2} = -\frac{\rho(y)}{\epsilon_s}, \qquad E = -\frac{d\phi}{dy}$$

  where ϕ is the electric potential, E the electric field and $\epsilon_s = \epsilon_r\epsilon_0$ the permittivity of the semiconductor.
- **One-sided junction:** the gate is p^+ ($N_a \gg N_d$). The junction holds equal and opposite charges on its two sides, $eN_a x_p = eN_d x_n$, so the depletion width on the gate side, $x_p = x_n N_d/N_a$, is tiny. Practically all of the depletion layer, and all of the voltage, lies on the channel side: $h \approx x_n$.

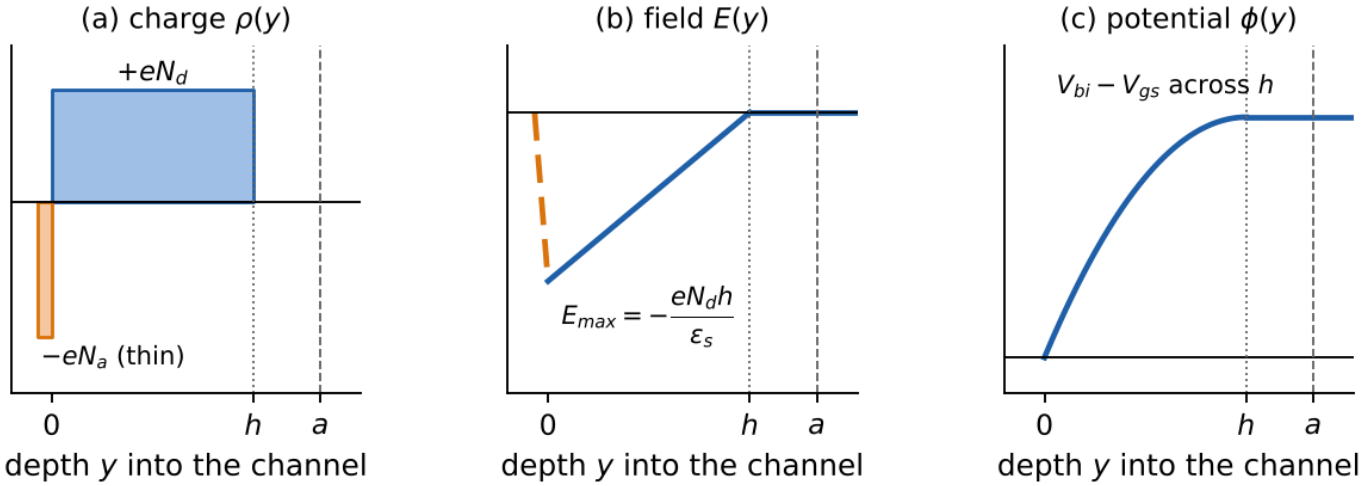


Figure 36: Charge, field and potential across the p+n gate junction, from the junction (y = 0) into the channel

Reading the figure. Panel (a): the shaded rectangle of height $+eN_d$ and width h is the donor charge in the channel; the thin shaded sliver at $y < 0$ is the equal and opposite acceptor charge in the p^+ gate. Panel (b): integrating the rectangle gives a field that rises linearly (solid blue line) from its largest magnitude at the junction to zero at $y = h$; the dashed orange segment is the very steep part in the gate. Panel (c): integrating the field gives a potential that rises along a parabola (solid blue curve) by $V_{bi} - V_{gs}$ across the layer, then stays flat in the neutral channel. Pinch-off is the case where the dotted line at h reaches the dashed line at a.

2. Derivation / Explanation

Step 1: Integrate Poisson's equation once (the field)

In the depletion layer $0 < y < h$:

$$\frac{dE}{dy} = \frac{\rho}{\epsilon_s} = \frac{eN_d}{\epsilon_s}$$

The field is zero at the edge $y = h$, where the neutral channel starts. Integrating from y to h:

$$E(y) = -\frac{eN_d}{\epsilon_s}(h - y), \qquad 0 \le y \le h$$

Its largest magnitude is at the junction: $|E_{max}| = eN_d h/\epsilon_s$.

Step 2: Integrate again (the potential)

With $E = -d\phi/dy$ and $\phi(0) = 0$ at the junction:

$$\phi(y) = -\int_0^y E\,dy' = \frac{eN_d}{\epsilon_s}\left(hy - \frac{y^2}{2}\right)$$

The total potential across the depletion layer is its value at $y = h$:

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$$\Delta\phi = \phi(h) = \frac{eN_d h^2}{2\epsilon_s}$$

Step 3: What sets the total potential

The total potential across the gate junction is the built-in potential V_{bi} plus the reverse bias. With the source grounded ($V_d = 0$) and a gate voltage V_{gs} (negative for reverse bias), the reverse bias is $-V_{gs}$, so

$$\Delta\phi = V_{bi} - V_{gs}$$

Equating the two expressions for $\Delta\phi$ and solving for h:

$$\boxed{h = \left[\frac{2\epsilon_s\,(V_{bi} - V_{gs})}{eN_d}\right]^{1/2}}$$

With a drain voltage, the channel potential $V(x)$ adds to the reverse bias at each point: $h(x) = [2\epsilon_s\,(V_{bi} - V_{gs} + V(x))\,/(eN_d)]^{1/2}$, which gives the wedge of 5(a).

Step 4: Pinch-off condition

The channel closes when the depletion layer fills it, $h = a$:

$$a = \left[\frac{2\epsilon_s\,(V_{bi} - V_{gs})}{eN_d}\right]^{1/2} \quad\Longrightarrow\quad V_{bi} - V_{gs} = \frac{ea^2 N_d}{2\epsilon_s}$$

The right-hand side depends only on the channel. It is defined as the **internal pinch-off voltage**:

$$\boxed{V_{p0} = \frac{ea^2 N_d}{2\epsilon_s}}$$

Step 5: The pinch-off (gate) voltage

The gate voltage $V_{gs} = V_p$ that produces pinch-off is found from $V_{bi} - V_p = V_{p0}$:

$$\boxed{V_p = V_{bi} - V_{p0}}$$

For a normal n-channel JFET $V_{p0} > V_{bi}$, so V_p is negative (a reverse gate bias is needed). Units: $[\mathrm{C}\cdot\mathrm{cm}^2\cdot\mathrm{cm}^{-3}]\,/\,[\mathrm{F/cm}] = \mathrm{C/F} = \mathrm{V}$ ✔.

3. Answer

Structure and assumptions: n-channel JFET with a p^+ gate ($N_a \gg N_d$), uniform channel doping N_d, channel thickness a (half-width a for a symmetric two-gate JFET), source and drain grounded ($V_d = 0$). Depletion approximation: the depletion layer of thickness h contains only ionised donors, $\rho = +eN_d$; the p^+ side's depletion is negligible (one-sided junction).

Step 1: Poisson's equation in the depletion layer

Measure y from the junction into the channel. In one dimension Poisson's equation is $d^2\phi/dy^2 = -\rho/\epsilon_s$, and with $\rho = +eN_d$ and $E = -d\phi/dy$:

$$\frac{d^2\phi}{dy^2} = -\frac{eN_d}{\epsilon_s}, \qquad \frac{dE}{dy} = \frac{eN_d}{\epsilon_s}$$

Step 2: The electric field

Integrate dE/dy once: $E(y) = \dfrac{eN_d}{\epsilon_s}y + C_1$. At the edge of the depletion layer, $y = h$, the channel is neutral and $E(h) = 0$, so $C_1 = -eN_d h/\epsilon_s$:

$$E(y) = -\frac{eN_d}{\epsilon_s}(h - y)$$

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Step 3: The potential

Integrate again, $\phi = -\int E\,dy$:

$$\phi(y) = \frac{eN_d}{\epsilon_s}\left(hy - \frac{y^2}{2}\right) + C_2$$

Take the junction as the reference, $\phi(0) = 0$, so $C_2 = 0$. At $y = h$:

$$\phi(h) = \frac{eN_d}{\epsilon_s}\left(h^2 - \frac{h^2}{2}\right) = \frac{eN_d h^2}{2\epsilon_s}$$

Step 4: The depletion thickness

The potential across the depletion layer is the total junction potential: the built-in potential plus the reverse bias. With the gate at V_{gs} (negative for an n-channel JFET) and the channel at 0, the reverse bias is $-V_{gs}$, so the total is $V_{bi} - V_{gs}$:

$$\frac{eN_d h^2}{2\epsilon_s} = V_{bi} - V_{gs} \quad \Rightarrow \quad h = \left[\frac{2\epsilon_s\,(V_{bi} - V_{gs})}{eN_d}\right]^{1/2}$$

Step 5: Pinch-off

The open channel thickness is $a - h$. It becomes zero when $h = a$. Put $h = a$ in Step 4:

$$V_{bi} - V_{gs} = \frac{ea^2 N_d}{2\epsilon_s}$$

$$\boxed{V_{p0} = \frac{ea^2 N_d}{2\epsilon_s}}$$

V_{p0}, the **internal pinch-off voltage**, is the total potential (built-in plus applied) across the gate junction at pinch-off. The corresponding **pinch-off gate voltage** is the V_{gs} that makes this true:

$$\boxed{V_p = V_{bi} - V_{p0}}$$

where e is the electron charge, N_d the channel doping, a the channel thickness (or half-width) and ϵ_s the semiconductor permittivity.

Steps to remember

Poisson with $\rho = +eN_d \to E = 0$ at $y = h \to \phi(h) = eN_d h^2/2\epsilon_s \to$ equals $V_{bi} - V_{gs} \to h \to$ pinch-off at $h = a \to V_{p0} = ea^2 N_d/2\epsilon_s$.

4. Keyword Explanation

1. Depletion approximation

The assumption that a depletion layer has no free carriers and ends sharply; inside it the charge is just the fixed dopant ions.

2. One-sided junction

A pn junction with one side much more heavily doped. Its depletion layer and voltage lie almost entirely on the lightly doped side.

3. Internal pinch-off voltage V_{p0}

The total gate-junction potential (built-in plus reverse bias) at which the depletion layer fills the channel: $V_{p0} = ea^2 N_d/(2\epsilon_s)$.

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5. Notes and Remarks

- **Internal vs. gate pinch-off voltage:** V_{p0} is positive and is a property of the channel. $V_p = V_{bi} - V_{p0}$ is the voltage you apply to the gate. Neamen uses both; some books call V_{p0} simply "the pinch-off voltage", so watch the definition.
- **Sensitivity:** $V_{p0} \propto a^2$. A 10 % error in the channel thickness changes V_{p0} by about 21 %, which is why JFET thresholds are hard to control.
- **Symmetric JFET:** the same formula holds with a the half-width, since each gate depletes half the channel.
- **p-channel JFET:** swap $N_d \to N_a$ (channel doping); then $V_p = V_{p0} - V_{bi}$, positive.

5.(c) Consider a GaAs n-channel pn JFET at $T = 300$ K with $N_a = 1 \times 10^{18}$ cm^{-3}, $N_d = 3 \times 10^{15}$ cm^{-3}, $\epsilon_s = 13.1$, and $a = 0.70$ μm. Determine the forward-bias gate voltage required to open a channel region that is 0.10 μm thick with zero drain voltage. Marks: 7

1. Theory

The basics

A **JFET** is a thin conducting channel of thickness a under a gate junction. The junction's **depletion layer**, of thickness h, has no free carriers, so only the part $a - h$ conducts. The gate voltage changes h, and so controls the current.

1. What is being asked? (Core Idea)

This is the JFET of 5(a) made of GaAs. With zero drain voltage the depletion layer has the same thickness h everywhere, given by the formula of 5(b):

$$h = \left[\frac{2\epsilon_s\,(V_{bi} - V_{gs})}{eN_d}\right]^{1/2}$$

The open (conducting) channel is $a - h$. We want it to be 0.10 µm, so the depletion layer must be $h = a - 0.10$ μm. We then ask: what gate voltage gives that h?

2. The built-in voltage of the gate junction

A pn junction has a built-in potential (Neamen):

$$V_{bi} = \frac{kT}{e}\ln\left(\frac{N_a N_d}{n_i^2}\right)$$

where $kT/e = 0.0259$ V at 300 K and n_i is the intrinsic carrier concentration (1.8×10^6 cm^{-3} for GaAs). GaAs has a wide bandgap (1.42 eV), so n_i is tiny and V_{bi} is large, about 1.25 V here.

3. Why might the gate need a forward bias?

Even at $V_{gs} = 0$ the junction has its built-in voltage, so a depletion layer already exists. If the channel is thin or lightly doped, this zero-bias depletion layer can already be thicker than the channel: the channel is closed with no gate voltage. Such a device is **normally off** (an **enhancement-mode** JFET). To open the channel, the gate must be **forward biased** ($V_{gs} > 0$), which lowers the total junction voltage $V_{bi} - V_{gs}$ and shrinks the depletion layer. In the formula this just means a positive V_{gs}.

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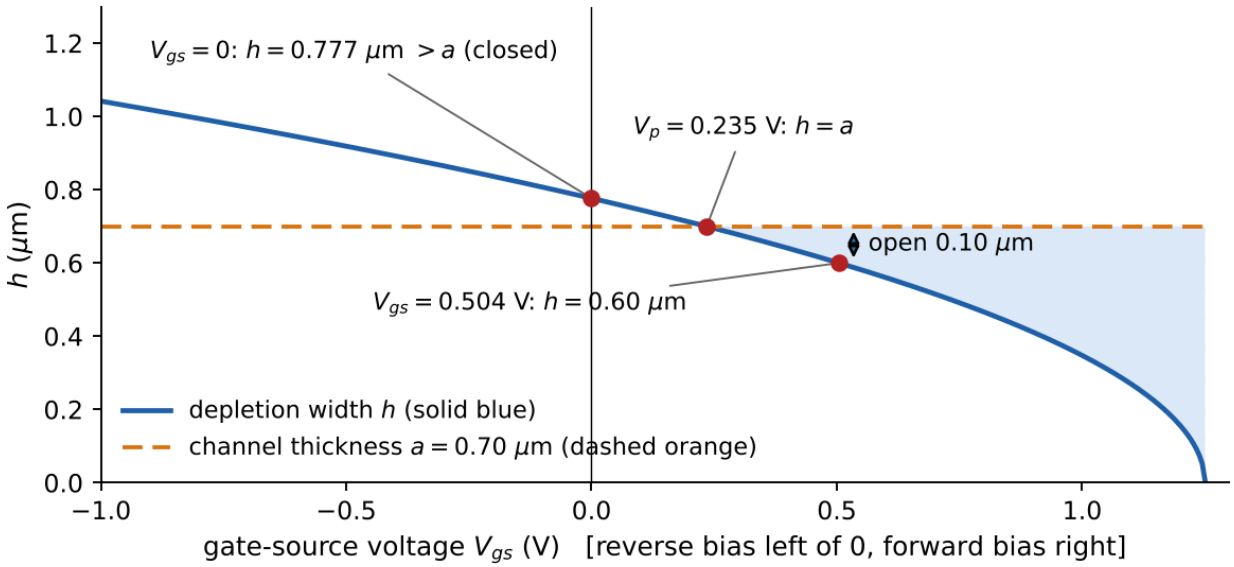


Figure 37: Depletion width against gate voltage for the GaAs JFET, with zero bias, the pinch-off voltage and the required gate voltage marked

Reading the figure. The solid blue curve is the depletion width h; the dashed orange line is the channel thickness $a = 0.70$ μm. Where the solid blue curve is above the dashed orange line the channel is closed; the dotted shaded area below the line is the open channel $a - h$. At $V_{gs} = 0$ the marked point lies above the line ($h = 0.777$ μm), so the channel is closed. The curve crosses the line at $V_p = +0.235$ V, and reaches $h = 0.60$ μm (an open channel of 0.10 μm) at $V_{gs} = 0.504$ V.

2. Answer

Given:

- GaAs, $T = 300$ K, $kT/e = 0.0259$ V, $n_i = 1.8 \times 10^6$ cm^{-3} (Neamen)
- Gate: $N_a = 1 \times 10^{18}$ cm^{-3}; channel: $N_d = 3 \times 10^{15}$ cm^{-3}
- $\epsilon_s = 13.1\,\epsilon_0 = 13.1 \times 8.85 \times 10^{-14} = 1.1594 \times 10^{-12}$ F/cm (the paper's "$\epsilon_s = 13.1$" is the relative permittivity)
- $a = 0.70$ μm $= 0.70 \times 10^{-4}$ cm
- Required open channel: 0.10 μm, $V_d = 0$

Relationships (from 5(b)):

$$V_{bi} = \frac{kT}{e}\ln\left(\frac{N_a N_d}{n_i^2}\right), \qquad h = \left[\frac{2\epsilon_s\,(V_{bi} - V_{gs})}{eN_d}\right]^{1/2}, \qquad V_{p0} = \frac{ea^2 N_d}{2\epsilon_s}$$

Step 1: Built-in potential

$$V_{bi} = 0.0259\,\ln\left[\frac{(10^{18})(3\times10^{15})}{(1.8\times10^{6})^2}\right] = 0.0259\,\ln\left(9.259\times10^{20}\right) = 0.0259 \times 48.277$$

$$\boxed{V_{bi} = 1.250\ \text{V}}$$

Step 2: Internal pinch-off voltage (is the channel open at zero bias?)

$$V_{p0} = \frac{(1.602\times10^{-19})(0.70\times10^{-4})^2(3\times10^{15})}{2(1.1594\times10^{-12})} = \frac{2.3549\times10^{-12}}{2.3187\times10^{-12}}$$

$$\boxed{V_{p0} = 1.016\ \text{V}}$$

$$V_p = V_{bi} - V_{p0} = 1.250 - 1.016 = +0.235\ \text{V}$$

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$V_{p0} < V_{bi}$, so the pinch-off voltage is **positive**: at $V_{gs} = 0$ the built-in voltage alone depletes $h = [2\epsilon_s V_{bi}/(eN_d)]^{1/2} = 0.777\ \mu\text{m} > a$. The channel is closed at zero gate bias (normally off, enhancement mode), so a **forward** gate bias is needed.

Step 3: Required depletion width

$$h = a - 0.10\ \mu\text{m} = 0.70 - 0.10 = 0.60\ \mu\text{m} = 0.60 \times 10^{-4}\ \text{cm}$$

Step 4: Gate voltage

From the depletion-width formula:

$$V_{bi} - V_{gs} = \frac{eN_d h^2}{2\epsilon_s} = \frac{(1.602 \times 10^{-19})(3 \times 10^{15})(0.60 \times 10^{-4})^2}{2(1.1594 \times 10^{-12})} = 0.7462\ \text{V}$$

$$V_{gs} = V_{bi} - 0.7462 = 1.2504 - 0.7462$$

$$\boxed{V_{gs} = +0.504\ \text{V (forward bias)}}$$

A forward gate bias of about 0.50 V shrinks the depletion layer from 0.777 µm to 0.60 µm and opens a 0.10 µm channel.

Final Answer:

- $V_{bi} = 1.250$ V, $V_{p0} = 1.016$ V, $V_p = +0.235$ V (normally-off device)
- Required gate voltage: $V_{gs} \approx 0.50$ V forward bias

3. Keyword Explanation

1. Enhancement-mode (normally-off) JFET

A JFET whose channel is already closed at $V_{gs} = 0$ because the built-in depletion exceeds the channel thickness; a forward gate bias is needed to open it. Its pinch-off (threshold) voltage is positive.

2. Built-in potential V_{bi}

The potential barrier of a pn junction at zero applied voltage, set by the doping on both sides.

4. Notes and Remarks

- **Gate current:** a forward-biased gate junction conducts. Here the forward bias (0.50 V) is well below the built-in voltage (1.25 V), and GaAs has a very small n_i, so the forward current of the gate diode is still small. That is why such enhancement-mode GaAs JFETs are run with gate voltages of only a few tenths of a volt; a much larger forward bias would let a large gate current flow and the gate would no longer control the channel without drawing current.
- **One-sided check:** with the full two-sided formula the channel-side depletion at $V_{gs} = 0.504$ V is $x_n = 0.5991$ µm and the gate side $x_p = 0.0018$ µm, so the one-sided approximation ($h = 0.60$ µm) is accurate to about 0.2 %.
- **Sensitivity:** near this point $dh/dV_{gs} = -\epsilon_s/(eN_d h) = -0.40\ \mu$m/V, so every 0.1 V of extra forward bias opens about 0.04 µm more channel.

5.(d) Sketch the ideal energy band diagram of a metal semiconductor junction in which $\Phi_m < \Phi_s$. Explain why this is an ohmic contact. Marks: 4

1. Theory

1. What is an ohmic contact? (Core Idea)

Every semiconductor device needs metal wires to bring current in and out. Where a metal touches a semiconductor, two things can happen:

- **Rectifying (Schottky) contact:** a potential barrier forms, and current flows easily in one direction only, like a diode (3(c), 3(a)).

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- **Ohmic contact:** no barrier (or a negligible one) forms. Current flows equally easily in both directions and is proportional to the voltage, $I = V/R_c$, with a small **contact resistance** R_c. It obeys Ohm's law, hence the name. Ohmic contacts are what we want for the source, drain, emitter, collector and other terminals of a device.

Which one forms depends on the **work functions**.

2. Work functions and the Fermi level

(Energy bands, the vacuum level and the electron affinity are built in 1(d).)

- The **Fermi level** E_F is the energy that marks how "full" a material is with electrons. Electrons always flow from where E_F is higher to where it is lower, until the two Fermi levels are equal (thermal equilibrium).
- The **work function** $e\phi$ is the energy from E_F up to the **vacuum level** (the energy of an electron just free of the material). For the metal it is $e\phi_m$.
- For a semiconductor, $e\phi_s = e\chi + (E_c - E_F)$, where $e\chi$ is the **electron affinity** (vacuum level to conduction band edge E_c).

So $\phi_m < \phi_s$ means the metal's Fermi level is **higher** than the semiconductor's (it is closer to the vacuum level).

3. What happens when they touch ($\phi_m < \phi_s$, n-type)

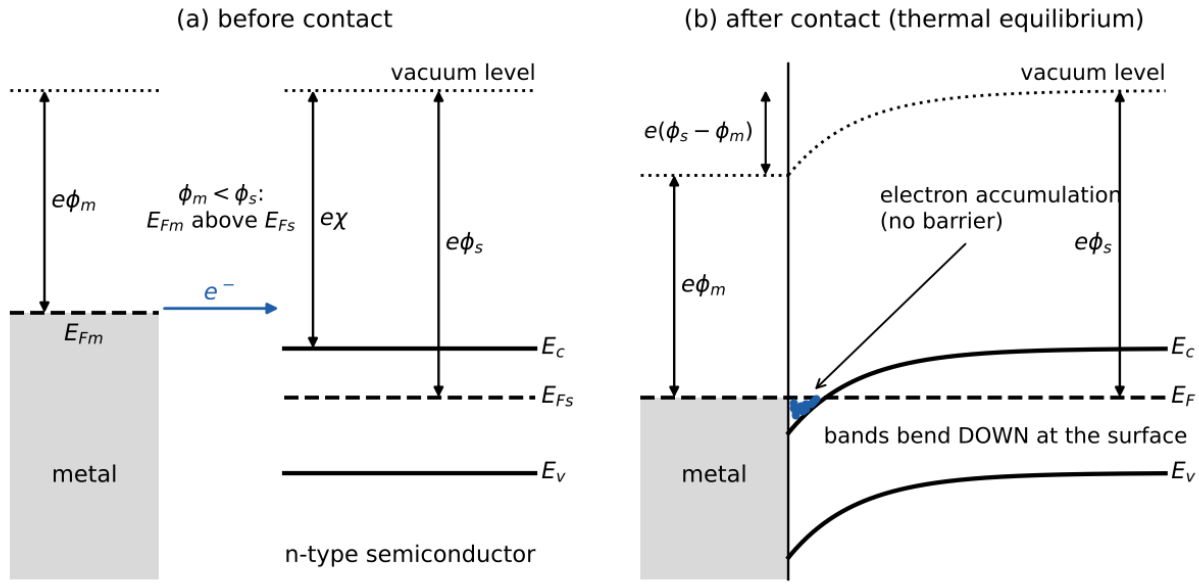


Figure 38: Ideal energy-band diagrams of a metal and an n-type semiconductor with $\phi_m < \phi_s$: (a) before contact, (b) after contact

Reading the figure. The dotted lines are the vacuum level, the solid lines E_c and E_v, the dashed lines the Fermi levels; the shaded block is the metal, filled with electrons up to its Fermi level.

- **(a) Before contact:** both materials share the same vacuum level. Because $\phi_m < \phi_s$, the metal Fermi level E_{Fm} lies above the semiconductor's E_{Fs}, so the electrons in the metal have higher energy (labelled arrow e^-).
- **(b) After contact:** electrons flow from the metal into the semiconductor until the Fermi levels line up (flat dashed E_F). The semiconductor gains extra electrons near its surface, an **accumulation layer** (the marked dots), and the metal is left with a thin positive surface charge. Seen by an electron, the surface is now a place of lower energy than the bulk: the bands bend *down* near the interface by $e(\phi_s - \phi_m)$. The conduction band at the surface comes close to (or below) E_F.

4. Why there is no barrier

Look at panel (b) from the point of view of an electron:

- **Semiconductor to metal:** an electron in the conduction band moves towards the interface. The band edge E_c goes **downhill** towards the metal, so nothing stops it.
- **Metal to semiconductor:** an electron at the metal's E_F meets the conduction band at the interface at or below its own energy. At most it sees the small step $E_c - E_F$ of the bulk semiconductor ($e\phi_n$, Neamen), which is small for a moderately or heavily doped semiconductor and is crossed easily by thermal energy.

Because neither direction has a real barrier, an applied voltage simply drives a current through the low resistance of the contact and the bulk, in either direction. The I-V curve is a straight line.

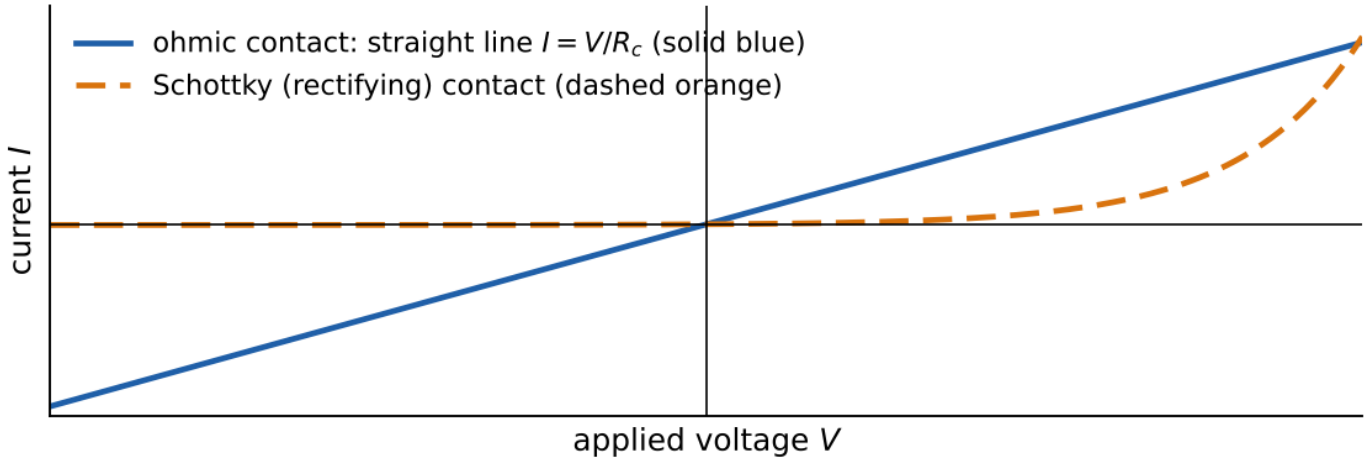


Figure 39: Current against voltage for an ohmic contact compared with a rectifying Schottky contact

Reading the figure. The solid blue straight line is the ohmic contact: the current is proportional to the voltage for both signs. The dashed orange curve is a rectifying Schottky contact: large current in forward bias, almost none in reverse.

2. Answer

Assumption: the semiconductor is n-type (the case in which $\phi_m < \phi_s$ gives an ohmic contact); ideal contact with no interface states.
Energy-band diagram: shown in the figure in the Theory. Before contact, $\phi_m < \phi_s$ puts the metal Fermi level E_{Fm} above the semiconductor Fermi level E_{Fs} ($e\phi_s = e\chi + E_c - E_F$). After contact, in thermal equilibrium:

- electrons flow from the metal into the semiconductor until the Fermi levels are equal;
- an **electron accumulation layer** forms at the semiconductor surface (the metal holds an equal positive surface charge);
- the semiconductor bands bend **downwards** towards the interface by $e(\phi_s - \phi_m)$; E_c at the surface comes close to or below E_F.

Why it is ohmic:

- Electrons moving from the semiconductor to the metal see E_c going downhill: **no barrier**.
- Electrons moving from the metal to the semiconductor see at most the small step $E_c - E_F = e\phi_n$ of the bulk, which is easily overcome; there is no depletion layer and no Schottky barrier.
- Majority carriers (electrons) therefore flow freely in both directions. The current is limited only by the small contact and bulk resistance, so $I \propto V$ for either polarity: a linear, low-resistance (ohmic) contact.

(For comparison, $\phi_m > \phi_s$ on n-type gives a depletion layer, upward band bending and a Schottky barrier $e\phi_{B0} = e(\phi_m - \chi)$: a rectifying contact.)

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3. Keyword Explanation

1. Work function

The energy needed to take an electron from the Fermi level to the vacuum level: $e\phi_m$ for a metal, $e\phi_s = e\chi + (E_c - E_F)$ for a semiconductor.

2. Accumulation layer

A thin layer at the semiconductor surface with more majority carriers than the bulk; it forms when the surface bands bend towards the majority-carrier band.

3. Specific contact resistance R_c

The resistance of a contact times its area ($\Omega \cdot \text{cm}^2$); a good ohmic contact has a small value.

4. Notes and Remarks

- **p-type is the opposite:** on a p-type semiconductor, $\phi_m > \phi_s$ gives the ohmic contact (bands bend up, hole accumulation) and $\phi_m < \phi_s$ gives a rectifying one.
- **Real contacts:** interface states pin the Fermi level at the surface (4(b)), so for most metals on Si and GaAs a barrier forms whatever ϕ_m is. Practical ohmic contacts are therefore made by **heavy doping** under the metal ($N_d \gtrsim 10^{19}$ cm^{-3}): the depletion layer becomes so thin that electrons **tunnel** through the barrier. The specific contact resistance then falls steeply with doping, $R_c \propto \exp\left[\frac{2\sqrt{\epsilon_s m_n^*}}{\hbar}\frac{\phi_{Bn}}{\sqrt{N_d}}\right]$ (Neamen). On GaAs, alloyed Au-Ge-Ni contacts are used for this.

5. Final Concise Answer

(a) In a single-gate n-channel JFET the gate-channel reverse bias grows from source to drain, so the depletion layer under the gate is a wedge, thickest at the drain. (i) $V_d < V_{d,sat}$: the channel is open along its whole length (linear, resistor-like region). (ii) $V_d = V_{d,sat} = V_g - V_p$: the depletion layer just touches the bottom of the channel at the drain end (pinched off at one point); I_D reaches $I_{D,sat}$. (iii) $V_d > V_{d,sat}$: the pinch-off point moves towards the source by ΔL, a short depleted region near the drain carries the current by its high field, and I_D stays at about $I_{D,sat}$ (slight rise from channel length modulation).

(b) Poisson's equation in the one-sided depletion layer ($\rho = eN_d$) gives $h = [2\epsilon_s(V_{bi} - V_{gs})/(eN_d)]^{1/2}$. Pinch-off is $h = a$, so the internal pinch-off voltage is $V_{p0} = ea^2N_d/(2\epsilon_s)$, and the pinch-off gate voltage is $V_p = V_{bi} - V_{p0}$.

(c) $V_{bi} = 1.250$ V, $V_{p0} = 1.016$ V, $V_p = +0.235$ V: the channel is closed at zero bias ($h = 0.777$ µm $> a$). For a 0.10 µm open channel, $h = 0.60$ µm, $V_{bi} - V_{gs} = 0.746$ V, so $V_{gs} \approx +0.50$ V (forward bias).

(d) With $\phi_m < \phi_s$ on n-type, electrons flow from the metal into the semiconductor, forming an electron accumulation layer; the bands bend down by $e(\phi_s - \phi_m)$. Electrons see no barrier in either direction, so the contact has a linear, low-resistance I-V characteristic: it is ohmic.

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6. (a) Draw the equivalent circuit for basic Ebers-Moll model as used in BJT. Also derive the expression of $V_{CE}(sat)$ from this model. Marks: 9

(b) Explain high level injection and current crowding effects in a BJT with necessary figures. How can the problem of current crowding be solved? Marks: 8

(c) Consider a silicon n-channel JFET at $T = 300$ K with the following parameters: $N_a = 1 \times 10^{18}$ cm^{-3}, $N_d = 1 \times 10^{16}$ cm^{-3}, $a = 0.75$ μm, $L = 10$ μm, $W = 30$ μm, and $\mu_n = 1000$ cm^2/(V s)

(i) Calculate the maximum current in an n-channel JFET.

(ii) Determine the maximum transconductance of an n-channel depletion mode JFET biased in the saturation region. $I_{P1} = 0.522$ mA, $V_{bi} = 0.814$ V, and $V_{po} = 4.35$ V. Marks: 8

6.(a) Draw the equivalent circuit for basic Ebers-Moll model as used in BJT. Also derive the expression of $V_{CE}(sat)$ from this model. Marks: 9

1. Theory

1. What is a bipolar junction transistor? (Core Idea)

A **bipolar junction transistor** (BJT) is three doped regions of semiconductor in a row. In an **npn** transistor they are:

- the **emitter** (E): heavily doped n-type (n^+). Its job is to *emit* (inject) electrons;
- the **base** (B): a very thin p-type layer in the middle. It controls the flow;
- the **collector** (C): n-type. Its job is to *collect* the electrons that cross the base.

So the device has two pn junctions back to back: the **B-E junction** and the **B-C junction**. It is called *bipolar* because both electrons and holes carry current.

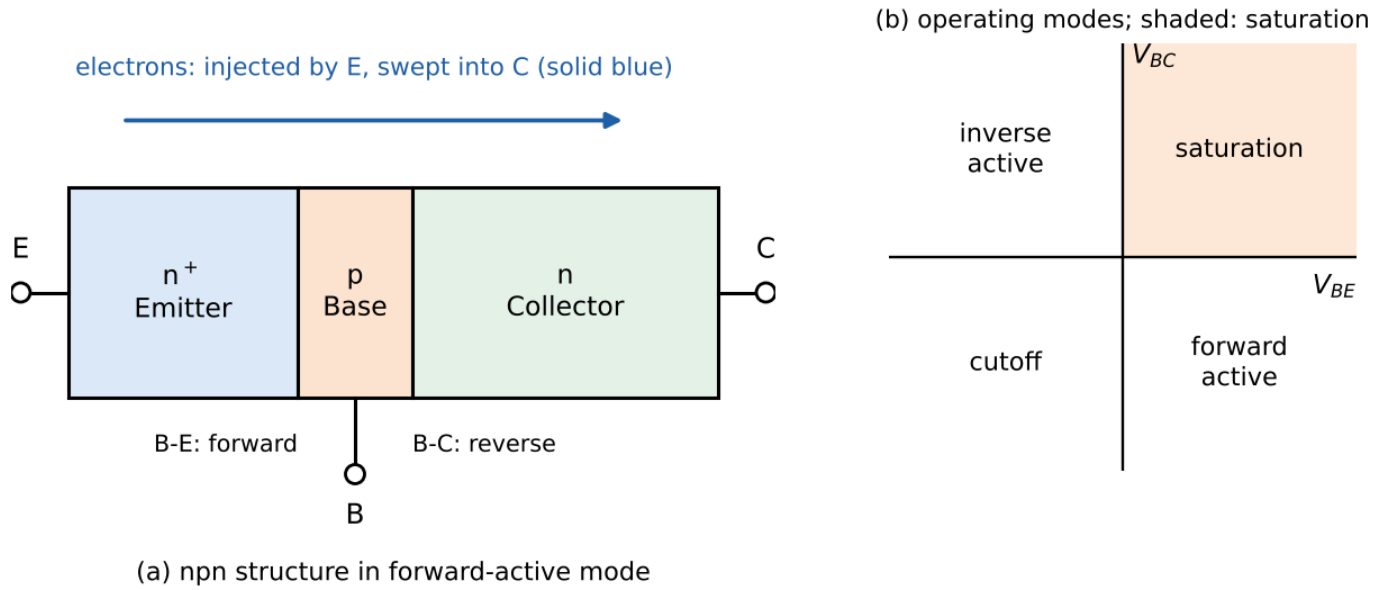


Figure 40: An npn transistor: (a) the three regions in forward-active mode, (b) the four operating modes set by the two junction voltages

Reading the figure. In panel (a) the B-E junction is forward biased, so the emitter pushes electrons into the thin base. Almost none of them meet a hole in the base before they reach the reverse-biased B-C junction, whose field sweeps them into the collector (the solid blue arrow). Panel (b) shows that the two junction voltages V_{BE} and V_{BC} decide the **mode**: each junction can be forward biased (voltage

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positive for npn) or reverse biased, so there are four modes. The shaded quadrant, where both junctions are forward biased, is **saturation**.

2. The four modes in words

- **Forward active** ($V_{BE} > 0$, $V_{BC} < 0$): the amplifier mode. I_C is controlled by V_{BE} (or by I_B) and hardly depends on V_{CE}.
- **Saturation** ($V_{BE} > 0$, $V_{BC} > 0$): the "switch on" mode. Both junctions inject, the collector current is limited by the outside circuit, and V_{CE} is very small, about 0.1–0.2 V. This small voltage is $V_{CE}(sat)$.
- **Cutoff** ($V_{BE} < 0$, $V_{BC} < 0$): the "switch off" mode; almost no current.
- **Inverse active** ($V_{BE} < 0$, $V_{BC} > 0$): the transistor used backwards; it works badly because the collector is not designed to emit.

Since $V_{CE} = V_{BE} - V_{BC}$, a positive V_{BC} (saturation) makes V_{CE} smaller than V_{BE}.

3. The current gains alpha and beta

In forward-active mode, a fraction α_F of the current that the B-E junction carries arrives at the collector. The rest is lost in the base (recombination) or is the hole current that the base injects back into the emitter. A good transistor has α_F close to 1 (0.98 to 0.995).

$$\alpha_F = \frac{I_C}{I_E}, \qquad \beta_F = \frac{I_C}{I_B} = \frac{\alpha_F}{1 - \alpha_F}$$

where

- α_F is the forward **common-base current gain**
- β_F is the forward **common-emitter current gain**; $\alpha_F = 0.99$ gives $\beta_F = 99$

Used backwards (inverse active), the collector emits and the emitter collects. The fraction that makes it is the **inverse common-base current gain** α_R. It is small, typically 0.1 to 0.5, because the collector is lightly doped and much larger than the emitter.

4. The pn-junction diode equation

Each junction alone is a diode. The current through a pn junction with voltage V across it (positive = forward) is

$$I = I_S \left[\exp\left(\frac{V}{V_t}\right) - 1 \right]$$

where

- I_S is the **saturation current** of the junction (very small, about 10^{-14} A)
- $V_t = kT/e = 0.0259$ V at 300 K is the **thermal voltage**

For a forward voltage of a few tenths of a volt, $\exp(V/V_t)$ is huge and the -1 can be dropped. For a reverse voltage, the exponential is tiny and $I \approx -I_S$.

5. The idea of the Ebers-Moll model

The BJT is two diodes that *talk to each other* through the thin base. Ebers and Moll wrote this as:

- each junction is a diode with its own current, I_F for the B-E junction and I_R for the B-C junction;
- a fraction α_F of the B-E diode current I_F reaches the collector: a current source $\alpha_F I_F$ across the B-C junction;
- a fraction α_R of the B-C diode current I_R reaches the emitter: a current source $\alpha_R I_R$ across the B-E junction.

This one model works in **all four modes**, because nothing in it assumes which junction is forward biased. That is why we can use it for saturation, where the simple forward-active picture fails.
The two diode saturation currents are linked by the **reciprocity relation**:

$$\alpha_F I_{ES} = \alpha_R I_{CS}$$

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where I_{ES} is the B-E saturation current measured with the B-C junction shorted, and I_{CS} the B-C saturation current with the B-E junction shorted. It follows from the symmetry of the minority-carrier flow in the base (both products describe the same diffusion through the base). Since $\alpha_R < \alpha_F$, $I_{CS} > I_{ES}$: the collector junction is the larger one.

2. Answer

Ebers-Moll equivalent circuit (npn). The model has two diodes back to back between the terminals, each shunted by a current-controlled current source that represents the carriers that cross the base from the other junction.

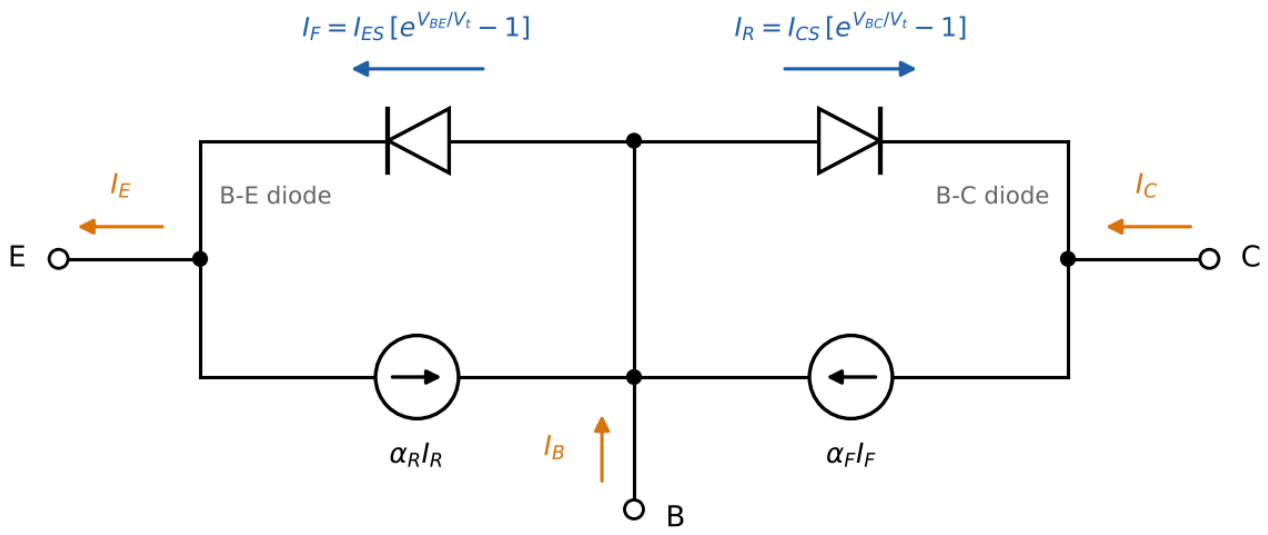


Figure 41: Ebers-Moll equivalent circuit of an npn transistor: B-E diode with the source $\alpha_R I_R$ in parallel, B-C diode with the source $\alpha_F I_F$ in parallel

Reading the figure. The internal base node is in the middle. On the left, the B-E diode (anode at the base) carries I_F towards the emitter, and the current source $\alpha_R I_R$ in parallel with it carries current from the emitter node towards the base node. On the right, the B-C diode carries I_R towards the collector, and the source $\alpha_F I_F$ carries current from the collector node towards the base node, which is the transport of emitted electrons into the collector. The terminal currents I_E (out of E), I_C (into C) and I_B (into B) are the labelled orange arrows.

Diode currents:

$$I_F = I_{ES}\left[\exp\left(\frac{V_{BE}}{V_t}\right) - 1\right], \qquad I_R = I_{CS}\left[\exp\left(\frac{V_{BC}}{V_t}\right) - 1\right]$$

with the reciprocity relation $\alpha_F I_{ES} = \alpha_R I_{CS}$.

Terminal currents (Kirchhoff's current law at the emitter and collector nodes, Neamen's sign convention):

$$\begin{aligned} I_C &= \alpha_F I_F - I_R \\ I_E &= I_F - \alpha_R I_R \\ I_B &= I_E - I_C = (1-\alpha_F)\, I_F + (1-\alpha_R)\, I_R \end{aligned}$$

Derivation of $V_{CE}(sat)$

Step 1: Saturation condition

In saturation both junctions are forward biased by several V_t, so the -1 terms are negligible:

$$I_F \approx I_{ES}\exp\left(\frac{V_{BE}}{V_t}\right), \qquad I_R \approx I_{CS}\exp\left(\frac{V_{BC}}{V_t}\right)$$

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$$V_{BE} = V_t \ln \frac{I_F}{I_{ES}}, \qquad V_{BC} = V_t \ln \frac{I_R}{I_{CS}}$$

Step 2: Solve for I_F and I_R in terms of I_C and I_B

From the I_C equation, $I_R = \alpha_F I_F - I_C$. Put this into the I_B equation:

$$\begin{aligned} I_B &= (1-\alpha_F) I_F + (1-\alpha_R)(\alpha_F I_F - I_C) \\ &= (1 - \alpha_F \alpha_R)\, I_F - (1-\alpha_R)\, I_C \end{aligned}$$

$$I_F = \frac{I_B + (1-\alpha_R)\, I_C}{1 - \alpha_F \alpha_R}$$

and then

$$I_R = \alpha_F I_F - I_C = \frac{\alpha_F I_B - (1-\alpha_F)\, I_C}{1 - \alpha_F \alpha_R}$$

Step 3: Collector-emitter voltage

Since $V_{CE} = V_{BE} - V_{BC}$:

$$V_{CE}(sat) = V_t \ln \frac{I_F}{I_{ES}} - V_t \ln \frac{I_R}{I_{CS}} = V_t \ln \left[\frac{I_F}{I_R} \cdot \frac{I_{CS}}{I_{ES}} \right]$$

Step 4: Use reciprocity and Step 2

From $\alpha_F I_{ES} = \alpha_R I_{CS}$, $I_{CS}/I_{ES} = \alpha_F/\alpha_R$. Divide the two results of Step 2; the common denominator $(1 - \alpha_F \alpha_R)$ cancels:

$$\frac{I_F}{I_R} = \frac{I_B + (1-\alpha_R)\, I_C}{\alpha_F I_B - (1-\alpha_F)\, I_C}$$

Put both into Step 3:

$$\boxed{V_{CE}(sat) = V_t \ln \left[\frac{[I_C(1-\alpha_R) + I_B]\, \alpha_F}{[\alpha_F I_B - (1-\alpha_F)\, I_C]\, \alpha_R} \right]}$$

Steps to remember

Saturation: drop the -1 terms $\rightarrow$ V_{BE}, V_{BC} as logarithms $\rightarrow$ from the I_C and I_B equations solve for I_F, I_R $\rightarrow$ $V_{CE} = V_{BE} - V_{BC}$ $\rightarrow$ reciprocity $I_{CS}/I_{ES} = \alpha_F/\alpha_R$ $\rightarrow$ $V_{CE}(sat)$.

Validity: the logarithm needs $\alpha_F I_B > (1-\alpha_F) I_C$, that is $I_C < \beta_F I_B$: the base current must be larger than the forward-active value I_C/β_F. That is exactly the condition for the transistor to be saturated.

Numerical illustration (values assumed, the question gives none): $\alpha_F = 0.99$, $\alpha_R = 0.10$, $I_C = 1$ mA, $I_B = 0.05$ mA, $V_t = 0.0259$ V.

$$\begin{aligned} V_{CE}(sat) &= 0.0259 \ln \left[\frac{[1(1-0.10) + 0.05]\,(0.99)}{[(0.99)(0.05) - (1-0.99)(1)]\,(0.10)} \right] \\ &= 0.0259 \ln \frac{0.9405}{0.00395} = 0.0259 \ln(238.1) = 0.0259 \times 5.473 \end{aligned}$$

$$\boxed{V_{CE}(sat) = 0.142 \text{ V}}$$

So a saturated transistor carrying 1 mA has only about 0.14 V across it: a nearly closed switch.

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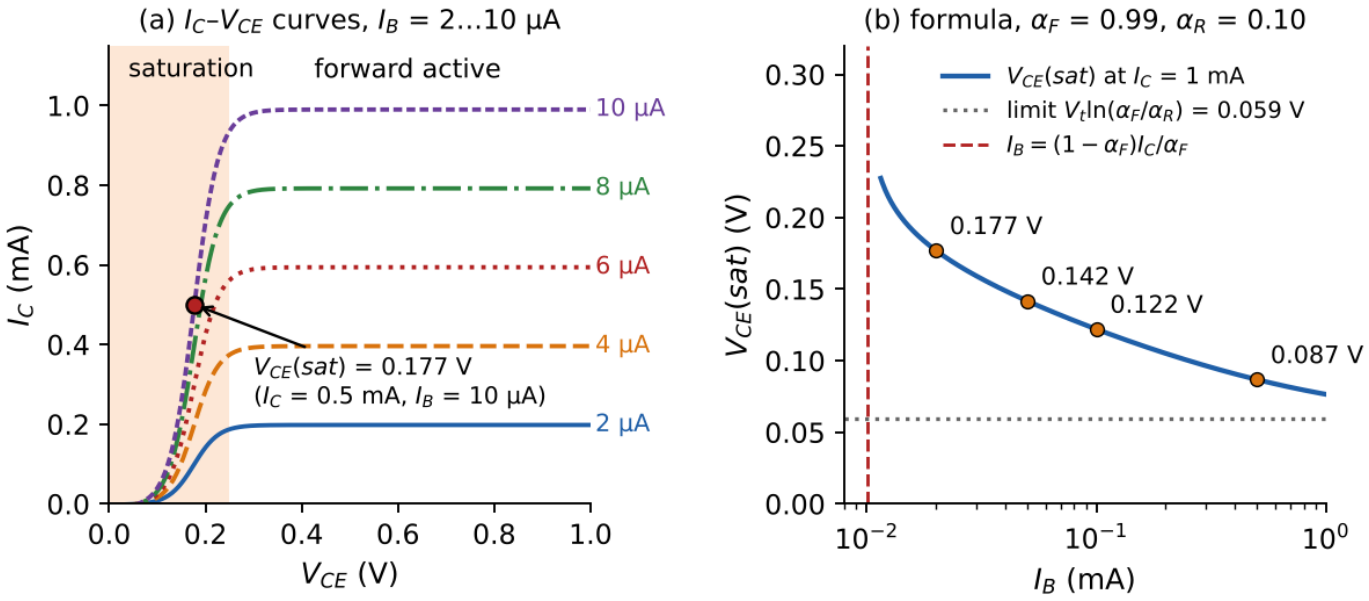


Figure 42: Saturation of the BJT: (a) Ebers-Moll output curves with the saturation region shaded, (b) $V_{CE}(sat)$ against base current from the derived formula

Reading the figure. Panel (a) is drawn from the full Ebers-Moll equations with $\alpha_F = 0.99$ and $\alpha_R = 0.10$, one curve per base current (solid blue 2 µA, dashed orange 4 µA, dotted red 6 µA, dash-dot green 8 µA, short-dashed purple 10 µA). To the right, each curve is flat at $I_C = \beta_F I_B$: forward active. In the shaded region on the left, the B-C junction is also forward biased, its current I_R subtracts from I_C, and the curves fall steeply to zero: this is saturation. The marked point is the formula's $V_{CE}(sat) = 0.177$ V for $I_C = 0.5$ mA and $I_B = 10$ µA. In panel (b), the solid blue curve is the boxed formula at $I_C = 1$ mA: more base drive (deeper saturation) gives a smaller $V_{CE}(sat)$, which approaches the dotted grey limit 0.059 V; the dashed red line is the smallest I_B (0.0101 mA) for which the transistor is saturated at all.

I_B (mA)	I_C/I_B	$V_{CE}(sat)$ (V)
0.02	50	0.177
0.05	20	0.142
0.10	10	0.122
0.50	2	0.087
$\to \infty$	$\to 0$	$V_t \ln(\alpha_F/\alpha_R) = 0.059$

Table 5: V_CE(sat) at I_C = 1 mA for alpha_F = 0.99, alpha_R = 0.10

3. Keyword Explanation

1. Saturation

The mode with both junctions forward biased. The transistor acts as a closed switch; V_{CE} is a small $V_{CE}(sat)$ and $I_C < \beta_F I_B$.

2. Reciprocity relation

$\alpha_F I_{ES} = \alpha_R I_{CS}$. It cuts the four Ebers-Moll parameters to three independent ones.

3. Inverse current gain α_R

The common-base gain with emitter and collector swapped. It is small because the collector is lightly doped and has a large area.

4. Forced beta

The ratio I_C/I_B set by the outside circuit in saturation. It is always less than β_F; the smaller it is, the deeper the saturation and the lower $V_{CE}(sat)$.

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4. Notes and Remarks

- **Only the ratio matters:** dividing the top and bottom of the formula by I_B shows that $V_{CE}(sat)$ depends only on I_C/I_B. That is why $I_C = 0.5$ mA, $I_B = 10$ µA in the figure gives the same 0.177 V as $I_C = 1$ mA, $I_B = 0.02$ mA in the table.
- **The lower limit:** as $I_B \to \infty$, the bracket tends to α_F/α_R, so $V_{CE}(sat) \to V_t \ln(\alpha_F/\alpha_R) = 0.0259 \ln(9.9) = 0.059$ V. A transistor with a larger α_R (more symmetric) can saturate closer to 0 V.
- **Real devices** show a few tenths of a volt more, because the collector and emitter body resistances add an ohmic drop $I_C(r_c + r_e)$ that the ideal model leaves out.
- **Other sign conventions:** some books take I_E as flowing *into* the emitter, which flips the sign of the I_E equation; the $V_{CE}(sat)$ result is the same.

6.(b) Explain high level injection and current crowding effects in a BJT with necessary figures. How can the problem of current crowding be solved? Marks: 8

1. Theory

1. Majority and minority carriers in the base (Core Idea)

The base of an npn transistor is p-type, doped with N_B acceptors per cm^3. So it has about N_B free **holes** (the **majority carriers**) and very few free **electrons** (the **minority carriers**). When the B-E junction is forward biased, the emitter **injects** electrons into the base. Their concentration at the emitter edge of the base is

$$n_p(0) = n_{p0} \exp\left(\frac{V_{BE}}{V_t}\right), \qquad n_{p0} = \frac{n_i^2}{N_B}$$

where n_{p0} is the equilibrium electron concentration in the base. These electrons diffuse across the base to the collector; that is the collector current. See 6(a) for the transistor and its modes.

2. Low-level and high-level injection

- **Low-level injection:** $n_p(0) \ll N_B$. The injected electrons are a tiny addition; the hole concentration stays N_B. All the textbook formulas ($I_C \propto \exp(V_{BE}/V_t)$, constant β) assume this.
- **High-level injection:** at large V_{BE} (large current), $n_p(0)$ grows until it is comparable with N_B. Now the injected electrons are *not* small. To keep the base electrically neutral, an equal extra number of holes must come in from the base contact, so $p_p(0) = N_B + n_p(0)$ rises above N_B.

3. Why the gain falls at high current

Two things go wrong at high injection:

- **Collector current grows more slowly.** The product $n_p(0)\, p_p(0) = n_i^2 \exp(V_{BE}/V_t)$ still holds. When $p_p(0) \approx n_p(0)$, this gives $n_p(0) \approx n_i \exp(V_{BE}/2V_t)$. So I_C rises as $\exp(V_{BE}/2V_t)$ instead of $\exp(V_{BE}/V_t)$: the slope on a log plot halves.
- **Injection efficiency falls.** The base now behaves as if it were more heavily doped (its hole concentration is larger). A more heavily doped base injects more holes back into the emitter. These holes are base current that does not reach the collector, so the **emitter injection efficiency** γ falls, and with it α and $\beta = I_C/I_B$.

At very low currents β also falls, for a different reason: recombination in the B-E depletion region adds a base current that grows as $\exp(V_{BE}/2V_t)$ and dominates when the useful current is small.

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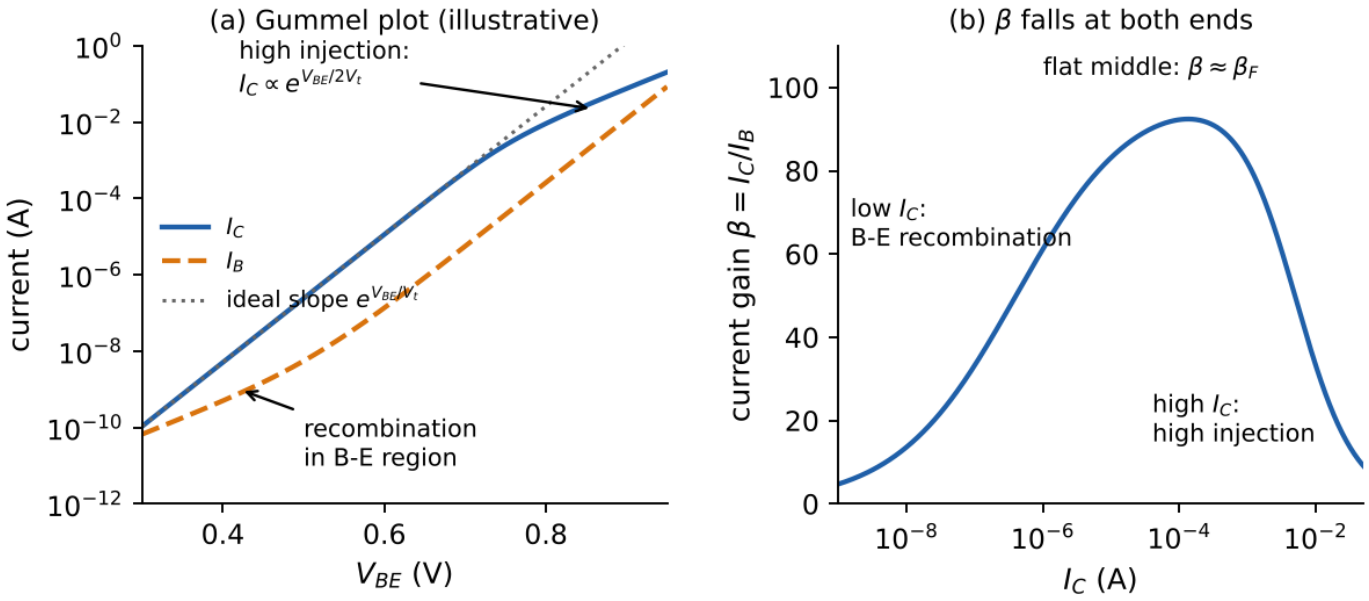


Figure 43: High-level injection: (a) Gummel plot of $\log I_C$ and $\log I_B$ against V_{BE}, (b) current gain β against I_C

Reading the figure. Both panels are drawn from a simple illustrative model, not from a given device. In panel (a), the solid blue I_C follows the dotted grey ideal $\exp(V_{BE}/V_t)$ line over many decades and then bends to half the slope at high V_{BE}: high injection. The dashed orange I_B is bent at low V_{BE} by B-E recombination. The vertical gap between the two curves is $\log \beta$. Panel (b) shows the result as the solid blue β curve: flat in the middle, falling at low I_C (recombination) and falling at high I_C (high injection).

4. What is current crowding?

The base current must reach the active base under the emitter. It enters at the base contact, which sits *beside* the emitter, and flows **sideways** (laterally) through the thin base layer. The thin base has a large resistance, so this sideways current causes an IR voltage drop.

The emitter sits at one voltage. So the actual voltage across the B-E junction is largest at the emitter **edge** (nearest the base contact) and smaller towards the emitter **centre** (furthest from it). Because the injected current depends exponentially on that voltage,

$$J_E(x) \propto \exp\left(\frac{V_{BE}(x)}{V_t}\right)$$

a small drop makes a large difference: a drop of only $V_t \ln 10 = 59.6$ mV reduces the local current density ten times. So the emitter current **crowds** at the edges of the emitter. This is called **emitter current crowding**.

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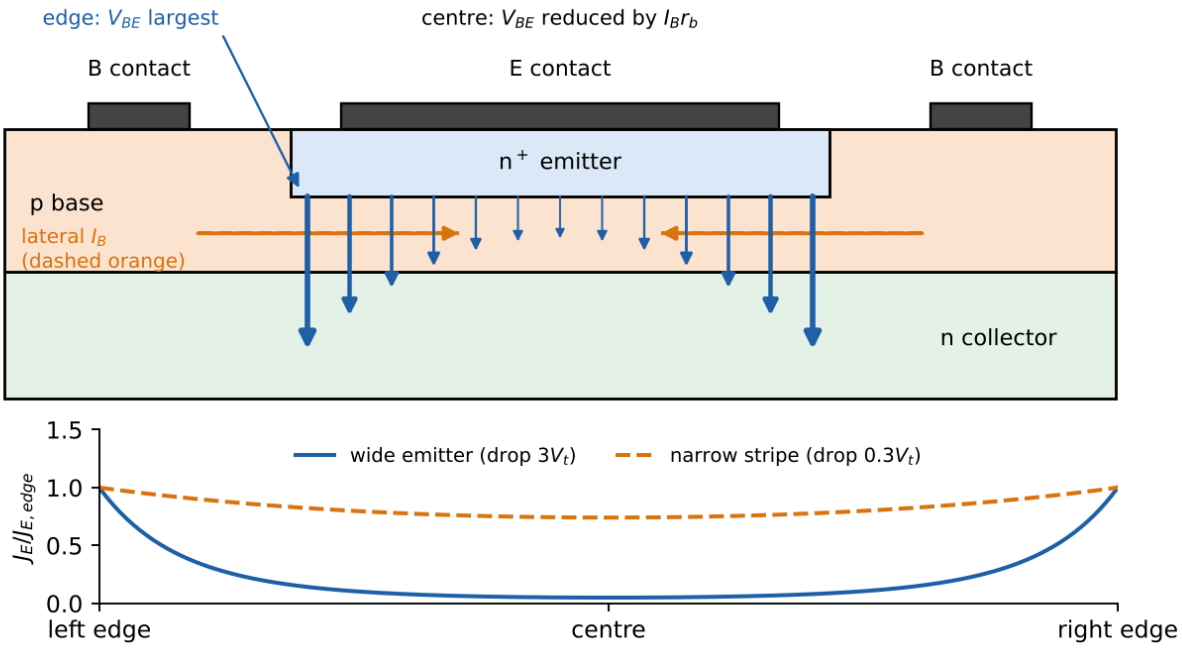


Figure 44: Emitter current crowding: lateral base current under the emitter and the resulting emitter current density across the emitter

Reading the figure. In the cross-section, the base current (dashed orange arrows) flows in from the two base contacts under the emitter. The solid blue arrows show the injected electron current: thick and long at the two edges of the emitter, thin and short at the centre. The lower panel plots the current density across the emitter. For a wide emitter with a lateral drop of $3V_t$ (solid blue), the centre carries only about 5 % of the edge current density: most of the emitter area is wasted. For a narrow stripe with a drop of $0.3V_t$ (dashed orange) the current is nearly uniform.

5. Why current crowding is harmful

- **Wasted area:** the centre of the emitter carries little current, but still adds capacitance.
- **Local high injection:** the edges carry a high current density, so they reach high-level injection (and its fall of β) while the average current is still modest.
- **Hot spots:** heat is made at the edges. A hotter spot conducts more (the B-E voltage for a given current falls by about 2 mV/K), so it takes even more current: **thermal runaway** and **second breakdown** can destroy power transistors.

2. Answer

(i) High-level injection

When the B-E junction is strongly forward biased, the injected minority electron concentration at the emitter edge of the base, $n_p(0) = n_{p0}\exp(V_{BE}/V_t)$, becomes comparable with the base doping N_B. This is **high-level injection**.

- To keep charge neutrality, majority holes enter the base in equal number: $p_p(0) = N_B + n_p(0) > N_B$. The effective base doping rises.
- From $n_p p_p = n_i^2 \exp(V_{BE}/V_t)$ with $p_p \approx n_p$: $n_p(0) \approx n_i \exp(V_{BE}/2V_t)$, so

$$I_C \propto \exp\left(\frac{V_{BE}}{2V_t}\right) \quad \text{(high injection)} \qquad \text{instead of} \qquad I_C \propto \exp\left(\frac{V_{BE}}{V_t}\right)$$

- The higher effective base doping increases hole injection from the base into the emitter, so the **emitter injection efficiency** falls and the **current gain β falls at high I_C** (Gummel plot and β–I_C curve in the Theory: the solid blue I_C line bends to half slope and β drops at the high-current end).

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(ii) Current crowding

- The base current flows **laterally** from the base contact through the thin, resistive base under the emitter. Its IR drop makes the forward bias V_{BE} largest at the emitter edges and smallest at the centre.
- Since $J_E \propto \exp(V_{BE}/V_t)$, the emitter current concentrates at the **emitter edges** next to the base contacts (cross-section figure in the Theory: thick solid blue arrows at the edges).
- **Effects:** the emitter centre is wasted; the edges go into local high injection (lower β) and overheat, causing **hot spots**, thermal runaway and **second breakdown** in power transistors.

(iii) Remedies for current crowding

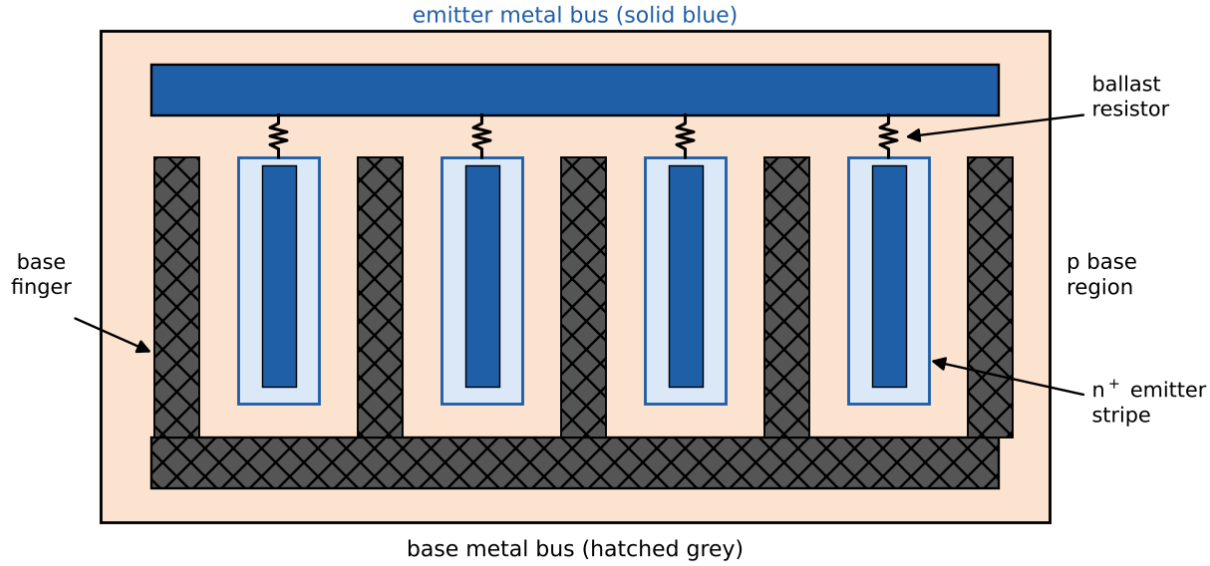


Figure 45: Interdigitated emitter-base layout of a power BJT, top view, with ballast resistors in the emitter fingers

Reading the figure. The top view shows many narrow n^+ emitter stripes (light blue with a solid blue metal strip) hanging from the emitter bus, interleaved with hatched grey base fingers rising from the base bus. Every emitter edge is close to a base contact, so the lateral path is short. The small zig-zag resistor at the top of each emitter finger is a ballast resistor.

- **Interdigitated (comb) geometry:** split one wide emitter into many **narrow emitter stripes** placed between base contact fingers. The lateral path, and so the IR drop, is small, and the current is nearly uniform.
- **Large emitter periphery-to-area ratio:** since the current flows at the edges anyway, design for a lot of edge (long, narrow stripes, or overlay and mesh emitters).
- **Lower base resistance:** heavier doping of the inactive (extrinsic) base under and around the contacts, and base contacts on both sides of each stripe.
- **Emitter ballast resistors:** a small resistor in series with each emitter finger. A finger that starts to hog current gets a larger drop across its resistor, which lowers its own V_{BE}: negative feedback that shares the current evenly and prevents hot spots.

3. Keyword Explanation

1. Emitter injection efficiency

The fraction of the emitter current carried by electrons injected into the base (npn), rather than by holes injected from the base into the emitter. It falls when the base becomes effectively more heavily doped.

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2. Gummel plot

A plot of $\log I_C$ and $\log I_B$ against V_{BE}. Departures from the ideal slope show recombination (low current) and high injection (high current).

3. Second breakdown

A sudden fall in collector voltage and a rise in current at a hot spot, caused by local heating that feeds on itself. It often destroys the transistor.

4. Ballast resistor

A small series resistor in each emitter segment that gives local negative feedback and forces equal current sharing.

4. Notes and Remarks

- **Kirk effect:** at high current densities the electron charge in the B-C depletion region also pushes the base edge into the collector (base widening). It is another cause of β falling at high current, separate from high injection in the base.
- **Crowding and high injection help each other:** crowding raises the local current density at the edges, so high injection starts there first.
- **Number check:** a lateral drop of only about 60 mV ($V_t \ln 10$) already makes the edge current density ten times the centre value, so even a small base resistance matters at large currents.

6.(c) Consider a silicon n-channel JFET at $T = 300$ K with the following parameters: $N_a = 1 \times 10^{18}$ cm^{-3}, $N_d = 1 \times 10^{16}$ cm^{-3}, $a = 0.75$ μm, $L = 10$ μm, $W = 30$ μm, and $\mu_n = 1000$ cm^2/(V s) (i) Calculate the maximum current in an n-channel JFET. (ii) Determine the maximum transconductance of an n-channel depletion mode JFET biased in the saturation region. $I_{P1} = 0.522$ mA, $V_{bi} = 0.814$ V, and $V_{po} = 4.35$ V. Marks: 8

1. Theory

1. The JFET in one paragraph (Core Idea)

A **junction field-effect transistor** (JFET) is an n-type **channel** of thickness a, length L and width W, with a heavily doped p$^+$ **gate** along it. Current flows from **source** to **drain** through the channel. The p$^+$n gate junction is reverse biased, so a **depletion region** grows from the gate into the channel and squeezes it. A more negative gate voltage V_{GS} means a thicker depletion region, a thinner channel and less current. When the depletion region fills the whole channel, the channel is **pinched off**. The channel profiles are drawn in 5(a), and the **internal pinch-off voltage** is derived in 5(b):

$$V_{p0} = \frac{e\, a^2 N_d}{2\varepsilon_s}$$

where V_{p0} is the total junction potential ($V_{bi} - V_{GS}$, plus the drain part) that just depletes the full channel thickness a.

2. The built-in potential

Even at zero bias, the gate junction has a **built-in potential** V_{bi}, so the channel is partly depleted already:

$$V_{bi} = V_t \ln\left(\frac{N_a N_d}{n_i^2}\right)$$

The total reverse voltage across the gate junction at the source end is $V_{bi} - V_{GS}$. Pinch-off at the source end happens when $V_{bi} - V_{GS} = V_{p0}$, so the **pinch-off gate voltage** is $V_P = V_{bi} - V_{p0}$.

3. The current scale I_{P1}

If the channel were never depleted, its conductance would be

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$$G_{01} = \frac{e\,\mu_n N_d\, a\, W}{L}$$

Neamen's long-channel analysis of the JFET gives the drain current in units of a reference current

$$I_{P1} = \frac{\mu_n (eN_d)^2\, W a^3}{6\,\varepsilon_s L} = \frac{G_{01} V_{p0}}{3}$$

Neamen calls I_{P1} the **pinch-off current**. It is an upper bound: it is the current scale the device would have if the channel kept its full thickness a. It sets the size of every current the JFET carries.

4. Saturation current and transconductance

When the drain end of the channel is pinched off, the current stops rising with V_{DS} (saturation, 5(a)). Its value then depends only on V_{GS} (Neamen):

$$I_{D1}(sat) = I_{P1}\left[1 - 3\left(\frac{V_{bi} - V_{GS}}{V_{p0}}\right)\left(1 - \frac{2}{3}\sqrt{\frac{V_{bi} - V_{GS}}{V_{p0}}}\right)\right]$$

The **transconductance** tells how strongly the gate controls the current: $g_{ms} = \partial I_{D1}(sat)/\partial V_{GS}$. Write $r = (V_{bi} - V_{GS})/V_{p0}$, so the bracket is $1 - 3r + 2r^{3/2}$ and $dr/dV_{GS} = -1/V_{p0}$:

$$g_{ms} = I_{P1}\left(-3 + 3r^{1/2}\right)\left(-\frac{1}{V_{p0}}\right) = \frac{3I_{P1}}{V_{p0}}\left(1 - \sqrt{\frac{V_{bi} - V_{GS}}{V_{p0}}}\right)$$

Both $I_{D1}(sat)$ and g_{ms} grow as V_{GS} rises towards 0 (the channel opens). For a depletion-mode JFET the gate must stay reverse biased (a forward-biased gate draws current), so the largest allowed value is $V_{GS} = 0$: both the **maximum current** and the **maximum transconductance** are at $V_{GS} = 0$.

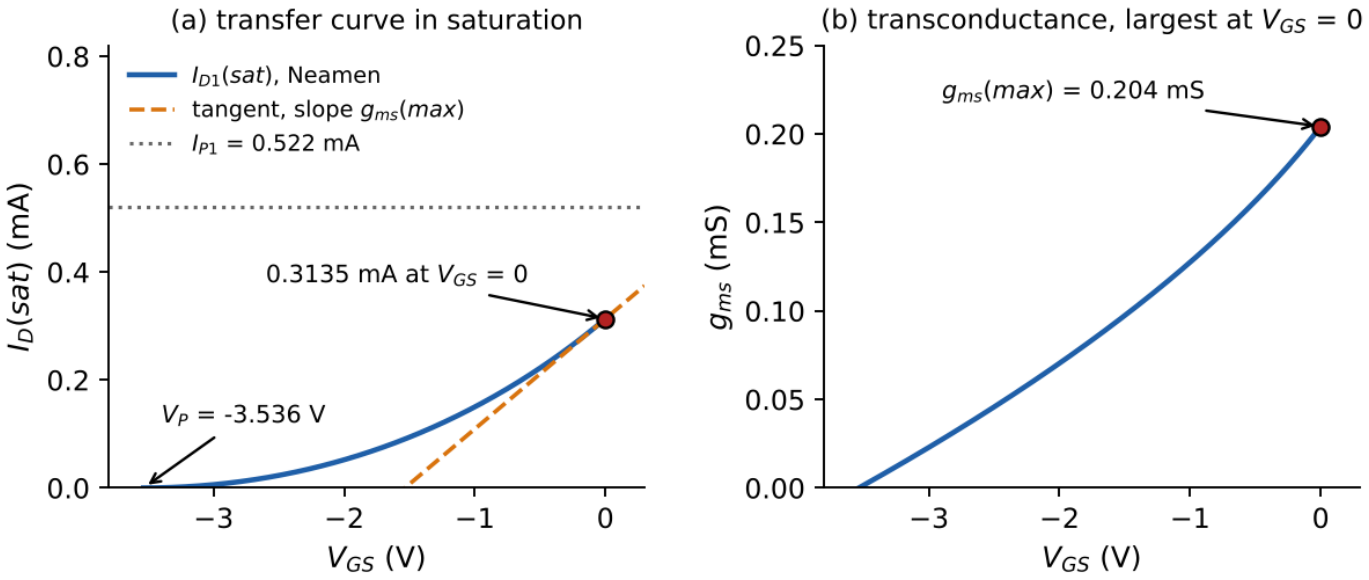


Figure 46: JFET in saturation: (a) transfer curve $I_D(sat)$ against V_{GS} with its tangent at $V_{GS} = 0$, (b) transconductance against V_{GS}

Reading the figure. In panel (a), the solid blue curve is Neamen's $I_{D1}(sat)$ with the given values. It starts at zero at the pinch-off gate voltage $V_P = -3.536$ V and rises to 0.3135 mA at $V_{GS} = 0$ (marked point). The dotted grey line is $I_{P1} = 0.522$ mA; the real saturation current never reaches it, because even at $V_{GS} = 0$ the built-in potential depletes part of the channel. The dashed orange line is the tangent at $V_{GS} = 0$; its slope is $g_{ms}(max)$. Panel (b) plots that slope against V_{GS} (solid blue): it is zero at pinch-off and largest, 0.204 mS, at $V_{GS} = 0$.

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2. Answer

Given:

- Silicon, $T = 300$ K: $\varepsilon_s = 11.7\varepsilon_0 = 11.7\times 8.85\times 10^{-14} = 1.0355\times 10^{-12}$ F/cm, $n_i = 1.5\times 10^{10}$ cm^{-3}, $V_t = 0.0259$ V, $e = 1.602\times 10^{-19}$ C
- $N_a = 10^{18}$ cm^{-3} (p$^+$ gate), $N_d = 10^{16}$ cm^{-3} (n channel)
- $a = 0.75\ \mu$m $= 0.75\times 10^{-4}$ cm, $L = 10\ \mu$m $= 10\times 10^{-4}$ cm, $W = 30\ \mu$m $= 30\times 10^{-4}$ cm
- $\mu_n = 1000$ cm^2/V s
- Given in (ii): $I_{P1} = 0.522$ mA, $V_{bi} = 0.814$ V, $V_{p0} = 4.35$ V

(i) Maximum current

Step 1: Pinch-off current I_{P1}.

$$I_{P1} = \frac{\mu_n (eN_d)^2\, W a^3}{6\,\varepsilon_s L}$$

$$eN_d = (1.602\times 10^{-19})(10^{16}) = 1.602\times 10^{-3}\ \text{C/cm}^3$$

$$I_{P1} = \frac{(1000)(1.602\times 10^{-3})^2(30\times 10^{-4})(0.75\times 10^{-4})^3}{6\,(1.0355\times 10^{-12})(10\times 10^{-4})}$$

$$= \frac{3.248\times 10^{-18}}{6.213\times 10^{-15}} = 5.228\times 10^{-4}\ \text{A}$$

$$\boxed{I_{P1} = 0.522\ \text{mA}}$$

This is the value the question quotes in (ii), and it is the maximum-current scale of the JFET in Neamen's treatment (the current of a fully open channel). Units: (cm^2/V s)(C^2/cm^6)(cm)(cm^3)/(F/cm·cm) = C^2/(V s F) = C/s = A.

Step 2: Check of V_{bi} and V_{p0} (needed for the actual current):

$$V_{bi} = 0.0259\ln\frac{(10^{18})(10^{16})}{(1.5\times 10^{10})^2} = 0.0259\times 31.43 = 0.814\ \text{V}$$

$$V_{p0} = \frac{e\,a^2 N_d}{2\varepsilon_s} = \frac{(1.602\times 10^{-19})(0.75\times 10^{-4})^2(10^{16})}{2(1.0355\times 10^{-12})} = \frac{9.011\times 10^{-12}}{2.071\times 10^{-12}} = 4.35\ \text{V}$$

Both agree with the values given in (ii).

Step 3: Largest saturation current actually carried ($V_{GS} = 0$). With $V_{bi}/V_{p0} = 0.814/4.35 = 0.1871$ and $\sqrt{0.1871} = 0.4326$:

$$I_{D1}(sat)\big|_{V_{GS}=0} = I_{P1}\left[1 - 3(0.1871)\left(1 - \tfrac{2}{3}(0.4326)\right)\right]$$

$$= 0.522\,[1 - 0.5614\times 0.7116] = 0.522\times 0.6005$$

$$\boxed{I_{D1}(sat)_{max} = 0.3135\ \text{mA}}$$

The built-in potential already depletes part of the channel at $V_{GS} = 0$, so the real maximum saturation current is about 60 % of I_{P1}.

(ii) Maximum transconductance

$$g_{ms} = \frac{3I_{P1}}{V_{p0}}\left(1 - \sqrt{\frac{V_{bi} - V_{GS}}{V_{p0}}}\right)$$

It is largest at $V_{GS} = 0$ (the most open channel allowed with the gate not forward biased):

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$$g_{ms}(max) = \frac{3(0.522\text{ mA})}{4.35\text{ V}}\left(1-\sqrt{\frac{0.814}{4.35}}\right)$$
$$= (0.360\text{ mA/V})(1-0.4326) = (0.360)(0.5674)$$
$$\boxed{g_{ms}(max) = 0.204\text{ mA/V} = 0.204\text{ mS}}$$

A 1 V change of gate voltage near $V_{GS}=0$ changes the drain current by about 0.2 mA.

Final Summary:

- $I_{P1} = 0.522$ mA (maximum current scale, Neamen's pinch-off current)
- Largest saturation current, at $V_{GS}=0$: $I_{D1}(sat) = 0.3135$ mA
- $g_{ms}(max) = 0.204$ mS at $V_{GS}=0$; pinch-off gate voltage $V_P = V_{bi} - V_{p0} = -3.536$ V

3. Keyword Explanation

1. Pinch-off current I_{P1}

$\mu_n(eN_d)^2Wa^3/(6\varepsilon_s L) = G_{01}V_{p0}/3$, the current scale of an undepleted channel. All JFET currents are a fraction of it.

2. Transconductance g_{ms}

$\partial I_D/\partial V_{GS}$ in saturation, in siemens (A/V). It is the gain parameter of the JFET as an amplifier.

3. Depletion-mode JFET

A JFET that conducts at $V_{GS}=0$ and is turned off by a negative gate voltage (n-channel). Its gate is never forward biased in normal use.

4. Notes and Remarks

- **Two readings of "maximum current":** the question's own data ($I_{P1}=0.522$ mA given in (ii)) show that (i) means I_{P1}, as in Neamen's example. Physically the device cannot quite carry it: the largest saturation current is $I_{D1}(sat)$ at $V_{GS}=0$, 0.3135 mA. Both are given above; write both in the exam.
- **A neat check:** $3I_{P1}/V_{p0} = G_{01} = e\mu_n N_d aW/L = 0.360$ mS, the conductance of the undepleted channel. So $g_{ms}(max) = G_{01}(1-\sqrt{V_{bi}/V_{p0}})$: the transconductance can never exceed the full-channel conductance.
- **Per unit width:** $g_{ms}(max)/W = 0.204\text{ mS}/30\ \mu\text{m} = 6.8$ mS/mm, a typical figure for a silicon JFET with this doping.
- **Rounding:** with the unrounded values ($I_{P1}=0.5228$ mA, $V_{bi}=0.8139$ V, $V_{p0}=4.3514$ V), $g_{ms}(max)=0.2046$ mS; the given rounded data give 0.2043 mS. Both are 0.20 mS to two figures.

5. Final Concise Answer

(a) The Ebers-Moll model of an npn BJT is a B-E diode $I_F = I_{ES}[\exp(V_{BE}/V_t)-1]$ and a B-C diode $I_R = I_{CS}[\exp(V_{BC}/V_t)-1]$ back to back, with a current source $\alpha_F I_F$ across the B-C diode and $\alpha_R I_R$ across the B-E diode, and $\alpha_F I_{ES} = \alpha_R I_{CS}$. Then $I_C = \alpha_F I_F - I_R$, $I_E = I_F - \alpha_R I_R$, $I_B = I_E - I_C$. In saturation (both junctions forward), solving for I_F, I_R and using $V_{CE} = V_{BE} - V_{BC}$ gives $V_{CE}(sat) = V_t \ln\{[I_C(1-\alpha_R)+I_B]\alpha_F/([\alpha_F I_B - (1-\alpha_F)I_C]\alpha_R)\}$, valid for $I_C < \beta_F I_B$. Example ($\alpha_F=0.99$, $\alpha_R=0.1$, $I_C=1$ mA, $I_B=0.05$ mA): $V_{CE}(sat)=0.142$ V; lower limit $V_t\ln(\alpha_F/\alpha_R)=0.059$ V.

(b) High-level injection: when the injected electron density in the base approaches N_B, extra holes enter for neutrality, the effective base doping rises, injection efficiency falls, I_C rises only as $\exp(V_{BE}/2V_t)$ and β falls at high current. Current crowding: the lateral base current's IR drop makes V_{BE} largest at the emitter edges, so the emitter current crowds at the edges, giving local high injection, hot spots and second breakdown. Remedies: interdigitated narrow emitter stripes between base fingers (high periphery-to-area ratio), low base resistance, and emitter ballast resistors.

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(c) (i) $I_{P1} = \mu_n (eN_d)^2 W a^3/(6\varepsilon_s L) = 0.522$ mA (maximum current scale); the largest saturation current actually carried, at $V_{GS} = 0$, is $I_{D1}(sat) = 0.3135$ mA. (ii) $g_{ms}(max) = (3I_{P1}/V_{p0})(1 - \sqrt{V_{bi}/V_{p0}}) = 0.204$ mS at $V_{GS} = 0$.

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FORMULA SHEET

EEE459

Compound Semiconductor and Hetero-Junction Devices

Formula Sheet

Session 2020–21 paper

Every formula behind the 4th Year 1st Semester Final Examination, 2024

TOPICS

1 Carrier transport — 4 formulas

2 Bands and density of states — 7 formulas

3 Compound alloys — 5 formulas

4 Heterojunctions — 8 formulas

5 Metal–semiconductor contacts — 9 formulas

6 Graded junction — 3 formulas

7 JFET — 8 formulas

8 Bipolar transistor — 4 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

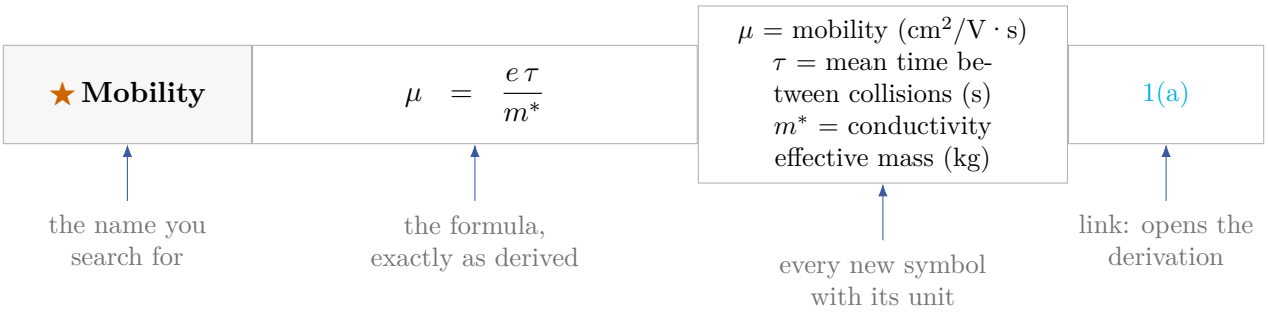
Prepared by Nur Mohammad

October 2026

BACK OF THE COVER

Before you use the sheet

How to read a row



In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 Carrier transport 4

2 Bands and density of states 4

3 Compound alloys 5

4 Heterojunctions 5

5 Metal–semiconductor contacts 6

6 Graded junction 6

7 JFET 7

8 Bipolar transistor 7

9 Key formulas 8

10 Symbol index 9

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Conventions (Neamen): e is the magnitude of the electron charge, potentials (ϕ, V) are in volts and energies (E, χe) in eV; $kT/e = V_t = 0.0259$ V at 300 K; $\epsilon_s = \epsilon_r \epsilon_0$ with $\epsilon_0 = 8.85 \times 10^{-14}$ F/cm; $n_i = 1.5 \times 10^{10}$ cm^{-3} (Si) and 1.8×10^6 cm^{-3} (GaAs). Subscript 1 is the narrow-gap side of a heterojunction. References point to the solutions of the 2020–21 paper.

Columns: Title · Equation · Notation · Ref. (reference).

1. Carrier transport:

Title	Equation	Notation	Ref.
★ **Mobility**	$\mu = \dfrac{e\,\tau}{m^*}$	μ = mobility (cm^2/V · s) τ = mean time between collisions (s) m^* = conductivity effective mass (kg)	1(a)
Drift velocity, low field	$v_d = \mu E$	v_d = drift velocity (cm/s) E = electric field (V/cm)	1(b)
Saturation velocity	$v_d \to v_{sat} \approx 10^7$ cm/s	reached in Si above about 3×10^4 V/cm	1(b)
Gunn oscillation	$f = \dfrac{v_d}{L}$	f = transit-time frequency (Hz) L = length of the GaAs sample (cm)	1(b)

Table 1: Carrier transport

2. Bands and density of states:

Title	Equation	Notation	Ref.
★ **Parabolic band**	$E = \dfrac{\hbar^2 k^2}{2m^*}$	E = energy from the band edge (J) k = wave number (m^{-1}) $\hbar$ = reduced Planck constant (J · s)	1(c)
Slope and curvature	$\dfrac{dE}{dk} = \dfrac{\hbar^2 k}{m^*}$, $\dfrac{d^2E}{dk^2} = \dfrac{\hbar^2}{m^*}$	the lighter the carrier, the faster E varies with k	1(c)
Si hole masses	$m_{hh,lh}[100] = \dfrac{m_0}{\gamma_1 \mp 2\gamma_2}$, $m_{hh,lh}[111] = \dfrac{m_0}{\gamma_1 \mp 2\gamma_3}$	$\gamma_1 = 4.22$, $\gamma_2 = 0.39$, $\gamma_3 = 1.44$ (Luttinger parameters of Si)	1(c)
★ **Density of states, 3D**	$g_{3D}(E) = \dfrac{4\pi(2m^*)^{3/2}}{h^3}\sqrt{E - E_c}$	g = states per unit energy per unit volume E_c = conduction-band edge	3(b)
Density of states, 2D	$g_{2D}(E) = \dfrac{m^*}{\pi\hbar^2}$ per subband	constant: a staircase, one step per subband E_n	3(b)
Density of states, 1D	$g_{1D}(E) \propto \dfrac{1}{\sqrt{E - E_n}}$	a spike at each subband edge E_n	3(b)
★ **Emission wavelength**	$\lambda\ (\mu\text{m}) = \dfrac{1.24}{E_g\ (\text{eV})}$	λ = wavelength of a photon of energy E_g	2(a)

Table 2: Bands and density of states

3. Compound alloys:

Title	Equation	Notation	Ref.
★ **Vegard's law, ternary**	$a(A_xB_{1-x}C) = x\,a_{AC} + (1-x)\,a_{BC}$	a = lattice constant (Å) x = mole fraction; the band gap is interpolated the same way (no bowing)	2(d)
★ **Quaternary** $In_xGa_{1-x}As_ySb_{1-y}$	$P = xy\,P_{InAs} + x(1-y)\,P_{InSb} + (1-x)y\,P_{GaAs} + (1-x)(1-y)\,P_{GaSb}$	P = lattice constant or band gap of the alloy, from the four binaries	2(d)
AlGaAs band gap	$E_g(Al_xGa_{1-x}As) = 1.424 + 1.247x$ eV	valid for $x < 0.45$ (direct gap)	1(d)
★ **Lattice misfit**	$f = \dfrac{a_{layer} - a_{sub}}{a_{sub}}$	a_{layer} = layer lattice constant a_{sub} = substrate lattice constant	2(b)
Critical thickness, estimate	$h_c \sim \dfrac{a_{sub}}{2\lvert f\rvert}$	order of magnitude only; thicker layers relax by misfit dislocations	2(b)

Table 3: Compound alloys

4. Heterojunctions:

Title	Equation	Notation	Ref.
★ **Band offsets (Anderson)**	$\Delta E_c = e(\chi_n - \chi_w),\ \Delta E_g = \Delta E_c + \Delta E_v$	χ_n, χ_w = electron affinities of the narrow- and wide-gap materials (V) ΔE_c, ΔE_v = conduction- and valence-band offsets (eV)	1(d)
Hole concentration	$p = N_v \exp\left[-\dfrac{E_F - E_v}{kT}\right]$	N_v = valence-band effective density of states (cm^{-3})	1(d)
★ **Built-in potential, pN**	$eV_{bi} = -\Delta E_v + kT \ln\left(\dfrac{p_P\,N_{vN}}{p_N\,N_{vP}}\right)$	p_P, p_N = hole concentrations in the P and N regions N_{vP}, N_{vN} = their valence-band densities of states	1(d)
Field, N side	$E_1(x) = \dfrac{eN_{d1}}{\epsilon_1}(x + x_n)$	$-x_n \le x \le 0$, N side on the left, field along $+x$; N_{d1} = donors, ϵ_1 = permittivity of side 1	4(a)
Field, P side	$E_2(x) = \dfrac{eN_{a2}}{\epsilon_2}(x_p - x)$	$0 \le x \le x_p$; N_{a2} = acceptors, ϵ_2 = permittivity of side 2	4(a)
★ **Interface condition**	$\epsilon_1 E_1(0) = \epsilon_2 E_2(0) \Rightarrow N_{d1}x_n = N_{a2}x_p$	the displacement field is continuous; E jumps if $\epsilon_1 \neq \epsilon_2$	4(a)
Potential split	$V_{bi1} = \dfrac{eN_{d1}x_n^2}{2\epsilon_1},\ V_{bi2} = \dfrac{eN_{a2}x_p^2}{2\epsilon_2},\ V_{bi} = V_{bi1} + V_{bi2}$	V_{bi1}, V_{bi2} = parts of V_{bi} dropped on each side	4(a)
Depletion width, N side	$x_n = \left[\dfrac{2\epsilon_1\epsilon_2 N_{a2} V_{bi}}{eN_{d1}(\epsilon_1 N_{d1} + \epsilon_2 N_{a2})}\right]^{1/2}$	x_p: swap $N_{d1} \leftrightarrow N_{a2}$ in the fraction's doping factors	4(a)

Table 4: Heterojunctions

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5. Metal–semiconductor contacts:

Title	Equation	Notation	Ref.
★ **Barrier height (ideal)**	$\phi_{B0} = \phi_m - \chi$	ϕ_m = metal work function (V) χ = electron affinity (V)	3(c)
★ **Built-in potential**	$V_{bi} = \phi_{B0} - \phi_n,\ \phi_n = \frac{kT}{e} \ln \frac{N_c}{N_d}$	ϕ_n = depth of E_F below E_c (V) N_c = conduction-band density of states (cm^{-3})	3(c)
Depletion width	$x_n = \left[\frac{2\epsilon_s(V_{bi}+V_R)}{eN_d}\right]^{1/2}$	V_R = reverse bias (V); $V_R = -V_a$ in forward bias	3(c)
★ **Maximum field**	$E_{max} = \frac{eN_d x_n}{\epsilon_s} = \frac{2(V_{bi}+V_R)}{x_n}$	at the metal–semiconductor interface (V/cm)	3(c)
Junction capacitance	$C' = \frac{\epsilon_s}{x_n} = \left[\frac{e\epsilon_s N_d}{2(V_{bi}+V_R)}\right]^{1/2}$	C' = capacitance per unit area ($\mathrm{F/cm}^2$)	3(c)
★ **Image-force lowering**	$\Delta\phi = \sqrt{\frac{eE}{4\pi\epsilon_s}}$	$\Delta\phi$ = barrier lowering (V) E = field at the interface (V/cm)	3(d)
Position of the peak	$x_m = \sqrt{\frac{e}{16\pi\epsilon_s E}}$	x_m = distance of the barrier maximum from the metal (cm)	3(d)
Barrier with interface states	$\phi_{B0} = \gamma(\phi_m - \chi) + (1-\gamma)\left(\frac{E_g}{e} - \phi_0\right)$	$\gamma = 1$ without interface states, 0 when the Fermi level is pinned ϕ_0 = neutral level of the interface states above E_v (V)	4(b)
Ohmic contact (ideal)	$\phi_m < \phi_s$ (n-type), $\phi_m > \phi_s$ (p-type)	ϕ_s = semiconductor work function (V)	5(d)

Table 5: Metal–semiconductor contacts

6. Graded junction:

Title	Equation	Notation	Ref.
Net doping	$N_d - N_a = a\,x$	a = grading constant (cm^{-4})	4(c)
★ **Depletion width**	$W = \left[\frac{12\epsilon_s(V_{bi}+V_R)}{ea}\right]^{1/3}$	W = total depletion width (cm)	4(c)
Junction capacitance	$C' = \left[\frac{ea\epsilon_s^2}{12(V_{bi}+V_R)}\right]^{1/3}$	varies as $(V_{bi}+V_R)^{-1/3}$ (abrupt: $-1/2$)	4(c)

Table 6: Graded junction

7. JFET:

Title	Equation	Notation	Ref.
pn built-in potential	$V_{bi} = V_t \ln \dfrac{N_a N_d}{n_i^2}$	N_a = gate acceptors, N_d = channel donors (cm^{-3})	5(c)
★ **Space-charge width**	$h = \left[\dfrac{2\epsilon_s(V_{bi} - V_{GS})}{eN_d}\right]^{1/2}$	h = depletion width into the channel (cm) V_{GS} = gate–source voltage (V)	5(b)
★ **Internal pinch-off voltage**	$V_{p0} = \dfrac{ea^2 N_d}{2\epsilon_s}$	a = metallurgical channel thickness (cm)	5(b)
Pinch-off voltage	$V_p = V_{bi} - V_{p0}$	gate voltage that just closes the channel at zero drain bias	5(b)
Saturation drain voltage	$V_{DS}(sat) = V_{p0} - (V_{bi} - V_{GS})$	the channel pinches off at the drain end	5(a)
★ **Pinch-off current**	$I_{P1} = \dfrac{\mu_n (eN_d)^2 W a^3}{6\epsilon_s L}$	W = channel width, L = channel length (cm) μ_n = electron mobility ($\mathrm{cm}^2/\mathrm{V}\cdot\mathrm{s}$)	6(c)
Saturation current	$I_{D1}(sat) = I_{P1}\left[1 - 3\left(\dfrac{V_{bi} - V_{GS}}{V_{p0}}\right)\left(1 - \dfrac{2}{3}\sqrt{\dfrac{V_{bi} - V_{GS}}{V_{p0}}}\right)\right]$	largest at $V_{GS} = 0$	6(c)
★ **Transconductance**	$g_{ms} = \dfrac{3I_{P1}}{V_{p0}}\left(1 - \sqrt{\dfrac{V_{bi} - V_{GS}}{V_{p0}}}\right)$	g_{ms} = saturation transconductance (S); largest at $V_{GS} = 0$	6(c)

Table 7: JFET

8. Bipolar transistor:

Title	Equation	Notation	Ref.
★ **Ebers–Moll currents**	$I_C = \alpha_F I_F - I_R,\ I_E = I_F - \alpha_R I_R,\ I_B = I_E - I_C$	α_F, α_R = forward and reverse common-base gains	6(a)
Diode currents	$I_F = I_{ES}\left[e^{V_{BE}/V_t} - 1\right],\ I_R = I_{CS}\left[e^{V_{BC}/V_t} - 1\right]$	I_{ES}, I_{CS} = reverse saturation currents of the B–E and B–C junctions (A)	6(a)
Reciprocity	$\alpha_F I_{ES} = \alpha_R I_{CS}$		6(a)
★ **Saturation voltage**	$V_{CE}(sat) = V_t \ln\left[\dfrac{I_C(1-\alpha_R) + I_B}{\alpha_F I_B - (1-\alpha_F) I_C} \cdot \dfrac{\alpha_F}{\alpha_R}\right]$	needs $\alpha_F I_B > (1-\alpha_F) I_C$	6(a)

Table 8: Bipolar transistor

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9. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Mobility	$\mu = \dfrac{e\,\tau}{m^*}$
Parabolic band	$E = \dfrac{\hbar^2 k^2}{2m^*}$
Density of states, 3D	$g_{3D}(E) = \dfrac{4\pi(2m^*)^{3/2}}{h^3}\sqrt{E - E_c}$
Emission wavelength	$\lambda\ (\mu\text{m}) = \dfrac{1.24}{E_g\ (\text{eV})}$
Vegard's law, ternary	$a(A_xB_{1-x}C) = x\,a_{AC} + (1-x)\,a_{BC}$
Quaternary $In_xGa_{1-x}As_ySb_{1-y}$	$P = xy\,P_{InAs} + x(1-y)\,P_{InSb} + (1-x)y\,P_{GaAs} + (1-x)(1-y)\,P_{GaSb}$
Lattice misfit	$f = \dfrac{a_{layer} - a_{sub}}{a_{sub}}$
Band offsets (Anderson)	$\Delta E_c = e(\chi_n - \chi_w),\ \Delta E_g = \Delta E_c + \Delta E_v$
Built-in potential, pN	$eV_{bi} = -\Delta E_v + kT\ln\left(\dfrac{p_P\,N_{vN}}{p_N\,N_{vP}}\right)$
Interface condition	$\epsilon_1 E_1(0) = \epsilon_2 E_2(0) \Rightarrow N_{d1}x_n = N_{a2}x_p$
Barrier height (ideal)	$\phi_{B0} = \phi_m - \chi$
Built-in potential	$V_{bi} = \phi_{B0} - \phi_n,\ \phi_n = \dfrac{kT}{e}\ln\dfrac{N_c}{N_d}$
Maximum field	$E_{max} = \dfrac{eN_dx_n}{\epsilon_s} = \dfrac{2(V_{bi}+V_R)}{x_n}$
Image-force lowering	$\Delta\phi = \sqrt{\dfrac{eE}{4\pi\epsilon_s}}$
Depletion width	$W = \left[\dfrac{12\epsilon_s(V_{bi}+V_R)}{ea}\right]^{1/3}$
Space-charge width	$h = \left[\dfrac{2\epsilon_s(V_{bi}-V_{GS})}{eN_d}\right]^{1/2}$
Internal pinch-off voltage	$V_{p0} = \dfrac{ea^2N_d}{2\epsilon_s}$
Pinch-off current	$I_{P1} = \dfrac{\mu_n(eN_d)^2Wa^3}{6\epsilon_sL}$
Transconductance	$g_{ms} = \dfrac{3I_{P1}}{V_{p0}}\left(1 - \sqrt{\dfrac{V_{bi}-V_{GS}}{V_{p0}}}\right)$
Ebers–Moll currents	$I_C = \alpha_FI_F - I_R,\ I_E = I_F - \alpha_RI_R,\ I_B = I_E - I_C$
Saturation voltage	$V_{CE}(sat) = V_t\ln\left[\dfrac{I_C(1-\alpha_R)+I_B}{\alpha_FI_B-(1-\alpha_F)I_C}\cdot\dfrac{\alpha_F}{\alpha_R}\right]$

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10. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
μ	mobility ($\mathrm{cm}^2/\mathrm{V}\cdot\mathrm{s}$)	Mobility
τ	mean time between collisions (s)	Mobility
m^*	conductivity effective mass (kg)	Mobility
v_d	drift velocity (cm/s)	Drift velocity, low field
E	electric field (V/cm)	Drift velocity, low field
f	transit-time frequency (Hz)	Gunn oscillation
L	length of the GaAs sample (cm)	Gunn oscillation
k	wave number (m^{-1})	Parabolic band
$\hbar$	reduced Planck constant ($\mathrm{J}\cdot\mathrm{s}$)	Parabolic band
g	states per unit energy per unit volume	Density of states, 3D
E_c	conduction-band edge	Density of states, 3D
λ	wavelength of a photon of energy E_g	Emission wavelength
a	lattice constant (Å)	Vegard's law, ternary
x	mole fraction; the band gap is interpolated the same way (no bowing)	Vegard's law, ternary
P	lattice constant or band gap of the alloy, from the four binaries	Quaternary $In_xGa_{1-x}As_ySb_{1-y}$
a_{layer}	layer lattice constant	Lattice misfit
a_{sub}	substrate lattice constant	Lattice misfit
χ_n, χ_w	electron affinities of the narrow- and wide-gap materials (V)	Band offsets (Anderson)
ΔE_c, ΔE_v	conduction- and valence-band offsets (eV)	Band offsets (Anderson)
N_v	valence-band effective density of states (cm^{-3})	Hole concentration
p_P, p_N	hole concentrations in the P and N regions	Built-in potential, pN
N_{vP}, N_{vN}	their valence-band densities of states	Built-in potential, pN
$0 \le x \le x_p$; N_{a2}	acceptors, ϵ_2 = permittivity of side 2	Field, P side
V_{bi1}, V_{bi2}	parts of V_{bi} dropped on each side	Potential split
ϕ_m	metal work function (V)	Barrier height (ideal)
χ	electron affinity (V)	Barrier height (ideal)
ϕ_n	depth of E_F below E_c (V)	Built-in potential
N_c	conduction-band density of states (cm^{-3})	Built-in potential
V_R	reverse bias (V); $V_R = -V_a$ in forward bias	Depletion width
C'	capacitance per unit area (F/cm^2)	Junction capacitance
$\Delta\phi$	barrier lowering (V)	Image-force lowering
x_m	distance of the barrier maximum from the metal (cm)	Position of the peak
γ	1 without interface states, 0 when the Fermi level is pinned	Barrier with interface states

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Symbol	Meaning	First used in
ϕ_0	neutral level of the interface states above E_v (V)	Barrier with interface states
ϕ_s	semiconductor work function (V)	Ohmic contact (ideal)
W	total depletion width (cm)	Depletion width
N_a	gate acceptors, N_d = channel donors (cm^{-3})	pn built-in potential
h	depletion width into the channel (cm)	Space-charge width
V_{GS}	gate–source voltage (V)	Space-charge width
μ_n	electron mobility ($\text{cm}^2/\text{V}\cdot\text{s}$)	Pinch-off current
g_{ms}	saturation transconductance (S); largest at $V_{GS}=0$	Transconductance
α_F, α_R	forward and reverse common-base gains	Ebers–Moll currents
I_{ES}, I_{CS}	reverse saturation currents of the B–E and B–C junctions (A)	Diode currents

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